
How Many Iterations to Jailbreak? Dynamic Budget Allocation for Multi-Turn LLM Evaluation

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Abstract

Evaluating and predicting the performance of large language models (LLMs) in multi-turn conversational settings is critical yet computationally expensive; key events—e.g., jailbreaks or successful task completion by an agent—often emerge only after repeated interactions. These events might be rare, and under any feasible computational budget, remain unobserved. Recent conformal survival frameworks construct reliable lower predictive bounds (LPBs) on the number of iterations to trigger the event of interest, but rely on static budget allocation that is inefficient in multi-turn setups. To address this, we introduce *Dynamic Allocation via Projected Optimization* (DAPRO), the first theoretically valid dynamic budget allocation framework for bounding the time-to-event in multi-turn LLM interactions. We prove that DAPRO satisfies the budget constraint and provides distribution-free, finite-sample coverage guarantees without requiring the conditional independence between censoring and event times assumed by prior conformal survival approaches. A key theoretical contribution is a novel coverage bound that scales with the square root of the mean censoring weight rather than the worst-case weight, yielding provably tighter guarantees than prior work. Furthermore, DAPRO can be employed to obtain unbiased, low-variance estimates of population-level evaluation metrics, such as the jailbreak rate, under limited computing resources. Comprehensive experiments across agentic task success, adversarial jailbreaks, toxic content generation, and RAG hallucinations using LLMs such as Llama 3.1 and Qwen 2.5 demonstrate that DAPRO consistently achieves coverage closer to the nominal level with lower variance than static baselines, while satisfying the budget constraint.

1 Introduction

Large language models (LLMs) are increasingly deployed in high-stakes applications ranging from healthcare and education to customer service [1, 2]. As their use expands, evaluating their safety and utility becomes essential. We define an unsafe response as content that contains toxic language [3], reveals confidential data [4], or produces factual hallucinations [5, 6]. For example, in educational AI tutors or customer service chatbots [7, 8], users rarely elicit harmful behavior with a single query; instead, they iteratively tweak prompts to bypass guardrails [9, 10]. To evaluate safety in these multi-turn adversarial settings, researchers must simulate these interactions under a global computational budget constraint $B \in \mathbb{N}$, e.g., 1,000 exchanges across all conversations. These conversations are monitored by an independent auditor, such as an LLM-as-a-judge [11]. Since these events can be rare,



Figure 1: Illustration of our framework: (i) collecting data via dynamic budget allocation; (ii) calibrating a pre-trained model; and (iii) deploying the model at inference time to serve as a guardrail.¹

particularly for well-aligned models [12], recently introduced *time-to-unsafe-sampling*. In our setting, this metric represents the number of conversational interactions required to elicit an unsafe response given an initial prompt. Utilizing this metric, one can frame LLM prompt risk assessment as a survival analysis task, which is advantageous for two reasons. First, the metric is well-defined even when LLM outputs are time-dependent or adversarial, such as in jailbreaking [9]. Second, it allows constructing a calibrated *lower predictive bound* (LPB) $\hat{L}(X_{\text{test}})$ for the unknown time-to-unsafe-sampling T_{test} of a new test prompt X_{test} with formal, finite-sample coverage guarantees $\mathbb{P}(T_{\text{test}} \geq \hat{L}(X_{\text{test}})) \geq 1 - \alpha$, where $1 - \alpha$ is the user-specified target coverage rate, e.g., $1 - \alpha = 90\%$. Informally, one can reliably expect the LLM to survive at least $\hat{L}(X_{\text{test}}) - 1$ exchanges before yielding unsafe content. With this reliability guarantee, this LPB can be used for safety evaluation and as a proactive guardrail, as a lower value indicates a higher safety risk. While we frame our discussion around safety, this framework applies to evaluating utility, such as constructing an upper predictive bound (UPB) for the *time-to-success* of an agent assisting a target model (see Figure 7 in Appendix B). For simplicity of exposition, we focus primarily on the safety scenario and LPBs hereafter.

To construct this LPB, we must first acquire a dataset of annotated conversations under the budget B , generated by running an attacking algorithm starting from a set of initial prompts, and continuously evaluating the outputs of the target model using the auditor. To satisfy the computational budget constraint, we set a censoring time C_i for each conversation i , which limits the maximum allowed conversation length. [12] proposes an optimized budget allocation strategy that minimizes the variance of the resulting LPB. However, this strategy sets static censoring times that are fixed *before any interaction occurs* and are not updated over time, even as the conversation unfolds. Consequently, this static method is not optimal for our multi-turn setup. Furthermore, it often under-utilizes the budget; when a conversation elicits unsafe content before its allocated budget is fully consumed, the remaining budget that was dedicated to that prompt is lost: it cannot be redistributed to other prompts that may benefit from additional sampling.

However, dynamically allocating the censoring times is mathematically challenging. First, to provide a theoretically valid coverage guarantee for the constructed LPBs, different conversations cannot actively share information or coordinate their decisions to stop or continue, since such communication breaks the independence assumptions required for this theoretical guarantee. Second, the online adaptation of C_i must satisfy the shared global budget constraint B , even though the trajectory and true length of each conversation are initially unknown. Third, the allocation must not merely be valid, but *optimal*, fully utilizing the budget to minimize the variance of the resulting LPBs.

In this work, we tackle these challenges and propose the first adaptive budget allocation framework that dynamically updates C_i while providing a theoretical coverage guarantee and satisfying the global budget constraint. To illustrate the necessity and efficiency of this dynamic adaptation, we conduct an experiment on the RealToxicityPrompts [3] dataset, where both the attacker and the target utilize the Qwen 2.5 14B Instruct [13] model. In Appendix E.4 we provide more details on this setup. Figure 2 shows that the static allocation strategy of [12] sets a static censoring time that does not change as the attacker-target interactions evolve. In contrast, our proposed strategy adapts the censoring times online as the conversation unfolds, achieving a coverage rate closer to the nominal coverage level with a lower variance. We emphasize that obtaining a lower variance is crucial in

¹This figure was generated using Google Gemini based on a prompt designed by the authors and subsequently refined.

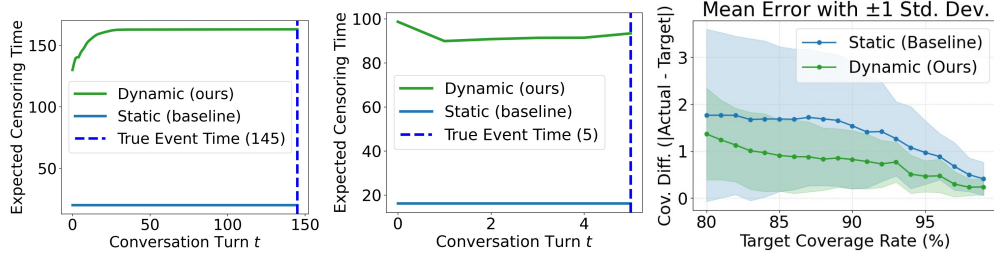


Figure 2: Adaptive vs. static budget allocation. By dynamically adjusting the budget over time (Left+Mid), our adaptive approach achieves lower empirical coverage error (Right).

practice, where only a single dataset is typically available. A lower variance means that our LPBs attain a coverage rate closer to the target level $1 - \alpha$ given the single dataset.

1.1 Our contribution

In this work, we introduce *Dynamic Allocation via PProjected Optimization* (DAPRO), a novel paradigm that treats budget allocation as a sequential decision. In contrast to prior work that sets static censoring times, DAPRO dynamically updates them in response to ongoing conversational interactions. By splitting the available data to first learn an optimal allocation policy and then adaptively deploying it, DAPRO invests computational resources in the most informative exchanges. This reduces the variance of the constructed predictive bounds while satisfying the global budget constraint. Specifically, our main contributions are:

- **Tighter Theoretical Guarantees:** We prove that DAPRO attains the desired coverage level with a bound strictly tighter than prior methods [12, 14]. Crucially, our novel proof technique removes the restrictive assumption that the censoring time C and true event time T must be conditionally independent given the prompt X . This result is of independent interest as it applies to a broad class of survival analysis tasks.
- **Budget Validity :** Under certain assumptions, we formally prove that our dynamic allocation strategy satisfies the global sampling budget constraint in expectation.
- **Population-Level Performance Evaluation Under Censoring:** Beyond individual bounds, DAPRO can efficiently evaluate LLM at the population level, e.g., providing unbiased estimates of the unsafe event rate, reducing the costs of LLM calls (Appendix F.6).
- **Comprehensive Empirical Evaluation:** We validate DAPRO across diverse tasks, including agentic task success, toxic content, adversarial jailbreaks, and RAG-based hallucinations, demonstrating our proposal consistently achieves empirical coverage closer to the nominal level and utilizes the budget more efficiently compared to existing baselines.

Software implementing the proposed method and reproducing our experiments is available at <https://github.com/Shai128/dapro>

1.2 Problem setup

We consider a set of base objectives $\{V_i\}_{i=1}^N$, e.g., harmful requests, from which an attacker model $\mathcal{A}$ generates an initial adversarial prompt for each objective. We define our calibration dataset as $\mathcal{D}_{\text{cal}} = \{X_i\}_{i=1}^N$, where $X_i = \mathcal{A}(V_i)$ is the initial attack prompt for the i -th objective. We assume that the training data $\mathcal{D}_{\text{train}}$, which is used to fit the predictive model, has a separate budget constraint. The combined dataset is denoted as $\mathcal{D}_{\text{all}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{cal}}$. For each prompt in $\mathcal{D}_{\text{cal}}$, an attacker $\mathcal{A}$ interacts sequentially with a target model $\mathcal{G}$ to elicit an unsafe response.

At step $t = 1$, the history is simply the initial prompt: $H_i(1) = \{A_i(1)\}$, where $A_i(1) := X_i = \mathcal{A}(V_i)$. At any step $t \geq 1$, the target model processes this prompt and outputs a response: $R_i(t) = \mathcal{G}(A_i(t))$. Next, we employ an audit function $\mathcal{J}$ to obtain a binary label $Y_i(t) = \mathcal{J}(A_i(t), R_i(t)) \in \{0, 1\}$, where $Y_i(t) = 1$ if $R_i(t)$ is an unsafe response, and $Y_i(t) = 0$ otherwise. If the attack is successful, we terminate this process. Otherwise, we continue to step $t + 1$, where the attacker

generates $A_i(t+1) = \mathcal{A}(H_i(t) \cup \{(R_i(t), Y_i(t))\})$, and we update the interaction history: $H_i(t+1) = H_i(t) \cup \{(R_i(t), Y_i(t), A_i(t+1))\}$. For notational convenience, we set $H_i = \{H_i(t)\}_{t=1}^{t_{\max}}$, where $t_{\max}$ is the maximal sequence length.

We define the uncensored time-to-event as the minimum number of steps required to achieve an unsafe response: $T_i = \min\{t \geq 1 : Y_i(t) = 1\}$. The censored time-to-event is thus $\tilde{T}_i = \min(T_i, C_i)$. Since the budget expended on the i -th sample is $\tilde{T}_i$, the budget constraint can be formulated as:

$$\mathbb{E} \left[\sum_{i=1}^N \tilde{T}_i \right] \leq B. \quad (1)$$

Our goal is to construct an LPB $\hat{L}(X_{\text{test}})$ given an initial test prompt X_{test} for the test event time T_{test} . Importantly, at test time, we construct the LPB *without* running any attacker-target exchanges, relying solely on the initial prompt X_{test} . This design choice reflects two practical considerations. First, a test-time sampling budget may be extremely limited or entirely unavailable. Second, and more critically, when the target model is an autonomous agent operating in a real deployment, we must assess the safety of a prompt *before* the agent has any opportunity to act, as any unsafe action taken by the agent might be irreversible. Finally, given a target coverage level $1 - \alpha \in (0, 1)$ and a tolerance level $\delta \in (0, 1)$, the LPB should satisfy:

$$\mathbb{P} \left(\hat{L}(X_{\text{test}}) \leq T_{\text{test}} \mid \mathcal{D}_{\text{all}} \right) \geq 1 - \alpha, \quad (2)$$

with probability at least $1 - \delta$ over the randomness of the data $\mathcal{D}_{\text{all}}$. We remark that the probability in (2) is taken over random draws of $X_{\text{test}}, T_{\text{test}}$. An LPB $\hat{L}$ satisfying this requirement is called a Probably Approximately Correct (PAC) LPB at level α with tolerance δ .

In Section 3, we introduce our approach for designing the censoring mechanism and allocating the sampling budget to satisfy the budget constraint (1). After obtaining the censoring and survival times, we can construct a calibrated LPB with the approach introduced in [12], which is presented hereafter.

2 Background

Our approach builds on the LLM safety evaluation framework of [12], which refines conformalized survival analysis [14] to construct a valid LPB for the time-to-unsafe under a known censoring mechanism. While we adopt their LPB construction, their budget allocation strategy is static, which is inefficient in our multi-turn setup. We briefly outline their LPB construction procedure below.

Suppose we are given quantile estimates $\hat{q}_\tau(x)$ for the time-to-event from a pre-trained model. Since the raw estimates $\hat{q}_\alpha(x)$ might not be sufficiently accurate, using them directly as LPBs might not achieve $1 - \alpha$ coverage. Thus, we find an adjusted quantile level $\hat{\tau}$ such that the resulting LPB $\hat{q}_{\hat{\tau}}(x)$ satisfies the coverage requirement. We obtain this calibrated quantile level by weighting the errors of the raw quantile estimates to account for the distribution shift induced by the censoring:

$$\hat{\alpha}(\tau) = \frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} w_\tau(i) \mathbb{I}\{\tilde{T}_i < \hat{q}_\tau(X_i) \leq C_i\}, \quad (3)$$

where the weights are formulated as: $w_\tau(i) = \mathbb{P}[\hat{q}_\tau(X_i) \leq C_i \mid \{X_j\}_{j \in \mathcal{I}_{\text{cal}}}]^{-1}$. Notice that these weights are known since the mechanism is actively designed by the algorithm. The calibrated LPB is then $\hat{L}(x) = \hat{q}_{\hat{\tau}}(x)$, where $\hat{\tau}$ is defined as the largest level achieving valid coverage:

$$\hat{\tau} = \sup\{\tau \in \mathcal{T} : \sup_{\tau' < \tau} \hat{\alpha}(\tau') \leq \alpha\},$$

where $\mathcal{T}$ is the search space for τ . As proven in [12], this LPB holds a PAC-type coverage guarantee, where the coverage gap scales with the maximal weight $w_\tau(i)$, such that larger weights lead to a higher variance. Motivated by this insight, [12] introduces two refinements to reduce variance that we adopt. First, the raw estimates are capped at $M \leq t_{\max}$, yielding trimmed estimates $\hat{f}_\tau(x) := \min(\hat{q}_\tau(x), M)$. This trimming bounds the maximal weight, and thus controls the tightness of the PAC-type coverage guarantee. Furthermore, by setting $\hat{L}(x) \leq M \leq t_{\max}$, we can determine

whether T_{test} falls below or above the LPB, so we can correctly evaluate the empirical coverage rate of our LPBs. Thus, throughout this work, we assume $M \leq t_{\text{max}}$. Second, since a conversation i only contributes to $\hat{\alpha}(\tau)$ when $\hat{f}_\tau(X_i) \leq C_i$, the authors propose fixing a prior quantile level τ_{prior} , which represents the prior belief that $\hat{\tau} \in [0, \tau_{\text{prior}}]$. For instance, for $1 - \alpha = 90\%$ and a reasonably accurate model, $\tau_{\text{prior}} = 30\%$ is appropriate since the true α quantile should lie below 30%. They design C_i to maximize $\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i)$, where $\hat{f}_{\text{prior}}(\cdot) := \hat{f}_{\tau_{\text{prior}}}(\cdot)$. This, in turn, maximizes this probability for all $\tau \leq \tau_{\text{prior}}$ by the monotonicity of $\hat{f}_\tau$. Thus, the search space is reduced to $\mathcal{T} = [0, \tau_{\text{prior}}]$. For completeness, this procedure is summarized in Algorithm 1 in Appendix D.1.

Due to space constraints, we include an extended discussion of additional related works in conformal prediction, survival analysis, and LLM safety in Appendix A.

3 Proposed Method

The proposed adaptive budget allocation strategy operates in two phases. Before these phases, we first randomly partition our calibration points, indexed by $\mathcal{I}_{\text{cal}} = \{1, \dots, N\}$, into two disjoint sets, indexed by $\mathcal{I}_{\text{cal1}}$ and $\mathcal{I}_{\text{cal2}}$, of sizes N_1 and N_2 , respectively. We re-index the samples in the first split as $i \in \{1, \dots, N_1\}$ for notational convenience.

In the first phase, we learn an optimal acquisition policy on $\mathcal{I}_{\text{cal1}}$. Our allocation strategy relies on a scoring function $S_t : \mathcal{H}_t \rightarrow \mathbb{R}$ that maps the conversation trajectory observed up to time t to a real-valued score. A higher score indicates a stronger signal to spend a unit of budget and advance the attack to the next step. For example, this score could be the estimated probability of observing an event at the current step, or the expected number of additional iterations required to trigger one. This scoring function can be trained over the training data $\mathcal{D}_{\text{train}}$. Our proposed policy operates iteratively: at each step t , we compute the current risk score $S_i(t)$ and map it to a continuation probability $P_i(t) \in [0, 1]$. Then, with probability $P_i(t)$, we advance to the next step, i.e., acquire a new conversational interaction; otherwise, the acquisition for this sample is terminated. This score-to-probability mapping is optimized over $\mathcal{I}_{\text{cal1}}$ to minimize the variance of our resulting LPBs while satisfying a global budget constraint, similarly to [12]. In the second phase, we deploy this policy on the remaining indices $\mathcal{I}_{\text{cal2}}$ to acquire new interactions. By leveraging the interactions observed in $\mathcal{I}_{\text{cal1}}$, we selectively terminate conversations that are unlikely to contribute to reducing the LPB variance.

3.1 Phase I: Learning the Optimal Acquisition Policy on $\mathcal{I}_{\text{cal1}}$

To learn an optimal mapping from scores to probabilities, we must first expend budget to observe the full conversation history for all samples in $\mathcal{I}_{\text{cal1}}$, until we observe an event, or reach the prior $\hat{f}_{\text{prior}}(X_i)$. For all $i \in \mathcal{I}_{\text{cal1}}$, we set the censoring times as $C_i = \hat{f}_{\text{prior}}(X_i)$, and expend $b_i := \min(T_i, \hat{f}_{\text{prior}}(X_i))$ budget units. We require the total budget B is sufficient for this observation phase, such that $B > \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i$. Thus, the weights of these samples are simply $w_\tau(i) = 1, \forall i \in \mathcal{I}_{\text{cal1}}$. The budget per sample remaining for Phase II is therefore $\bar{B}_2 := \frac{1}{|\mathcal{I}_{\text{cal2}}|} (B - \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i)$. Next, we compute the scores for all observed time steps across the subset $\mathcal{I}_{\text{cal1}}$:

$$S_i(t) = S_t(H_i(t)), \quad \forall t \in \{1, \dots, b_i\}, \quad i \in \mathcal{I}_{\text{cal1}}.$$

We use these scores to assign a continuation probability $P_i(t) \in [0, 1]$ to each step based on the scores $S_i(t)$ that minimizes the variance of our produced LPBs. For $t > b_i$, we set $P_i(t) = 0$ since advancing beyond b_i does not contribute to (3). Following the theoretical coverage bound we develop in Section 3.3, an optimal probability matrix should minimize the inverse of $\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i)$, without exceeding the budget units that remain for the second step. We also require that these probabilities are monotonic with respect to the scores $S_i(t)$, so we can derive a sound mapping from the scores to probabilities. By the construction of our policy, we get $\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i) =$

$\prod_{t=1}^{b_i} P_i(t)$, and therefore the optimization problem is formulated as:

$$\begin{aligned} \min_{P \in [0,1]^{N_1 \times t_{\max}}} \quad & \frac{1}{N_1} \sum_{i=1}^{N_1} \frac{1}{\prod_{t=1}^{b_i} P_i(t)} \\ \text{s.t.} \quad & \frac{1}{N_1} \sum_{i=1}^{N_1} \mathcal{B}(P_i) \leq \bar{B}_2, \quad P_i(t) \leq P_j(t) \iff S_i(t) \leq S_j(t), \quad \forall t, i, j. \end{aligned}$$

Here, $\mathcal{B}(P_i)$ is the expected budget expended on the i -th sample: $\mathcal{B}(P_i) = \sum_{t=1}^{b_i} \prod_{j=1}^t P_i(j)$. We solve this optimization problem using coordinate descent, yielding an optimal sequence of continuation probabilities $\{P_i(t)\}_{t=1}^{b_i}$ for each sample $i \in \mathcal{I}_{\text{cal1}}$. To generalize this mapping to the unseen points in $\mathcal{I}_{\text{cal2}}$, we fit a model M_t over the paired tuples $\{S_i(t), P_i(t)\}_{i \in \mathcal{I}_{\text{cal1}}}$ to transport the risk score into an optimal continuation probability.

3.2 Phase II: Adaptive Acquisition on $\mathcal{I}_{\text{cal2}}$

Armed with the sequence of score-to-probability mapping models $\{M_t\}_{t=1}^{t_{\max}}$, we adaptively acquire interactions for the second split, $\mathcal{I}_{\text{cal2}}$. We initialize the process at $t = 1$ and define a continuation indicator $\xi_i(0) = 1$. At each step t , we compute the score $S_i(t) = \mathcal{S}_t(H_i(t))$ and transport it into a continuation probability $P_i(t) = M_t(S_i(t))$. We then draw a Bernoulli random variable with parameter $P_i(t)$ to determine if the interaction should proceed. Formally, the continuation indicator at step t updates as $\xi_i(t) = \xi_i(t-1)$ with probability $M_t(S_i(t))$, and $\xi_i(t) = 0$ otherwise. If $\xi_i(t) = 1$, we advance to the next step: we acquire the new conversation exchange $H_i(t+1)$, set $t \leftarrow t+1$, and repeat the process. The interaction terminates when $\xi_i(t) = 0$ or we reach the predefined boundary $t = \min(\hat{f}_{\text{prior}}(X_i), T_i)$. Finally, we define the censoring time for each sample $i \in \mathcal{I}_{\text{cal2}}$ as the last successful step: $C_i = \hat{f}_{\text{prior}}(X_i)$ if $\xi_i(T_i) = 1$ or $\xi_i(\hat{f}_{\text{prior}}(X_i)) = 1$, and $C_i = \max\{t \in \{0, 1, \dots, t_{\max}\} : \xi_i(t) = 1\}$ otherwise. When an event occurs at step $T_i \leq \hat{f}_{\text{prior}}(X_i)$, we terminate the procedure early at this step, and set $C_i = \hat{f}_{\text{prior}}(X_i)$ to indicate that the budget for this sample is fully accounted for, as the event has been directly observed. Consequently, for samples $i \in \mathcal{I}_{\text{cal2}}$ that successfully reach their target sequence length, i.e., $\hat{f}_{\text{prior}}(X_i) \leq C_i$, the probability of achieving this event is the product of the sequential continuation probabilities:

$$\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i \mid X_i, H_i, T_i, \{(X_j, H_j, T_j)\}_{j \in \mathcal{I}_{\text{cal1}}}, \mathcal{D}_{\text{train}}) = \prod_{t=1}^{\min(\hat{f}_{\text{prior}}(X_i), T_i)} P_i(t).$$

The censored event times are given by $\tilde{T}_i = \min(C_i, T_i)$. Lastly, we employ Algorithm 1 using the samples we acquired $\{(C_i, w(i) = \mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i)^{-1}, \tilde{T}_i)\}_{i \in \mathcal{I}_{\text{cal}}}$. Note that while the true probabilities $\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i)$ are unknown for samples where $C_i < \hat{f}_{\text{prior}}(X_i)$, Algorithm 1 evaluates weights for samples satisfying $C_i \geq \hat{f}_{\text{prior}}(X_i)$. That is, the unobserved probabilities are unnecessary for the LPB construction. For convenience, this procedure is summarized in Algorithm 5 in Appendix D.5. In what follows, we prove that this process generates an LPB that achieves the desired coverage rate while satisfying the budget constraint.

3.3 Theoretical guarantees

In this section, we analyze the coverage rate and budget used by our proposed approach. First, we show that the LPB constructed by Algorithm 5 holds a PAC-type coverage guarantee.²

Theorem 1 (Coverage validity (informal)). *Fix a miscoverage level $\alpha \in (0, 1)$ and a tolerance level $\delta \in (0, 1)$. Suppose that $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ and $(X_{\text{test}}, T_{\text{test}}, H_{\text{test}})$ are drawn i.i.d., and that $\hat{f}_{\tau}(x)$ is non-decreasing and continuous in τ . Further assume that there exists a constant $\bar{w} \geq 1$, such that almost surely: $\mathbb{E}[w_{\tau}(i) \mid \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal1}}}, \mathcal{D}_{\text{train}}] \leq \bar{w}, \forall i \in \mathcal{I}_{\text{cal2}}$. Then, with probability at least $1 - \delta$ over the draw of $\mathcal{D}_{\text{all}}$, the lower predictive bound $\hat{L}(x)$ generated by Algorithm 5 satisfies:*

$$\mathbb{P}\left[T_{\text{test}} \geq \hat{L}(X_{\text{test}}) \mid \mathcal{D}_{\text{all}}\right] \geq 1 - \alpha - \frac{\log(1/\delta)}{3|\mathcal{I}_{\text{cal}}|} - \sqrt{\frac{\log^2(1/\delta)}{9|\mathcal{I}_{\text{cal}}|^2} + \frac{2(\bar{w} - \alpha^2) \log(1/\delta)}{|\mathcal{I}_{\text{cal}}|}}.$$

²An analogous UPB guarantee holds by reversing the inequality, and is omitted for brevity.

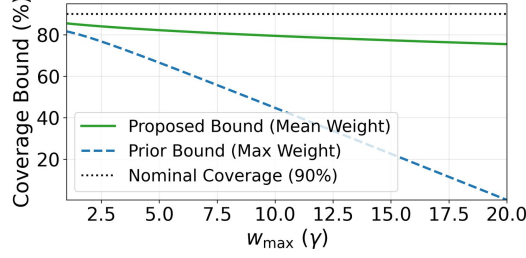


Figure 3: Theoretical coverage lower bounds of Theorem 1 (green) and [12, Theorem 4.1] (blue) as a function of the maximum weight, γ , assuming the weights are uniformly distributed.

We defer the proof of Theorem 1 to Appendix C.1. Although the theorem relies on an almost-sure upper bound $\bar{w}$ on the expected weight, we refer to this bound as the mean weight as optimizing the mean weight minimizes this upper bound. Furthermore, the almost-sure assumption can be replaced with a high-probability PAC guarantee over the calibration splits, and we omit it for clarity. While this proof builds on the theoretical foundations in [12, 14], we highlight several key implications that distinguish our result. First, this coverage guarantee holds in finite samples for any quantile estimator $\hat{q}_\tau(\cdot)$, any LLM, and any distribution of the data. Crucially, unlike prior conformal frameworks [12, 14], our result does not rely on the assumption of conditional independence between the censoring times C and event times T given the covariates X . DAPRO violates this assumption as it uses T for early stopping when an event is elicited. Second, our coverage gap scales as $\sqrt{\bar{w}/N}$, where $\bar{w}$ is the mean inverse-censoring weight bound. This is tighter than the bounds derived in [12, 14], which scale with $\sqrt{\gamma^2/N}$, where $\gamma \geq \bar{w}$ is the maximal, worst-case, weight across all calibration samples. Motivated by this insight, our Phase I optimization explicitly minimizes the mean weight to achieve coverage closer to the nominal level. Finally, we remark that this tighter bound is of independent interest, as our proof technique can be directly applied to the static procedure of [12]. Their original motivation for minimizing the mean weight relied on an oracle assumption, in which the true conditional quantiles are known and the variance of the miscoverage estimator scales linearly with the mean weight. Our analysis bridges this gap by formally demonstrating that, even in finite samples and without oracle quantiles, the coverage gap is bounded by the mean weight, providing a theoretical justification for their optimization objective.

To demonstrate the tightness of our proposed bound, we compare in Figure 3 our theoretical coverage guarantee with the one developed by [12]. We consider a realistic setup with $N = 3000$ calibration samples, a target coverage level of $1 - \alpha = 0.9$, and a tolerance of $\delta = 0.05$. For the sake of this illustration, we assume the inverse-censoring weights are uniformly distributed between 1 and a maximum value γ , yielding a mean weight of $\bar{w} = \frac{1}{2}(1 + \gamma)$. The figure shows that while the prior bound decreases rapidly as the maximum weight increases, our proposed bound remains significantly tighter and informative across the entire range.

We now show that when the estimates of our Phase I are sufficiently accurate, the expected budget consumed by Algorithm 5 satisfies the budget constraint, and we develop a finite-sample bound for the budget utilized in practice.

Theorem 2 (Budget validity (informal)). *Suppose that $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ are drawn i.i.d. Furthermore, assume that the empirical probabilities $\{P_i(t)\}_{t \in \{1, \dots, b_i\}, i \in \mathcal{I}_{\text{cal1}}}$ learned in Phase I approximate the optimal oracle allocation, and the models $\{M_t\}_{t=1}^{t_{\max}}$ are sufficiently accurate. Then, the expected budget used by Algorithm 5 is upper bounded by B . Additionally, with probability at least $1 - \delta$, the average budget per sample consumed by Algorithm 5 is bounded by:*

$$\frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} \tilde{T}_i \leq \frac{B}{|\mathcal{I}_{\text{cal}}|} + \frac{t_{\max} \log(1/\delta)}{3|\mathcal{I}_{\text{cal}}|} + \frac{1}{|\mathcal{I}_{\text{cal}}|} \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2|\mathcal{I}_{\text{cal2}}| t_{\max} \bar{B}_2 \log(1/\delta)}.$$

This theorem is formally stated and proved in Appendix C.2. In Appendix C.2.4, we relax the assumptions that the models are sufficiently accurate and develop an upper bound that accounts for the estimation errors of the score-to-probability mapping. While this budget bound scales with $O(t_{\max}^2/\sqrt{|\mathcal{I}_{\text{cal1}}|})$, it is highly conservative: it assumes that estimation errors at every step aggregate

and do not cancel each other. In practice, positive and negative errors mostly cancel, and Algorithm 5 satisfies the nominal budget constraint across our experiments. We therefore interpret this bound as a sample complexity result: it establishes the asymptotic consistency of our approach.

4 Experiments

We evaluate the coverage rate and budget consumption of our approach on attacker-target conversations generated using the PAIR algorithm [9]. We sample 10,000 prompts from each of the following datasets: RealToxicityPrompts [3] (where the target event $Y = 1$ is a toxic response) and the Anthropic Red Team dataset [15, 16] (where $Y = 1$ is a successful jailbreak). To demonstrate the framework’s applicability to utility evaluation, we also include the AutoIF dataset [17], a benchmark for complex instruction following. For AutoIF, the "attacker" is a helper agent, and the event ($Y = 1$) occurs when the target model successfully satisfies all instructions. To generate the interactions, we utilize Qwen 2.5 14B Instruct [13] as the attacker model, as its outputs had the lowest formatting error rate among the models we examined. For the target models, we use Qwen 2.5 14B Instruct, Llama 3.1 8B Instruct [18], Phi 4 Mini Instruct [19], and Gemma 3 4B Instruct [20]. We define the target event ($Y(t) = 1$) using an audit function: for the Toxicity dataset, following [12], we use Detoxify [21] with a threshold of 0.5 to identify toxic responses. For the Red-Team dataset, we employ Qwen 2.5 14B Instruct as an LLM-as-a-judge [11, 22], which scores responses on a 1–10 scale; a score of 10 denotes a successful jailbreak. Finally, for the AutoIF dataset, a successful instruction-following response event is verified programmatically. In Appendix F.3, we conduct an experiment with Llama-Guard [23] serving as the audit function, which outputs a binary success label. Additionally, Appendix F.4 evaluates DAPRO on a RAG-based dataset, in which the event is a hallucination. We cap the maximum sequence length at $t_{\max} = 200$. The full details regarding the dataset generation are provided in Appendix E.1.

We split the data into training (4000), calibration (3000), and test (3000) sets. We fix the training data and fit a transformer as the predictive model over it, using the full conversation history. We assume an unlimited budget during the training phase; this work focuses on the calibration phase budget only. At any given time t_1 , the model estimates the probability of an event occurring at any future time step $t_2 > t_1$. Specifically, it outputs a $t_{\max} \times (t_{\max} + 1)$ matrix containing $\mathbb{P}(Y(t_2) = 1 \mid T > t_1, H(t_1))$ and the probability of reaching the horizon without an unsafe event, $\mathbb{P}(T > t_{\max} \mid T > t_1, H(t_1))$. All methods use the same predictive models. In all experiments, we evaluate the methods over 50 independent random splits of the calibration and test sets. During each trial, the calibration algorithms are employed using the calibration data and evaluated on the test set. We set the average budget per sample as $B/|\mathcal{I}_{\text{cal}}| = 20$. Following [12], we set $\tau_{\text{prior}} = 0.56$, and $M = 200$.

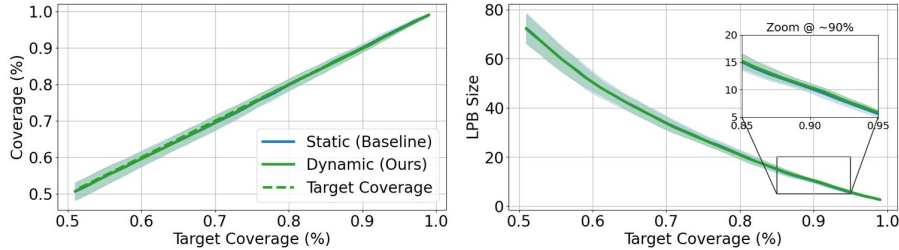


Figure 4: Coverage rate and LPB size of various methods over the RedTeam datasets with Qwen 2.5 serving as attacker, target, and judge.

We compare our proposal to an uncalibrated baseline, which constructs the LPB directly from the raw model predictions at the target α level, $\hat{q}_\alpha$, and to the static optimized allocation strategy proposed in [12]. In addition to our main method, we developed two alternative dynamic allocation strategies, which we introduce in Appendix D. The first is designed to greedily discover as many unsafe events as possible. The second one is a locally adaptive variant that optimizes the continuation probabilities step-by-step rather than globally. Since DAPRO attains a lower variance compared to these variants across our experiments, we focus our main presentation on it and defer the evaluation of these dynamic variants to Appendix F.

Figure 4 presents the empirical coverage rate and LPB size of each method as a function of the target coverage level on the RedTeam dataset with the Qwen 2.5 model serving as attacker, target, and judge. This figure demonstrates that all calibration methods attain the desired coverage level, while the uncalibrated model does not. This is anticipated by our theoretical results. Figure 5 presents the deviation from the target 90% coverage rate and average budget per sample used by each method on the Toxicity dataset. This figure reveals that our proposed method satisfies the budget constraint while attaining the lowest coverage deviation. In addition, Figure 6 details the same metrics on the AutoIF dataset for both an LPB (target coverage 90%) and an upper predictive bound (UPB, target coverage 70%). We evaluate the UPB at this lower target coverage since a 90% target pushes the bound too close to the maximum $t_{\max} = 200$, which artificially reduces the variance differences between methods. Across both bound types, the results reveal the same trend: DAPRO attains a coverage rate closer to the nominal level compared to the baselines.

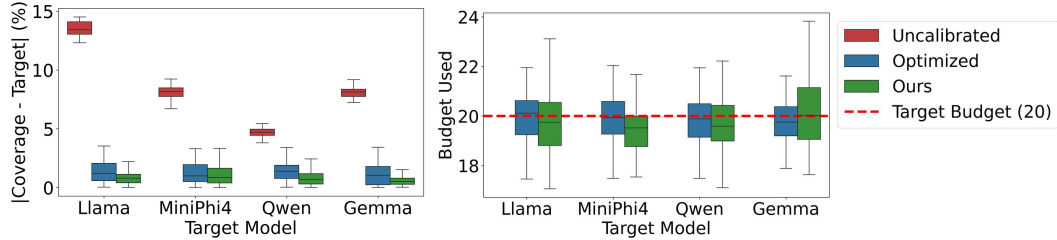


Figure 5: **Toxicity** dataset: absolute coverage deviation (left) and average budget utilized (right) across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample.

Additional experimental setups, including an analysis of the empirical mean weights and the number of observed unsafe events, are deferred to Appendix F. Furthermore, our ablation study in Appendix F.7 reveals that even when degrading its components, i.e., the score or first data-split size N_1 , DAPRO is robust and maintains a tighter coverage rate than the baseline. Finally, beyond constructing individual LPBs, our framework can be utilized to construct unbiased estimates of population-level evaluation metrics, such as the unsafe event rate. As demonstrated in Appendix F.6, DAPRO accurately recovers the true metric quantities with zero bias and a lower variance than existing baselines.

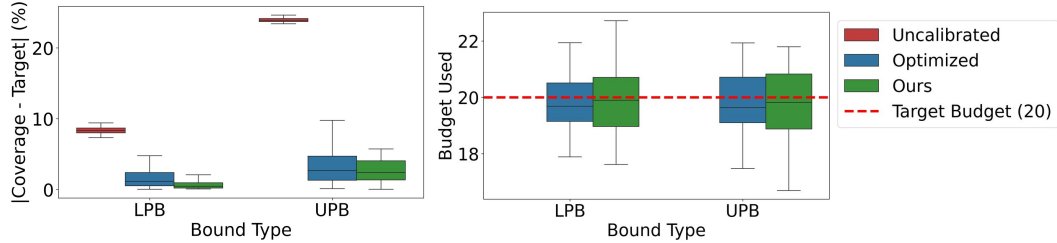


Figure 6: **AutoIF** dataset: coverage deviation (left) and budget utilized (right) over Qwen 2.5 target model. Target coverage rate: 90% for LPB and 70% for UPB, and $\bar{B} = 20$ budget per sample.

5 Discussion and impact statement

We introduced DAPRO, a novel dynamic budget allocation approach for LLM evaluation. Our approach can be employed to construct reliable LPBs for individual prompts and extract unbiased population-level metrics, such as the jailbreak rate. By updating censoring times online based on ongoing attacker-target interactions, our method achieves lower variance, as supported by our PAC-type coverage guarantee, which is tighter than prior approaches. Our comprehensive experiments on real datasets validate these theoretical claims, demonstrating tighter bounds compared to baselines.

While our method satisfies the budget constraint in our empirical simulations, our theoretical expected budget guarantee relies on the accuracy of the estimated score mapping. Establishing a finite-sample budget guarantee that does not rely on the accuracy of the model could be a promising research direction, possibly by relying on risk control theory [24, 25] or leave-one-out to handle scarce

data [26–28]. For applications requiring a strict budget guarantee, practitioners can utilize the variants presented in Appendices D.3 and D.4, which provide exact budget control without relying on the accuracy of a score mapping. Furthermore, the calibrated quantile level $\hat{\tau}$ in our current method is a single scalar applied identically to all test prompts. An intriguing extension would be to make $\hat{\tau}$ adaptive to the test prompt X_{test} , possibly leveraging the newly acquired calibration data to select a prompt-specific quantile level, e.g., by utilizing ideas from [29, 30]. This would allow the LPB to be tighter for predictable interactions, and more conservative for those with high variance.

We note that, like all conformal prediction-based methods, formal guarantees rely on the standard conformal assumption of exchangeability between calibration and test sets. Specifically, we assume the test attacker, target, judge models, and prompt distribution are the same as those used during calibration. In practice, these may shift over time as models are updated or new attack strategies emerge. Developing methods that are robust to such distributional shifts [31, 32] or audit annotation errors [33, 34] is an important direction for future work. Finally, while our framework provides rigorous guarantees to help developers proactively assess prompt risks and set usage limits, its insights into model failures could potentially be exploited by adversaries, akin to many developments in ML. Therefore, our proposal must be deployed as a complementary tool within a broader safety, alignment, and governance pipeline.

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A Additional Related work

Our work sits at the intersection of LLM safety evaluation, conformal prediction, and survival analysis. Prior efforts in safety evaluation mostly focus on detecting harmful content in model outputs [3, 35] or constructing guardrails at inference time [23, 36]. A complementary line of work takes an adversarial perspective, generating red-teaming attacks to probe model vulnerabilities [9, 10, 37–39]. These works aim to maximize attack success. On the defensive side, recent efforts in certified LLM safety aim to provide statistical guarantees against such attacks, typically by bounding the failure probability of a single prompt using techniques like randomized smoothing [40, 41]. In contrast, our goal is to provide rigorous statistical guarantees on prompt risk using the multi-turn interaction structure that adversarial settings naturally produce, and explicitly evaluating the time-to-event. While existing adversarial red-teaming approaches effectively investigate these multi-turn vulnerabilities, they do not provide statistical guarantees, as they do not account for the distribution shift induced by right-censoring.

While our framework was designed to provide rigorous safety bounds, it can be employed for sample-efficient LLM evaluation [42]. To reduce computational costs and API calls when evaluating large models, recent methods utilize active testing [43], multi-armed bandits [44], and adaptive dataset partitioning [45]. Our framework directly complements these efforts: by framing metric evaluation as a sequential, budget-constrained survival problem, DAPRO produces theoretically valid estimates with low variance under resource constraints.

Our theoretical framework builds upon conformal prediction [46, 47], a powerful framework for constructing prediction sets with finite-sample validity guarantees and without distributional assumptions. Split conformal prediction [48] and its extension to covariate shift via weighting [49], are utilized by the calibration procedure we adopt from [12]. Conformal prediction has recently been extended to survival analysis [14, 50, 51], where the goal is to construct lower predictive bounds on event times under censoring. Our work builds directly on [12], who apply conformalized survival analysis to LLM safety by introducing the time-to-unsafe-sampling metric and a static, variance-minimizing budget allocation scheme. We extend this framework to the adaptive setting where conversation evolves, and our coverage proof provides a tighter PAC-type guarantee [52–54].

Finally, the variance-minimization phase of our framework is related to active learning [55] and optimal experimental design [56], which study how to allocate a limited budget across observations to minimize estimator variance. In particular, [57] formalizes active data collection for statistical inference while maintaining valid guarantees, and [58] develops adaptive designs that minimize treatment effect estimation variance. Our contribution is adapting these ideas to the survival analysis and LLM safety setting, where observations arrive sequentially, and the allocation policy must be learned from a first data split before being projected on a second.

B Extension to LLM utility evaluation and upper predictive bounds (UPB)

While the main text focuses on safety risk assessment, our framework extends to evaluating LLM utility in multi-turn interactions. In this setting, the adversarial "attacker" is a "helper agent" aiming to guide the target model toward successful task completion, such as satisfying complex constraints, e.g., using a specific word multiple times, as in the AutoIF dataset [17]. Here, the event of interest ($Y = 1$) is defined as task success. Instead of bounding the time-to-unsafe with a lower predictive bound (LPB), we construct an upper predictive bound (UPB), denoted by $\hat{U}(X)$. The UPB provides a finite-sample guarantee that the time-to-success T will not exceed $\hat{U}(X)$ with probability $1 - \alpha$, i.e., $\mathbb{P}(T \leq \hat{U}(X)) \geq 1 - \alpha$. As illustrated in Figure 7, the dynamic allocation mechanism remains identical to the safety setting. Technically, a low UPB indicates that the task is highly feasible, requiring few interactions to succeed, whereas a UPB reaching the maximum horizon $t_{\max}$ flags the task as practically infeasible. The theoretical coverage guarantees derived for the LPB apply to the UPB by simply reversing the target inequality direction.



Figure 7: Illustration of our framework for LLM utility evaluation: (i) collecting data via dynamic budget allocation; (ii) calibrating a pre-trained model; and (iii) deploying the model at inference time to assess task feasibility.³

C Theory

In this section, we present our theoretical guarantees. For ease of notations, we abuse some of the notations made in the first part and set $\mathcal{D}_{\text{cal}} = \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal}}}$, $\mathcal{D}_{\text{cal1}} = \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal1}}}$ and $\mathcal{D}_{\text{cal2}} = \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal2}}}$. Hence, the combined data is denoted as $\mathcal{D}_{\text{all}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{cal}}$.

C.1 Coverage rate guarantee of Algorithm 5

Before presenting the coverage validity guarantee of our proposed algorithm, we first show that the weights it uses are correct.

Proposition 1 (Valid weights of Algorithm 5). *The weights used by Algorithm 5 equal to $w_\tau(i) = \mathbb{P}[C_i \geq \hat{f}_\tau(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]^{-1}$ for all $i \in \mathcal{I}_{\text{cal}}$ such that $C_i = \hat{f}_{\text{prior}}(X_i)$ and $\tau \in \mathcal{T} = [0, \tau_{\text{prior}}]$.*

Proof. First, for all $i \in \mathcal{I}_{\text{cal1}}$, $C_i = \hat{f}_{\text{prior}}(X_i)$ deterministically, and hence, $w_\tau(i) = 1 = \mathbb{P}[C_i \geq \hat{f}_\tau(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]^{-1}$. The above holds since $\hat{f}_\tau$ is non-decreasing in τ and $\tau \leq \tau_{\text{prior}}$, we have $\hat{f}_\tau(X_i) \leq \hat{f}_{\text{prior}}(X_i) = C_i$ for all $\tau \in \mathcal{T}$, giving $\mathbb{P}(C_i \geq \hat{f}_\tau(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) = 1$ deterministically, and hence $w_\tau(i) = 1$. For any $i \in \mathcal{I}_{\text{cal2}}$, the weights used by the algorithm are:

$$w_\tau(i) = \left(\prod_{t=1}^{\min(\hat{f}_{\text{prior}}(X_i), T_i)} P_i(t) \right)^{-1}.$$

Above, $\forall i \in \mathcal{I}_{\text{cal2}} : P_i(t) = M_t(S_i(t))$ are the outputs of the mapping models. By our construction, $P_i(t) \equiv \mathbb{P}(\xi_i(t) = 1 \mid \xi_i(t-1) = 1, X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$. Furthermore, $\xi_i(t) = \xi_i(t-1) \cdot Z_t^{(i)}$ where $Z_t^{(i)} \sim \text{Bern}(P_i(t))$ are mutually independent given $X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}$. Since the event $\mathbb{1}\{\xi_i(k) = 1\}$ is equivalent to $\{Z_1^{(i)} = 1, \dots, Z_k^{(i)} = 1\}$, the chain rule and mutual independence give:

$$\begin{aligned} \prod_{t=1}^{\min(\hat{f}_{\text{prior}}(X_i), T_i)} P_i(t) &= \prod_{t=1}^{\min(\hat{f}_{\text{prior}}(X_i), T_i)} \mathbb{P}(\xi_i(t) = 1 \mid \xi_i(t-1) = 1, X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\ &= \mathbb{P}(\xi_i(\min(\hat{f}_{\text{prior}}(X_i), T_i)) = 1 \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}). \end{aligned}$$

By the definition of C_i as:

$$C_i := \begin{cases} \hat{f}_{\text{prior}}(X_i) & \text{if } i \in \mathcal{I}_{\text{cal1}}, \\ \hat{f}_{\text{prior}}(X_i) & \text{if } i \in \mathcal{I}_{\text{cal2}}, \xi_i(\min(T_i, \hat{f}_{\text{prior}}(X_i))) = 1, \\ \max\{t \in \{0, \dots, t_{\text{max}}\} : \xi_i(t) = 1\} & \text{otherwise,} \end{cases}$$

³This figure was generated using Google Gemini based on a prompt designed by the authors and subsequently refined.

we get:

$$\begin{aligned}
\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\
&= \mathbb{P}(C_i = \hat{f}_{\text{prior}}(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\
&= \mathbb{P}(\xi_i(\min(\hat{f}_{\text{prior}}(X_i), T_i)) = 1 \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\
&= \prod_{t=1}^{\min(\hat{f}_{\text{prior}}(X_i), T_i)} P_i(t).
\end{aligned}$$

Observe that for $C_i = \hat{f}_{\text{prior}}(X_i)$ and $\tau \leq \tau_{\text{prior}}$ we get $C_i \geq \hat{f}_{\tau}(X_i) > 0$ by the monotonicity of $\hat{f}_{\tau}$ and therefore:

$$\mathbb{P}(\hat{f}_{\text{prior}}(X_i) \leq C_i \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) = \mathbb{P}(\hat{f}_{\tau}(X_i) \leq C_i \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}).$$

□

We now turn to present the main coverage rate guarantee.

Theorem 3. Fix a miscoverage level $\alpha \in (0, 1)$ and a tolerance level $\delta \in (0, 1)$. Suppose that $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ and $(X_{\text{test}}, T_{\text{test}}, H_{\text{test}})$ are drawn i.i.d., and that $\hat{f}_{\tau}(x)$ is non-decreasing and continuous in τ . Further assume that there exists a constant $\bar{w} \geq 1$, which may depend on $\mathcal{D}_{\text{train}}$ but not on $\mathcal{D}_{\text{cal1}}$, such that almost surely over $\mathcal{D}_{\text{cal1}}$:

$$\mathbb{E}[w_{\tau}(i) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \leq \bar{w} \quad \text{for all } i \in \mathcal{I}_{\text{cal2}}.$$

Define the coverage gap:

$$\Delta := \frac{\log(1/\delta)}{3|\mathcal{I}_{\text{cal}}|} + \sqrt{\frac{\log^2(1/\delta)}{9|\mathcal{I}_{\text{cal}}|^2} + \frac{2(\bar{w} - \alpha^2) \log(1/\delta)}{|\mathcal{I}_{\text{cal}}|}}.$$

Then, with probability at least $1 - \delta$ over the draw of $\mathcal{D}_{\text{all}} = (\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{cal}})$, the lower predictive bound $\hat{L}(x)$ generated by Algorithm 5 satisfies:

$$\mathbb{P}\left[T_{\text{test}} \geq \hat{L}(X_{\text{test}}) \mid \mathcal{D}_{\text{all}}\right] \geq 1 - \alpha - \Delta.$$

Proof. First, we note that, in practice, we do not compute the entire sequential continuation indicators $\xi_i(t)$. However, for the sake of this proof, indicators $\xi_i(t)$ can be viewed as an infinite sequence generated via $Z_t^i \sim \text{Ber}(M_t(S_i(t)))$, which is dependent only on the history H_i , features X_i , and calibration data $\mathcal{D}_{\text{cal1}}$. The early stopping at T_i is conducted for practical purposes to satisfy the budget constraint, but the underlying distribution of the censoring variable C_i relies solely on Bernoulli random variables that are perfectly independent of the target sequence length T_i . Thus, by the construction of the censoring times, and following the i.i.d. assumption, we have

1. By construction, for all $i \in \mathcal{I}_{\text{cal2}}$: $(\xi_i, T_i) \perp \{(\xi_j, T_j)\}_{j \in \mathcal{I}_{\text{cal2}} \setminus \{i\}} \mid \{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}}, \mathcal{D}_{\text{cal1}}$. Therefore, given $\{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}}$ and $\mathcal{D}_{\text{cal1}}$, the censoring times $\{C_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent, since C_i is a deterministic function of (ξ_i, T_i, X_i, H_i) .
2. The marginal law of (X_i, T_i, H_i) for $i \in \mathcal{I}_{\text{cal2}}$ is independent of $\mathcal{D}_{\text{cal1}}$ given $\mathcal{D}_{\text{train}}$, and $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$.

By Proposition 1, the weights used by the algorithm are correct: $w_{\tau}(i) = \mathbb{P}\left[C_i \geq \hat{f}_{\tau}(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}\right]^{-1}$ for all $i \in \mathcal{I}_{\text{cal}}$ such that $C_i = \hat{f}_{\text{prior}}(X_i)$ and $\tau \in \mathcal{T} = [0, \tau_{\text{prior}}]$. Every $i \in \mathcal{I}_{\text{cal}}$ such that $C_i \neq \hat{f}_{\text{prior}}(X_i)$, does not contribute to the weighted sum in (3), so we do not need access to its true weight. Therefore, all assumptions of Theorem 4 hold, and thus the LPB generated by Algorithm 5 is a valid PAC-type LPB with the desired bounds. □

We now present the general theorem for coverage validity guarantee.

Theorem 4 (General Coverage Validity). *Fix a miscoverage level $\alpha \in (0, 1)$ and a tolerance level $\delta \in (0, 1)$. Let $\mathcal{D}_{\text{train}}$ be the training set, and let $\mathcal{I}_{\text{cal}} = \mathcal{I}_{\text{cal1}} \sqcup \mathcal{I}_{\text{cal2}}$ be a disjoint partition of the calibration set indexes, with $|\mathcal{I}_{\text{cal1}}| + |\mathcal{I}_{\text{cal2}}| = |\mathcal{I}_{\text{cal}}|$. Suppose that $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ and $(X_{\text{test}}, T_{\text{test}}, H_{\text{test}})$ are drawn i.i.d., that $\hat{f}_\tau(x)$ is non-decreasing and continuous in τ , with $\hat{f}_\tau(x) > 0$ for all x and $\tau > 0$. We assume that for $i \in \mathcal{I}_{\text{cal1}}$ the weights are given by $w_\tau(i) = 1$, and*

$$w_\tau(i) = \frac{1}{\mathbb{P}[C_i \geq \hat{f}_\tau(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]} \quad \text{for } i \in \mathcal{I}_{\text{cal2}}.$$

Assume the following conditions hold:

1. **(Mutual independence)** For all $i \in \mathcal{I}_{\text{cal2}}$: $(C_i, T_i) \perp \{(C_j, T_j)\}_{j \in \mathcal{I}_{\text{cal2}} \setminus \{i\}} \mid \{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}, \mathcal{D}_{\text{cal1}}}$.
2. **(Marginal independence of calibration data)** The marginal law of (X_i, T_i, H_i) for $i \in \mathcal{I}_{\text{cal2}}$ is independent of $\mathcal{D}_{\text{cal1}}$ given $\mathcal{D}_{\text{train}}$, and $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$.
3. **(Bounded mean weight)** There exists a constant $\bar{w} \geq 1$, which may depend on $\mathcal{D}_{\text{train}}$ but not on $\mathcal{D}_{\text{cal1}}$, such that almost surely over $\mathcal{D}_{\text{cal1}}$:

$$\mathbb{E}[w_\tau(i) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \leq \bar{w} \quad \text{for all } i \in \mathcal{I}_{\text{cal2}}.$$

Define the estimated miscoverage rate and calibrated quantile level by:

$$\hat{\alpha}(\tau) := \frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} w_\tau(i) \mathbb{I}\{\tilde{T}_i < \hat{f}_\tau(X_i) \leq C_i\}, \quad \hat{\tau} := \sup\left\{\tau \in \mathcal{T} : \sup_{\tau' \leq \tau} \hat{\alpha}(\tau') \leq \alpha\right\}.$$

We remark that for $\hat{\tau}$ to be well defined, we assume there exists $\tau' \in \mathcal{T} = [0, \tau_{\text{prior}}]$ such that $\hat{\alpha}(\tau') \leq \alpha$. This assumption can be trivially satisfied by including $\tau = 0^+$ in $\mathcal{T}$ with $\hat{f}_0(x) = 0$, which yields $\hat{\alpha}(0) = 0 \leq \alpha$.

Define the coverage gap:

$$\Delta := \frac{\log(1/\delta)}{3|\mathcal{I}_{\text{cal}}|} + \sqrt{\frac{\log^2(1/\delta)}{9|\mathcal{I}_{\text{cal}}|^2} + \frac{2(\bar{w} - \alpha^2) \log(1/\delta)}{|\mathcal{I}_{\text{cal}}|}}.$$

Then, with probability at least $1 - \delta$ over the draw of $\mathcal{D}_{\text{all}} = (\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{cal}})$, the lower predictive bound $\hat{L}(x) = \hat{f}_{\hat{\tau}}(x)$ satisfies:

$$\mathbb{P}\left[T_{\text{test}} \geq \hat{L}(X_{\text{test}}) \mid \mathcal{D}_{\text{all}}\right] \geq 1 - \alpha - \Delta.$$

Proof. We define the oracle miscoverage level by:

$$\tau(\alpha + \Delta) = \sup\left\{\lambda \in \mathcal{T} : \mathbb{P}(T < \hat{f}_\lambda(X) \mid \mathcal{D}_{\text{train}}) \leq \alpha + \Delta\right\}.$$

The above set is non-empty since $\tau = 0$ satisfies $\mathbb{P}(T < \hat{f}_0(X) \mid \mathcal{D}_{\text{train}}) = \mathbb{P}(T < 0 \mid \mathcal{D}_{\text{train}}) = 0 \leq \alpha + \Delta$ by the convention $\hat{f}(x) = 0$.

Observe that it suffices to show $1 - \delta \leq \mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) \mid \mathcal{D}_{\text{train}})$, since on the event $\{\hat{\tau} \leq \tau(\alpha + \Delta)\}$, the monotonicity of $\hat{f}_\tau$ in τ and the left-continuity of $\tau \mapsto \mathbb{P}(T \geq \hat{f}_\tau(X) \mid \mathcal{D}_{\text{all}})$ give:

$$\mathbb{P}(T \geq \hat{f}_{\hat{\tau}}(X) \mid \mathcal{D}_{\text{all}}) \geq \mathbb{P}(T \geq \hat{f}_{\tau(\alpha + \Delta)}(X) \mid \mathcal{D}_{\text{all}}) \geq 1 - \alpha - \Delta.$$

It therefore remains to show $\mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) \mid \mathcal{D}_{\text{train}}) \geq 1 - \delta$. Fix $\varepsilon > 0$ and set $\lambda := \tau(\alpha + \Delta) + \varepsilon$. We claim that $\{\hat{\alpha}(\lambda) > \alpha\} \subseteq \{\hat{\tau} \leq \lambda\}$. Indeed, if $\hat{\alpha}(\lambda) > \alpha$, then for every $\tau \geq \lambda$, $\sup_{\tau' \leq \tau} \hat{\alpha}(\tau') \geq \hat{\alpha}(\lambda) > \alpha$, so no $\tau \geq \lambda$ belongs to the set $\{\tau \in \mathcal{T} : \sup_{\tau' \leq \tau} \hat{\alpha}(\tau') \leq \alpha\}$, hence $\hat{\tau} = \sup\{\tau \in \mathcal{T} : \sup_{\tau' \leq \tau} \hat{\alpha}(\tau') \leq \alpha\} \leq \lambda$. It therefore suffices to show:

$$\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) \leq \delta, \tag{4}$$

since this gives $\mathbb{P}(\hat{\tau} \leq \lambda \mid \mathcal{D}_{\text{train}}) \geq 1 - \delta$ for every $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0^+$ and applying continuity of the probability measure to the decreasing family of events $\{\hat{\tau} \leq \tau(\alpha + \Delta) + \varepsilon\}$ (whose intersection equals $\{\hat{\tau} \leq \tau(\alpha + \Delta)\}$) gives:

$$\mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) \mid \mathcal{D}_{\text{train}}) = \lim_{\varepsilon \rightarrow 0^+} \mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) + \varepsilon \mid \mathcal{D}_{\text{train}}) \geq 1 - \delta,$$

as required.

We define:

$$\begin{aligned} S_\lambda &:= \sum_{i \in \mathcal{I}_{\text{cal}}} w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}, \\ S_\lambda^{(1)} &:= \sum_{i \in \mathcal{I}_{\text{cal1}}} w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}, \\ S_\lambda^{(2)} &:= \sum_{i \in \mathcal{I}_{\text{cal2}}} w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}, \end{aligned}$$

so that $S_\lambda = S_\lambda^{(1)} + S_\lambda^{(2)}$ and $\hat{\alpha}(\lambda) = S_\lambda / |\mathcal{I}_{\text{cal}}|$. By construction of C_i for $i \in \mathcal{I}_{\text{cal1}}$: $C_i = \hat{f}_{\text{prior}}(X_i)$ deterministically and $w_\lambda(i) = 1$ (since $w_\lambda(i)^{-1} = \mathbb{P}(C_i \geq \hat{f}_\lambda(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) = 1$), so $\mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\} = 1$ deterministically, giving:

$$S_\lambda^{(1)} = \sum_{i \in \mathcal{I}_{\text{cal1}}} \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} =: N_\lambda^{(1)}.$$

Lastly, we set $A := \bar{w} - \alpha^2 \geq 1 - \alpha^2 > 0$ for $\alpha \in (0, 1)$.

Step 1: Chernoff bound given $\mathcal{D}_{\text{cal1}}$. By the tower property:

$$\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) = \mathbb{E} \left[\mathbb{P}(N_\lambda^{(1)} + S_\lambda^{(2)} \leq |\mathcal{I}_{\text{cal}}| \alpha \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \mid \mathcal{D}_{\text{train}} \right]. \quad (5)$$

Conditioning on $\mathcal{D}_{\text{cal1}}$ makes $N_\lambda^{(1)}$ a deterministic non-negative integer. Notice that $\bar{w}$ is also deterministic given $\mathcal{D}_{\text{train}}$ by its definition. Since $N_\lambda^{(1)}$ is deterministic given $\mathcal{D}_{\text{cal1}}$, Markov's inequality applied to the non-negative random variable $\exp(-t(N_\lambda^{(1)} + S_\lambda^{(2)} - |\mathcal{I}_{\text{cal}}| \alpha)) \geq 0$ gives, for any $t > 0$:

$$\mathbb{P}(N_\lambda^{(1)} + S_\lambda^{(2)} \leq |\mathcal{I}_{\text{cal}}| \alpha \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \leq e^{t|\mathcal{I}_{\text{cal}}| \alpha - t N_\lambda^{(1)}} \cdot \mathbb{E} \left[e^{-t S_\lambda^{(2)}} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right]. \quad (6)$$

Step 2: Bounding $\mathbb{E}[e^{-t S_\lambda^{(2)}} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]$. For $i \in \mathcal{I}_{\text{cal2}}$, define $V_i := w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}$, so $S_\lambda^{(2)} = \sum_{i \in \mathcal{I}_{\text{cal2}}} V_i$.

Mutual independence of $\{V_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$. By the i.i.d. assumption, $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$ (the global i.i.d. assumption gives cross- i independence, and the marginal law of each (X_i, T_i, H_i) is independent of $\mathcal{D}_{\text{cal1}}$ given $\mathcal{D}_{\text{train}}$). By the mutual independence assumption, given $\{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}}$ and $\mathcal{D}_{\text{cal1}}$, the censoring times $\{C_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent. Combining these via the tower property: for any measurable $f_i(X_i, T_i, H_i, C_i)$ and any finite $S \subseteq \mathcal{I}_{\text{cal2}}$,

$$\begin{aligned} & \mathbb{E} \left[\prod_{i \in S} f_i(X_i, T_i, H_i, C_i) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\prod_{i \in S} f_i(X_i, T_i, H_i, C_i) \mid \{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}}, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \\ &= \mathbb{E} \left[\prod_{i \in S} \mathbb{E}[f_i(X_i, T_i, H_i, C_i) \mid X_i, T_i, H_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \\ &= \prod_{i \in S} \mathbb{E}[f_i(X_i, T_i, H_i, C_i) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}], \end{aligned} \quad (7)$$

where the second equality uses mutual independence of $\{C_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ given $\{(X_k, H_k, T_k)\}_{k \in \mathcal{I}_{\text{cal2}}}$ and $\mathcal{D}_{\text{cal1}}$, and the third uses mutual independence of $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal2}}}$ given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$. Since V_i is a measurable function of (X_i, T_i, H_i, C_i) , (7) establishes mutual independence of $\{V_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$.

Mean of V_i (IPCW identity). By the tower property, conditioning first on $(X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$ and using the weight definition $w_\lambda(i)^{-1} = \mathbb{P}(C_i \geq \hat{f}_\lambda(X_i) \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$:

$$\begin{aligned} & \mathbb{E}[V_i \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \\ &= \mathbb{E} \left[w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \underbrace{\mathbb{E}[\mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\} \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]}_{=w_\lambda(i)^{-1}} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \\ &= \mathbb{E}[\mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] = \mathbb{P}(T < \hat{f}_\lambda(X) \mid \mathcal{D}_{\text{train}}) =: P_\lambda, \end{aligned}$$

where the last equality uses the i.i.d assumption.

Second moment and variance of V_i .

Since V_i is a product of three terms, we have:

$$\begin{aligned} V_i^2 &= \left(w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\} \right)^2 \\ &= w_\lambda(i)^2 \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\}^2 \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}^2 \\ &= w_\lambda(i)^2 \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mathbb{I}\{C_i \geq \hat{f}_\lambda(X_i)\}. \end{aligned}$$

By the tower property of conditional expectation:

$$\mathbb{E}[V_i^2 \mid X_i, H_i, T_i, \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] = w_\lambda(i)^2 \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} w_\lambda(i)^{-1} = w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\},$$

therefore:

$$\mathbb{E}[V_i^2 \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] = \mathbb{E}[w_\lambda(i) \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \leq \mathbb{E}[w_\lambda(i) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \leq \bar{w}.$$

One-sided Bernstein MGF bound for $S_\lambda^{(2)}$. Define mean-zero variables $Y_i^{(2)} := P_\lambda - V_i$ for $i \in \mathcal{I}_{\text{cal2}}$. Since $V_i \geq 0$, we have $Y_i^{(2)} \leq P_\lambda \leq 1$.

Since $Y_i^{(2)} := P_\lambda - V_i$ and $P_\lambda = \mathbb{E}[V_i \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}]$ is deterministic given $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}})$, we have:

$$\begin{aligned} \text{Var}(Y_i^{(2)} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) &= \text{Var}(P_\lambda - V_i \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\ &= \text{Var}(V_i \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}) \\ &= \mathbb{E}[V_i^2 \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] - P_\lambda^2 \\ &\leq \bar{w} - P_\lambda^2 \\ &\leq \bar{w} - \alpha^2 =: A \end{aligned}$$

where the second equality uses the fact that adding or subtracting a constant does not change variance and the last inequality follows from the definition of the supremum $\tau(\alpha + \Delta)$ and since $\lambda = \tau(\alpha + \Delta) + \varepsilon > \tau(\alpha + \Delta)$, we have $P_\lambda > \alpha + \Delta$.

We apply [59, Proposition 2.14] for $Y_i^{(2)}$ with $Y_i^{(2)} \leq 1$ a.s., and $\mathbb{E}[(Y_i^{(2)})^2 \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] = \text{Var}[Y_i^{(2)} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}] \leq A$ to get:

$$\mathbb{E} \left[\exp \left(t \sum_{i \in \mathcal{I}_{\text{cal2}}} Y_i^{(2)} \right) \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}} \right] \leq \exp \left(\frac{t^2 |\mathcal{I}_{\text{cal2}}| A / 2}{1 - t/3} \right), \quad 0 < t < 3.$$

Since $S_\lambda^{(2)} = |\mathcal{I}_{\text{cal2}}|P_\lambda - \sum_{i \in \mathcal{I}_{\text{cal2}}} Y_i^{(2)}$:

$$\mathbb{E}\left[e^{-tS_\lambda^{(2)}} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}\right] \leq \exp\left(-t|\mathcal{I}_{\text{cal2}}|P_\lambda + \frac{t^2|\mathcal{I}_{\text{cal2}}|A/2}{1-t/3}\right), \quad 0 < t < 3. \quad (8)$$

The right-hand side of (8) depends only on the deterministic quantities P_λ , A , $|\mathcal{I}_{\text{cal2}}|$, and t , and in particular does not depend on $\mathcal{D}_{\text{cal1}}$.

Step 3: Bounding $\mathbb{E}[e^{-tN_\lambda^{(1)}} \mid \mathcal{D}_{\text{train}}]$. Under our assumptions, $\{(X_i, T_i)\}_{i \in \mathcal{I}_{\text{cal1}}}$ are i.i.d. with marginal law independent of $\mathcal{D}_{\text{cal2}}$ given $\mathcal{D}_{\text{train}}$. Hence $Z_i^{(1)} := \mathbb{I}\{T_i < \hat{f}_\lambda(X_i)\} \sim \text{Bernoulli}(P_\lambda)$ are i.i.d. given $\mathcal{D}_{\text{train}}$, and $N_\lambda^{(1)} = \sum_{i \in \mathcal{I}_{\text{cal1}}} Z_i^{(1)}$.

Define mean-zero $U_i := P_\lambda - Z_i^{(1)}$ for $i \in \mathcal{I}_{\text{cal1}}$, so $\{U_i\}_{i \in \mathcal{I}_{\text{cal1}}}$ are i.i.d. given $\mathcal{D}_{\text{train}}$. Since $Z_i^{(1)} \geq 0$, $U_i \leq P_\lambda \leq 1$, so the one-sided bound $U_i \leq 1$ holds.

Variance of U_i bounded by A . $\text{Var}(U_i \mid \mathcal{D}_{\text{train}}) = P_\lambda(1 - P_\lambda) = P_\lambda - P_\lambda^2$. Since $w_\lambda(i) \geq 1$ implies $\bar{w} \geq 1 \geq P_\lambda$ (P_λ is a probability), and since $P_\lambda \geq \alpha + \Delta > \alpha$ implies $P_\lambda^2 > \alpha^2$:

$$P_\lambda(1 - P_\lambda) = P_\lambda - P_\lambda^2 \leq \bar{w} - \alpha^2 = A.$$

We apply [59, Proposition 2.14] with $b = 1$ and $\nu = A$ to each i.i.d. U_i , summing over $|\mathcal{I}_{\text{cal1}}|$ terms, and writing $N_\lambda^{(1)} = |\mathcal{I}_{\text{cal1}}|P_\lambda - \sum_{i \in \mathcal{I}_{\text{cal1}}} U_i$:

$$\begin{aligned} \mathbb{E}\left[e^{-tN_\lambda^{(1)}} \mid \mathcal{D}_{\text{train}}\right] &= e^{-t|\mathcal{I}_{\text{cal1}}|P_\lambda} \cdot \mathbb{E}\left[\exp\left(t \sum_{i \in \mathcal{I}_{\text{cal1}}} U_i\right) \mid \mathcal{D}_{\text{train}}\right] \\ &\leq \exp\left(-t|\mathcal{I}_{\text{cal1}}|P_\lambda + \frac{t^2|\mathcal{I}_{\text{cal1}}|A/2}{1-t/3}\right), \quad 0 < t < 3. \end{aligned} \quad (9)$$

Step 4: Combining the bounds. Applying (6) pointwise in $\mathcal{I}_{\text{cal1}}$, then taking the outer expectation over $\mathcal{I}_{\text{cal1}}$ via (5):

$$\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) \leq e^{t|\mathcal{I}_{\text{cal}}|\alpha} \cdot \mathbb{E}\left[e^{-tN_\lambda^{(1)}} \cdot \mathbb{E}\left[e^{-tS_\lambda^{(2)}} \mid \mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{train}}\right] \mid \mathcal{D}_{\text{train}}\right].$$

Since the upper bound in (8) is deterministic (does not depend on $\mathcal{D}_{\text{cal1}}$), it factors out of the outer expectation:

$$\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) \leq e^{t|\mathcal{I}_{\text{cal}}|\alpha} \cdot \exp\left(-t|\mathcal{I}_{\text{cal2}}|P_\lambda + \frac{t^2|\mathcal{I}_{\text{cal2}}|A/2}{1-t/3}\right) \cdot \mathbb{E}\left[e^{-tN_\lambda^{(1)}} \mid \mathcal{D}_{\text{train}}\right].$$

Applying (9) and using $|\mathcal{I}_{\text{cal1}}| + |\mathcal{I}_{\text{cal2}}| = |\mathcal{I}_{\text{cal}}|$:

$$\begin{aligned} \mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) &\leq \exp\left(t|\mathcal{I}_{\text{cal}}|\alpha - t|\mathcal{I}_{\text{cal2}}|P_\lambda - t|\mathcal{I}_{\text{cal1}}|P_\lambda + \frac{t^2(|\mathcal{I}_{\text{cal2}}| + |\mathcal{I}_{\text{cal1}}|)A/2}{1-t/3}\right) \\ &= \exp\left(-t|\mathcal{I}_{\text{cal}}|(P_\lambda - \alpha) + \frac{t^2|\mathcal{I}_{\text{cal}}|A/2}{1-t/3}\right) \\ &\leq \exp\left(-t|\mathcal{I}_{\text{cal}}|\Delta + \frac{t^2|\mathcal{I}_{\text{cal}}|A/2}{1-t/3}\right) =: \exp(\mathcal{E}(t)), \end{aligned}$$

for $0 < t < 3$, where the last inequality uses $P_\lambda - \alpha \geq \Delta$ (since $\lambda > \tau(\alpha + \Delta)$ implies $P_\lambda = \mathbb{P}(T < \hat{f}_\lambda(X) \mid \mathcal{D}_{\text{train}}) \geq \alpha + \Delta$, by definition of $\tau(\alpha + \Delta)$ and continuity of $\hat{f}_\tau$).

Step 5: Evaluation at the Bernstein point and solving for Δ . Set $N := |\mathcal{I}_{\text{cal}}|$, and recall $A = \bar{w} - \alpha^2 > 0$. We evaluate $\mathcal{E}$ at:

$$t^* := \frac{\Delta}{A + \Delta/3}.$$

Notice that $t^* \in (0, 3)$ since $\Delta, A > 0$. Substituting into $\mathcal{E}(t^*)$:

$$\begin{aligned}\mathcal{E}(t^*) &= -\frac{N\Delta^2}{A + \Delta/3} + \frac{NA}{2} \cdot \frac{\Delta^2}{(A + \Delta/3)^2} \cdot \frac{1}{A/(A + \Delta/3)} \\ &= -\frac{N\Delta^2}{A + \Delta/3} + \frac{NA}{2} \cdot \frac{\Delta^2}{A(A + \Delta/3)} \\ &= -\frac{N\Delta^2}{A + \Delta/3} + \frac{N\Delta^2}{2(A + \Delta/3)} \\ &= -\frac{N\Delta^2}{2(A + \Delta/3)}\end{aligned}$$

Therefore,

$$\exp(\mathcal{E}(t^*)) \leq \delta \iff \frac{N\Delta^2}{2(A + \Delta/3)} \geq \log\left(\frac{1}{\delta}\right). \quad (10)$$

Multiplying both sides by $2(A + \Delta/3) > 0$ and rearranging:

$$|\mathcal{I}_{\text{cal}}|\Delta^2 - \frac{2}{3}\log(1/\delta)\Delta - 2(\bar{w} - \alpha^2)\log\left(\frac{1}{\delta}\right) = 0.$$

This quadratic in Δ has positive leading coefficient $|\mathcal{I}_{\text{cal}}| > 0$ and negative constant term $-2A\log(1/\delta) < 0$. By Descartes' rule of signs, it has exactly one positive root:

$$\Delta := \frac{\log(1/\delta)}{3|\mathcal{I}_{\text{cal}}|} + \sqrt{\frac{\log^2(1/\delta)}{9|\mathcal{I}_{\text{cal}}|^2} + \frac{2(\bar{w} - \alpha^2)\log(1/\delta)}{|\mathcal{I}_{\text{cal}}|}}.$$

For the above Δ , condition (10) holds with equality, so $\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{D}_{\text{train}}) \leq \exp(\mathcal{E}(t^*)) \leq \delta$, satisfying (4).

Step 6: Conclusion. Since (4) holds for every $\varepsilon > 0$ with $\lambda = \tau(\alpha + \Delta) + \varepsilon$, we have $\mathbb{P}(\hat{\tau} \leq \lambda \mid \mathcal{D}_{\text{train}}) \geq 1 - \delta$ for every $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0^+$ and applying continuity of the probability measure to the decreasing family of events $\{\hat{\tau} \leq \tau(\alpha + \Delta) + \varepsilon\}$, whose intersection is $\{\hat{\tau} \leq \tau(\alpha + \Delta)\}$:

$$\mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) \mid \mathcal{D}_{\text{train}}) = \lim_{\varepsilon \rightarrow 0^+} \mathbb{P}(\hat{\tau} \leq \tau(\alpha + \Delta) + \varepsilon \mid \mathcal{D}_{\text{train}}) \geq 1 - \delta.$$

On the event $\{\hat{\tau} \leq \tau(\alpha + \Delta)\}$, monotonicity of $\hat{f}_{\hat{\tau}}$ in τ gives $\hat{f}_{\hat{\tau}}(x) \leq \hat{f}_{\tau(\alpha + \Delta)}(x)$ for all x , hence $\{T \geq \hat{f}_{\tau(\alpha + \Delta)}(X)\} \subseteq \{T \geq \hat{f}_{\hat{\tau}}(X)\}$. Since $\hat{f}_{\tau(\alpha + \Delta)}$ is $\mathcal{D}_{\text{train}}$ -measurable and $(X_{\text{test}}, T_{\text{test}})$ is independent of $(\mathcal{D}_{\text{cal1}}, \mathcal{D}_{\text{cal2}})$ given $\mathcal{D}_{\text{train}}$:

$$\mathbb{P}\left[T_{\text{test}} \geq \hat{f}_{\hat{\tau}}(X_{\text{test}}) \mid \mathcal{D}_{\text{all}}\right] \geq \mathbb{P}\left[T_{\text{test}} \geq \hat{f}_{\tau(\alpha + \Delta)}(X_{\text{test}}) \mid \mathcal{D}_{\text{all}}\right] \geq 1 - \alpha - \Delta,$$

where the last inequality uses the definition of $\tau(\alpha + \Delta)$ and left-continuity of $\tau \mapsto \mathbb{P}(T < \hat{f}_{\tau}(X) \mid \mathcal{D}_{\text{train}})$, which give $\mathbb{P}(T < \hat{f}_{\tau(\alpha + \Delta)}(X) \mid \mathcal{D}_{\text{all}}) \leq \alpha + \Delta$. Notice that $\mathbb{P}(T < \hat{f}_{\tau(\alpha + \Delta)}(X) \mid \mathcal{D}_{\text{all}}) = \mathbb{P}(T < \hat{f}_{\tau(\alpha + \Delta)}(X) \mid \mathcal{D}_{\text{train}})$ since $\hat{f}_{\tau(\alpha + \Delta)}$ is $\mathcal{D}_{\text{train}}$ -measurable and $(X_{\text{test}}, T_{\text{test}}) \perp\!\!\!\perp \mathcal{D}_{\text{cal}} \mid \mathcal{D}_{\text{train}}$. This holds with probability at least $1 - \delta$ over $\mathcal{D}_{\text{cal}}$ given $\mathcal{D}_{\text{train}}$, hence over $\mathcal{D}_{\text{all}}$, completing the proof. $\square$

C.2 Budget validity results

In this section, we prove that the expected budget used by Algorithm 5 is bounded by the nominal budget constraint B . Unless stated otherwise, all expectations and probabilities in this section are conditional on $\mathcal{D}_{\text{train}}$; we omit this explicit conditioning for notational convenience.

C.2.1 Assumptions

We begin by setting the following assumptions.

Assumption 1 (i.i.d. data). The pairs $\{(S_k, P_k)\}_{k=1}^N$ are independent and identically distributed, each drawn from a joint distribution ν on $\mathbb{R}^{t_{\max}} \times [0, 1]^{t_{\max}}$. We denote by μ the marginal distribution of S_k on $\mathbb{R}^{t_{\max}}$ and by μ_t the marginal distribution of the t -th coordinate $S_k(t)$ on $\mathbb{R}$, for $t \in \{1, \dots, t_{\max}\}$.

Assumption 2 (Split independence). The data is randomly split by π , which is a random variable, taking values in the collection of subsets of $\{1, \dots, N\}$ of size N_1 , distributed uniformly over all $\binom{N}{N_1}$ such subsets. The random split π is independent of the full dataset $\{(S_k, P_k)\}_{k=1}^N$.

Recall that for a given split π , Algorithm 5 splits the calibration data, indexed by $\{1, \dots, N\}$, as follows:

$$\begin{aligned}\mathcal{I}_{\text{cal1}} &:= \pi \subset \{1, \dots, N\}, & |\mathcal{I}_{\text{cal1}}| &= N_1, \\ \mathcal{I}_{\text{cal2}} &:= \{1, \dots, N\} \setminus \mathcal{I}_{\text{cal1}}, & |\mathcal{I}_{\text{cal2}}| &= N_2, \\ \mathcal{D}_{\text{cal1}} &:= \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal1}}} & & \text{(calibration dataset),} \\ S_{\text{cal2}} &:= S_j \quad \text{for a fixed } j \in \mathcal{I}_{\text{cal2}}.\end{aligned}$$

By symmetry of the i.i.d. assumption and the uniform split, the choice of j within $\mathcal{I}_{\text{cal2}}$ does not affect any expectation below. We re-index the calibration (cal1) samples as $i \in \{1, \dots, N_1\}$ for notational convenience.

Assumption 3 (Oracle mapping). For each $\beta \geq 0$ and each $t \in \{1, \dots, t_{\max}\}$, there exists a measurable function

$$M_t^*(\cdot; \beta) : \mathbb{R} \rightarrow [0, 1]$$

that is monotonically increasing in its first argument. For $s = (s(1), \dots, s(t_{\max})) \in \mathbb{R}^{t_{\max}}$ we write $M^*(s; \beta) := (M_1^*(s(1); \beta), \dots, M_{t_{\max}}^*(s(t_{\max}); \beta))$.

Recall that $\bar{B}_2 = (B - B_1)/N_2$ is the remaining budget per sample for Phase II, which is a deterministic function of $\mathcal{D}_{\text{cal1}}$. For a given realization of $\mathcal{D}_{\text{cal1}}$ (and hence a given value of $\bar{B}_2$), the oracle mapping $M^*(\cdot; \bar{B}_2)$ satisfies the following two conditions $\mathcal{D}_{\text{cal1}}$ -almost surely:

1. **(Pathwise oracle fit)** For all $i \in \{1, \dots, N_1\}$:

$$P_i(t) = M_t^*(S_i(t); \bar{B}_2) \quad \mathbb{P}\text{-almost surely.}$$

2. **(Pathwise population budget constraint)**

$$\mathbb{E}_{S \sim \mu}[\mathcal{B}(M^*(S; \bar{B}_2))] \leq \bar{B}_2.$$

Assumption 4 (Estimated mapping). Given $\mathcal{D}_{\text{cal1}}$, the estimated mapping $M_t(\cdot; \mathcal{D}_{\text{cal1}}) : \mathbb{R} \rightarrow [0, 1]$ is measurable and monotonically increasing for each $t \in \{1, \dots, t_{\max}\}$. Write $M(s; \mathcal{D}_{\text{cal1}}) := (M_1(s(1); \mathcal{D}_{\text{cal1}}), \dots, M_{t_{\max}}(s(t_{\max}); \mathcal{D}_{\text{cal1}}))$.

Assumption 5 (vanishing cross-term moments). For each $t \in \{1, \dots, t_{\max}\}$ and each non-empty subset $A \subseteq \{1, \dots, t\}$, the following holds for $\mathcal{D}_{\text{cal1}}$ -almost every realization:

$$\mathbb{E}_{S_{\text{cal2}}} \left[\prod_{k \in A} \varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}}) \cdot \prod_{k \in \{1, \dots, t\} \setminus A} M_k^*(S_{\text{cal2}}(k)) \mid \mathcal{D}_{\text{cal1}} \right] = 0,$$

where $\varepsilon_k(s; \mathcal{D}_{\text{cal1}}) = M_k(s; \mathcal{D}_{\text{cal1}}) - M_k^*(s; \bar{B}_2)$.

Conceptually, Assumption 5 requires that these mixed products have zero conditional expectation over S_{cal2} , given the calibration data $\mathcal{D}_{\text{cal1}}$. Intuitively, this implies that the projection errors $\{\varepsilon_k\}_k$ are unbiased and mutually uncorrelated across time steps. We emphasize that this condition is not a mere mathematical artifact of our proof; rather, it is required to guarantee that the projection inaccuracies do not inflate the expected budget consumed during Phase II.

In practice, there are two cases where this assumption is satisfied. First, it holds exactly in finite samples if the coordinates of the score vector S_{cal2} are mutually independent, and the estimated mapping is coordinate-wise unbiased, i.e., $\mathbb{E}_{S_{\text{cal2}}(k)}[\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})] = 0$ almost surely, for all k .

Alternatively, even without assuming independence or finite-sample unbiasedness, this assumption holds asymptotically assuming $M_t(\cdot \mid \mathcal{D}_{\text{cal1}})$ is a consistent estimator of M_t^* . Specifically, if

$\mathbb{E}_{S_{\text{cal2}}(j)}[|M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) - M_j^*(S_{\text{cal2}}(j))| \mid \mathcal{D}_{\text{cal1}}] \rightarrow 0$ as $N_1 \rightarrow \infty$, $\mathcal{D}_{\text{cal1}}$ -almost surely, then, the magnitude of the cross-terms for any fixed non-empty set A vanishes:

$$\left| \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j \in A} \varepsilon_j \cdot \prod_{j \notin A} M_j^*(S_{\text{cal2}}; \bar{B}_2) \mid \mathcal{D}_{\text{cal1}} \right] \right| \leq \prod_{j \in A} \mathbb{E}_{S_{\text{cal2}}(j)}[|\varepsilon_j| \mid \mathcal{D}_{\text{cal1}}] \rightarrow 0 \quad \text{as } N_1 \rightarrow \infty,$$

where the inequality uses the fact that $M_j^* \in [0, 1]$ and Hölder's inequality. Thus, Assumption 5 holds asymptotically for any fixed $\mathcal{D}_{\text{cal1}}$ and target budget $\bar{B}_2$, meaning the budget constraint can be satisfied with arbitrary precision given sufficient calibration data in Phase I.

C.2.2 Expected budget validity proofs

To establish the expected budget bound, we first prove the following lemmas.

Independence lemma

Lemma 1 (Independence of $\mathcal{D}_{\text{cal1}}$ and S_{cal2}). *Under Assumptions 1 and 2,*

$$\mathcal{D}_{\text{cal1}} \perp S_{\text{cal2}}.$$

Proof. Given the split π , $\mathcal{D}_{\text{cal1}}$ and S_{cal2} correspond to disjoint subsets of the i.i.d. array $\{(S_k, P_k)\}_{k=1}^N$. Disjoint subsets of an i.i.d. sequence are independent of each other. Since π is independent of the data (Assumption 2), conditioning on π does not introduce dependence, and so $\mathcal{D}_{\text{cal1}} \perp S_{\text{cal2}}$. $\square$

Properties of the budget function The budget function $\mathcal{B} : [0, 1]^{t_{\max}} \rightarrow \mathbb{R}$ is given by:

$$\mathcal{B}(P) = \sum_{t=1}^{\min(T_i, \hat{f}_{\text{prior}}(X_i))} \prod_{j=1}^t P(j), \quad P = (P(1), \dots, P(t_{\max})) \in [0, 1]^{t_{\max}}.$$

Lemma 2 (Properties of the budget function). *1. **Boundedness:** $0 \leq \mathcal{B}(P) \leq t_{\max}$ for all $P \in [0, 1]^{t_{\max}}$.*

*2. **Multilinearity:** $\mathcal{B}$ is a polynomial in $\{P(j)\}_{j=1}^{t_{\max}}$ that is linear in each coordinate $P(j)$ when all other coordinates are held fixed.*

*3. **Monotonicity:** $\mathcal{B}$ is monotonically increasing in each coordinate $P(j)$.*

Proof. (i) Each product $\prod_{j=1}^t P(j) \in [0, 1]$ and there are $t_{\max}$ terms. (ii) Direct inspection of the polynomial structure. (iii) For any j_0 , increasing $P(j_0)$ increases every term $\prod_{j=1}^t P(j)$ with $t \geq j_0$. $\square$

Estimation error does not affect expected loss

Lemma 3 (Cross-term cancellation). *Under Assumptions 1, 2, 3, 4, and 5, the following holds for $\mathcal{D}_{\text{cal1}}$ -almost every realization of the calibration data:*

$$\mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2)) \mid \mathcal{D}_{\text{cal1}}].$$

Here $\mathbb{E}_{S_{\text{cal2}}}[\cdot \mid \mathcal{D}_{\text{cal1}}]$ denotes the conditional expectation over S_{cal2} given $\mathcal{D}_{\text{cal1}}$. Since $\mathcal{D}_{\text{cal1}} \perp S_{\text{cal2}}$ (Lemma 1), this conditional expectation equals the marginal expectation over $S_{\text{cal2}} \sim \mu$, with $\mathcal{D}_{\text{cal1}}$ treated as a fixed parameter.

Proof. Fix a realization of $\mathcal{D}_{\text{cal1}}$ (a.s.). Define for each $j \in \{1, \dots, t_{\max}\}$:

$$p_j := M_j^*(S_{\text{cal2}}(j); \bar{B}_2), \quad e_j := \varepsilon_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) = M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) - M_j^*(S_{\text{cal2}}(j); \bar{B}_2).$$

Given $\mathcal{D}_{\text{cal1}}$, both p_j and e_j are bounded, measurable functions of $S_{\text{cal2}}(j)$ alone. For each $t \in$

$\{1, \dots, t_{\max}\}$, expand each factor as $M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) = p_j + e_j$ and expand the product by the distributive law:

$$\prod_{j=1}^t M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) = \prod_{j=1}^t (p_j + e_j) = \sum_{A \subseteq \{1, \dots, t\}} \left(\prod_{j \in A} e_j \right) \left(\prod_{j \in \{1, \dots, t\} \setminus A} p_j \right).$$

We prove the above by induction over t . For $t = 1$: $\prod_{j=1}^1 (p_j + e_j) = p_1 + e_1$, which satisfies this identity. Assume that the identity holds for some arbitrary integer $k \geq 1$. Then, for $t = k + 1$:

$$\prod_{j=1}^{k+1} (p_j + e_j) = \left(\prod_{j=1}^k (p_j + e_j) \right) (p_{k+1} + e_{k+1})$$

By our inductive assumption, we can substitute the expansion for the product up to k :

$$= \left(\sum_{A \subseteq S_k} \left(\prod_{j \in A} e_j \right) \left(\prod_{j \in S_k \setminus A} p_j \right) \right) (p_{k+1} + e_{k+1})$$

Distributing the terms p_{k+1} and e_{k+1} across the sum gives two separate sums:

$$\begin{aligned} &= \sum_{A \subseteq S_k} \left(\prod_{j \in A} e_j \right) \left(\prod_{j \in S_k \setminus A} p_j \right) p_{k+1} + \sum_{A \subseteq S_k} \left(\prod_{j \in A} e_j \right) \left(\prod_{j \in S_k \setminus A} p_j \right) e_{k+1} \\ &= \sum_{X \subseteq S_{k+1}} \left(\prod_{j \in X} e_j \right) \left(\prod_{j \in S_{k+1} \setminus X} p_j \right) \\ &= \prod_{j=1}^{k+1} (p_j + e_j). \end{aligned}$$

We now take the conditional expectation given $\mathcal{D}_{\text{cal1}}$.

$$\begin{aligned} &\mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) \mid \mathcal{D}_{\text{cal1}} \right] \\ &= \sum_{A \subseteq \{1, \dots, t\}} \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j \in A} e_j \cdot \prod_{j \in \{1, \dots, t\} \setminus A} p_j \mid \mathcal{D}_{\text{cal1}} \right]. \end{aligned} \quad (11)$$

Exchanging sum and expectation is valid since the sum is finite and all terms are bounded (Lemma 2(i)). Next, we isolate the $A = \emptyset$ term.

$$(11) = \underbrace{\mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t p_j \mid \mathcal{D}_{\text{cal1}} \right]}_{A=\emptyset} + \sum_{\emptyset \neq A \subseteq \{1, \dots, t\}} \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j \in A} e_j \cdot \prod_{j \in \{1, \dots, t\} \setminus A} p_j \mid \mathcal{D}_{\text{cal1}} \right]. \quad (12)$$

Now, for each non-empty $A \subseteq \{1, \dots, t\}$, we apply Assumption 5:

$$\mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j \in A} e_j \cdot \prod_{j \in \{1, \dots, t\} \setminus A} p_j \mid \mathcal{D}_{\text{cal1}} \right] = 0 \quad \mathcal{D}_{\text{cal1}}\text{-a.s.}$$

Hence the entire sum over non-empty A in (12) is zero, and:

$$\mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) \mid \mathcal{D}_{\text{cal1}} \right] = \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t M_j^*(S_{\text{cal2}}(j); \bar{B}_2) \mid \mathcal{D}_{\text{cal1}} \right]. \quad (13)$$

Summing (13) over $t \in \{1, \dots, t_{\max}\}$ ($P_i(t) = M_t^*(S_i; \bar{B}_2) = 0$ for $t > b_i$) and using linearity of conditional expectation:

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}); \bar{B}_2) \mid \mathcal{D}_{\text{cal1}}].$$

□

Oracle budget is bounded

Lemma 4 (Oracle budget bound). *Under Assumption 3, for $\mathcal{D}_{\text{cal1}}$ -almost every realization:*

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2)) \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2.$$

Proof. This follows directly from condition (ii) of Assumption 3, which states that $\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2)) \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2$ holds $\mathcal{D}_{\text{cal1}}$ -almost surely. □

Proof of the Holdout Budget Bound

Proposition 2 (Holdout sample budget bound). *Under Assumptions 1, 2, 3, 4, and 5, for $\mathcal{D}_{\text{cal1}}$ -almost every realization:*

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2.$$

Proof of Proposition 2. By Lemma 3 (cross-term cancellation), for $\mathcal{D}_{\text{cal1}}$ -almost every realization:

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2)) \mid \mathcal{D}_{\text{cal1}}].$$

By Lemma 4 (oracle budget bound), for $\mathcal{D}_{\text{cal1}}$ -almost every realization:

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2)) \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2.$$

Chaining these two relations gives the result. □

Proof of the Algorithm's Budget Validity

Theorem 5 (Algorithm's budget bound). *Under Assumptions 1, 2, 3, 4, and 5, the expected budget used by Algorithm 5 is bounded by B .*

Proof of Theorem 5. Condition on $\mathcal{D}_{\text{cal1}}$. Given $\mathcal{D}_{\text{cal1}}$, the Phase I budget $B_1 = \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i$ is a fixed constant, and $\bar{B}_2 = (B - B_1)/N_2$ is a fixed constant as well. The total realized budget is $B_1 + B_2$, where $B_2 = \sum_{i \in \mathcal{I}_{\text{cal2}}} U_i$ is random over Phase II draws, with $U_i = \sum_{t=1}^{b_i} \prod_{j=1}^t \xi_i(j)$.

By the tower property and the mutual independence of $\{U_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ conditioned on $\mathcal{D}_{\text{cal1}}$:

$$\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] = \sum_{i \in \mathcal{I}_{\text{cal2}}} \mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}].$$

For each $i \in \mathcal{I}_{\text{cal2}}$, by the tower property and $\mathcal{D}_{\text{cal1}} \perp\!\!\!\perp S_i$ (Lemma 1):

$$\mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_i} [\mathbb{E}[U_i \mid S_i, T_i, \mathcal{D}_{\text{cal1}}] \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}}))] ,$$

where the second equality uses $\mathbb{E}[U_i \mid S_i, T_i, \mathcal{D}_{\text{cal1}}] = \mathcal{B}(M(S_i; \mathcal{D}_{\text{cal1}}))$ (the expected stopping time equals the budget function evaluated at the continuation probabilities, given the Bernoulli draws are independent).

By Proposition 2, for $\mathcal{D}_{\text{cal1}}$ -almost every realization:

$$\mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2.$$

Therefore $\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] \leq N_2 \bar{B}_2$, and:

$$\begin{aligned} \mathbb{E}[B_1 + B_2 \mid \mathcal{D}_{\text{cal1}}] &\leq B_1 + N_2 \bar{B}_2 \\ &= B_1 + N_2 \cdot \frac{B - B_1}{N_2} \\ &= B. \end{aligned}$$

Taking the expectation over $\mathcal{D}_{\text{cal1}}$ gives $\mathbb{E}[B_1 + B_2] \leq B$. □

C.2.3 Finite-sample budget guarantee

In this section, we provide a finite-sample bound for the budget consumed by Algorithm 5. Throughout this section, all probabilities and expectations are conditioned on $\mathcal{D}_{\text{train}}$. For notational convenience, we omit this conditioning.

Theorem 6. *Under Assumptions 1, 2, 4, and 5, with probability at least $1 - \delta$, the average budget per sample consumed by Algorithm 5 is bounded by:*

$$\frac{1}{N} \sum_{i \in \mathcal{I}_{\text{cal}}} \tilde{T}_i \leq \frac{B}{N} + \frac{t_{\max} \log(1/\delta)}{3N} + \frac{1}{N} \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2N_2 t_{\max} \bar{B}_2 \log(1/\delta)}.$$

Proof. We denote $B_1 = \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i$, where $b_i = \min(T_i, \hat{f}_{\text{prior}}(X_i))$, and $U_i = \sum_{t=1}^{b_i} \prod_{j=1}^t \xi_i(j)$ for any $i \in \mathcal{I}_{\text{cal2}}$, so that $B_2 = \sum_{i \in \mathcal{I}_{\text{cal2}}} U_i$.

For a fixed $\mathcal{D}_{\text{cal1}}$, and fixed (S_i, T_i) , the continuation probabilities $M_j(S_i(j); \mathcal{D}_{\text{cal1}})$ are deterministic. The Bernoulli draws $\{\xi_i(j)\}_j$ are then mutually independent, so:

$$\mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}, S_i, T_i] = \sum_{t=1}^{b_i} \prod_{j=1}^t M_j(S_i(j); \mathcal{D}_{\text{cal1}}) = \mathcal{B}(M(S_i; \mathcal{D}_{\text{cal1}})).$$

Taking a further expectation over (S_i, T_i) and using $\mathcal{D}_{\text{cal1}} \perp\!\!\!\perp S_i$ (Lemma 1):

$$\mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}].$$

By Proposition 2, for $\mathcal{D}_{\text{cal1}}$ -almost every realization:

$$\mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2.$$

Hence:

$$\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] = \sum_{i \in \mathcal{I}_{\text{cal2}}} \mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] \leq N_2 \bar{B}_2.$$

Notice that the collection $\{U_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ is mutually independent given $\mathcal{D}_{\text{cal1}}$. Since $U_i \in [0, t_{\max}]$ almost surely, by Lemma 2, and $\mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] \leq \bar{B}_2$, the Bhatia–Davis inequality gives:

$$\text{Var}[U_i] \leq (t_{\max} - \mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}]) \cdot \mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}] \leq t_{\max} \bar{B}_2.$$

Observe that:

$$\mathbb{E} \left[\left(U_i - \frac{1}{N_2} \mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] \right)^2 \middle| \mathcal{D}_{\text{cal1}} \right] = \mathbb{E}[(U_i - \mathbb{E}[U_i \mid \mathcal{D}_{\text{cal1}}])^2 \mid \mathcal{D}_{\text{cal1}}] = \text{Var}(U_i \mid \mathcal{D}_{\text{cal1}}).$$

We now apply the Bernstein inequality to the sum $B_2 = \sum_{i \in \mathcal{I}_{\text{cal2}}} U_i$. Notice that $\{U_i\}_{i \in \mathcal{I}_{\text{cal2}}}$ are mutually independent variables given $\mathcal{D}_{\text{cal1}}$, where each $U_i \in [0, t_{\max}]$.

$$\begin{aligned} \mathbb{P}(B_2 - \mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] \geq \varepsilon \mid \mathcal{D}_{\text{cal1}}) &\leq \exp \left(- \frac{\varepsilon^2/2}{\sum_{i \in \mathcal{I}_{\text{cal2}}} \text{Var}(U_i \mid \mathcal{D}_{\text{cal1}}) + t_{\max} \varepsilon/3} \right) \\ &\leq \exp \left(- \frac{\varepsilon^2/2}{N_2 t_{\max} \bar{B}_2 + t_{\max} \varepsilon/3} \right). \end{aligned}$$

We now show that by setting the ε as given below, the probability above is bounded by δ :

$$\varepsilon = \frac{t_{\max} \log(1/\delta)}{3} + \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2N_2 t_{\max} \bar{B}_2 \log(1/\delta)}$$

For ease of notation, we denote $V = \sum_{i \in \mathcal{I}_{\text{cal2}}} t_{\max} \bar{B}_2$.

$$\varepsilon - \frac{t_{\max} \log(1/\delta)}{3} = \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2V \log(1/\delta)}$$

Squaring both sides yields:

$$\left(\varepsilon - \frac{t_{\max} \log(1/\delta)}{3}\right)^2 = \frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2V \log(1/\delta)$$

$$\varepsilon^2 - \frac{2t_{\max} \log(1/\delta)}{3} \varepsilon + \frac{t_{\max}^2 \log^2(1/\delta)}{9} = \frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2V \log(1/\delta)$$

Notice that the $\frac{t_{\max}^2 \log^2(1/\delta)}{9}$ term appears on both sides and cancels out:

$$\varepsilon^2 - \frac{2t_{\max} \log(1/\delta)}{3} \varepsilon = 2V \log(1/\delta)$$

Divide the entire equation by 2:

$$\frac{\varepsilon^2}{2} - \frac{t_{\max} \varepsilon}{3} \log(1/\delta) = V \log(1/\delta)$$

Move the ε term to the right side:

$$\frac{\varepsilon^2}{2} = V \log(1/\delta) + \frac{t_{\max} \varepsilon}{3} \log(1/\delta)$$

$$\frac{\varepsilon^2}{2} = \left(V + \frac{t_{\max} \varepsilon}{3}\right) \log(1/\delta)$$

The exponent in the original inequality bound is:

$$-\frac{\varepsilon^2/2}{V + t_{\max} \varepsilon/3}$$

Substituting the equivalent expression we just found for $\frac{\varepsilon^2}{2}$:

$$-\frac{\left(V + \frac{t_{\max} \varepsilon}{3}\right) \log(1/\delta)}{V + t_{\max} \varepsilon/3} = -\log(1/\delta)$$

Therefore,

$$\exp(-\log(1/\delta)) = \exp(\log(\delta)) = \delta$$

Putting it all together into a single block to match the original formulation:

$$\begin{aligned} \mathbb{P}(B_2 - \mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal}1}] \geq \varepsilon \mid \mathcal{D}_{\text{cal}1}) &\leq \exp\left(-\frac{\varepsilon^2/2}{\sum_{i \in \mathcal{I}_{\text{cal}2}} t_{\max} \bar{B}_2 + t_{\max} \varepsilon/3}\right) \\ &\leq \exp\left(-\frac{\varepsilon^2/2}{\sum_{i \in \mathcal{I}_{\text{cal}2}} t_{\max} \bar{B}_2 + t_{\max} \varepsilon/3}\right) \\ &= \exp\left(-\frac{(\sum_{i \in \mathcal{I}_{\text{cal}2}} t_{\max} \bar{B}_2 + \frac{t_{\max} \varepsilon}{3}) \log(1/\delta)}{\sum_{i \in \mathcal{I}_{\text{cal}2}} t_{\max} \bar{B}_2 + t_{\max} \varepsilon/3}\right) \\ &= \exp(-\log(1/\delta)) \\ &= \delta. \end{aligned}$$

Therefore, with probability at least $1 - \delta$, the budget used in Phase II is bounded by:

$$B_2 \leq N_2 \bar{B}_2 + \frac{t_{\max} \log(1/\delta)}{3} + \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2N_2 t_{\max} \bar{B}_2 \log(1/\delta)}.$$

By the construction of $\bar{B}_2$ of Algorithm 5, we have:

$$B_1 + N_2 \bar{B}_2 \leq B.$$

Therefore, the average budget per sample used by Algorithm 5 is bounded by:

$$\frac{1}{N} \sum_{i \in \mathcal{I}_{\text{cal}}} \tilde{T}_i \leq \frac{B}{N} + \frac{t_{\max} \log(1/\delta)}{3N} + \frac{1}{N} \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2N_2 t_{\max} \bar{B}_2 \log(1/\delta)}.$$

□

C.2.4 Budget Guarantee with Bounded Estimation Errors

We develop a bound for the utilized budget, replacing Assumption 5 with a weaker condition that bounds the mean absolute estimation error at each step instead of requiring that the estimation errors cancel out. Specifically, we rely on the following assumption:

Assumption 6 (Bounded Mean Absolute Estimation Error). For each $j \in \{1, \dots, t_{\max}\}$, there exists a constant $\eta_j \geq 0$ such that, almost surely over $\mathcal{D}_{\text{cal1}}$:

$$\mathbb{E}_{S_{\text{cal2}}(j)}[|\varepsilon_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}] \leq \eta_j,$$

where $\varepsilon_j(s; \mathcal{D}_{\text{cal1}}) := M_j(s; \mathcal{D}_{\text{cal1}}) - M_j^*(s; \bar{B}_2)$ is the pointwise estimation error at step j .

Observe that Assumption 5 requires the cross-term expectations $\mathbb{E}[\prod_{j \in A} \varepsilon_j(s; \mathcal{D}_{\text{cal1}}) \cdot \prod_{j \notin A} M_j^*(s; \bar{B}_2) \mid \mathcal{D}_{\text{cal1}}] = 0$ for every non-empty A , which implies $\eta_j = 0$ in case where S_{cal2} has independent coordinates. In contrast, Assumption 6 is strictly weaker: it requires only that the marginal mean absolute error at each coordinate is bounded, without making assumptions on the joint cross-term. In particular, it does not require that the score coordinates are independent. Furthermore, this assumption is reasonable for consistent estimators $M_j(\cdot; \mathcal{D}_{\text{cal1}})$ that satisfy $\mathbb{E}_{S_{\text{cal2}}(j)}[|\varepsilon_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}] \rightarrow 0$ as $N_1 \rightarrow \infty$, and thus $\eta_j \rightarrow 0$.

Theorem 7 (Expected Budget with Estimation Error). *Under Assumptions 1, 2, 3, 4 and 6, the expected total budget consumed by Algorithm 5 satisfies:*

$$\mathbb{E}_{\mathcal{D}_{\text{cal}}} \left[\sum_{i \in \mathcal{I}_{\text{cal}}} \tilde{T}_i \right] \leq B + N_2 \Gamma,$$

where $\Gamma = \sum_{k=1}^{t_{\max}} (t_{\max} - k + 1) \eta_k$.

To prove it, we first develop an upper bound on the expected budget used in Phase II:

Lemma 5. *Under Assumptions 1, 2, 3, 4, for $\mathcal{D}_{\text{cal1}}$ -almost every realization:*

$$\mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] \leq \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2))] + \Gamma,$$

where the total bias is:

$$\Gamma := \sum_{k=1}^{t_{\max}} (t_{\max} - k + 1) \eta_k.$$

Proof. We first show a telescoping product identity. For any two sequences of real numbers $\{a_j\}_{j=1}^t$ and $\{b_j\}_{j=1}^t$, the following identity holds:

$$\prod_{j=1}^t a_j - \prod_{j=1}^t b_j = \sum_{k=1}^t \left(\prod_{j=1}^{k-1} a_j \right) (a_k - b_k) \left(\prod_{j=k+1}^t b_j \right),$$

where empty products equal 1. We verify by induction. For $t = 1$: $a_1 - b_1 = (a_1 - b_1)$, which is trivially true. Assuming it holds for $t - 1$:

$$\begin{aligned} \prod_{j=1}^t a_j - \prod_{j=1}^t b_j &= a_t \prod_{j=1}^{t-1} a_j - b_t \prod_{j=1}^{t-1} b_j \\ &= (a_t - b_t) \prod_{j=1}^{t-1} a_j + b_t \left(\prod_{j=1}^{t-1} a_j - \prod_{j=1}^{t-1} b_j \right) \\ &= (a_t - b_t) \prod_{j=1}^{t-1} a_j + b_t \sum_{k=1}^{t-1} \left(\prod_{j=1}^{k-1} a_j \right) (a_k - b_k) \left(\prod_{j=k+1}^{t-1} b_j \right) \\ &= \sum_{k=1}^t \left(\prod_{j=1}^{k-1} a_j \right) (a_k - b_k) \left(\prod_{j=k+1}^t b_j \right), \end{aligned}$$

completing the induction.

We now apply this identity to the continuation probabilities. Set $a_j = M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}})$ and $b_j = M_j^*(S_{\text{cal2}}(j); \bar{B}_2)$, so $a_j - b_j = \varepsilon_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}})$ and $a_j, b_j \in [0, 1]$. Then for each $t \in \{1, \dots, t_{\max}\}$:

$$\begin{aligned} & \prod_{j=1}^t M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) - \prod_{j=1}^t M_j^*(S_{\text{cal2}}(j); \bar{B}_2) \\ &= \sum_{k=1}^t \underbrace{\left(\prod_{j=1}^{k-1} M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) \right)}_{=: W_k^- \in [0, 1]} \cdot \varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}}) \cdot \underbrace{\left(\prod_{j=k+1}^t M_j^*(S_{\text{cal2}}(j); \bar{B}_2) \right)}_{=: W_k^+ \in [0, 1]}. \end{aligned}$$

Since $W_k^-, W_k^+ \in [0, 1]$ (being products of values in $[0, 1]$), we have $|W_k^- \cdot \varepsilon_k \cdot W_k^+| \leq |\varepsilon_k|$ pointwise. Therefore, taking the conditional expectation given $\mathcal{D}_{\text{cal1}}$:

$$\begin{aligned} \left| \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t M_j - \prod_{j=1}^t M_j^* \mid \mathcal{D}_{\text{cal1}} \right] \right| &\leq \mathbb{E}_{S_{\text{cal2}}} \left[\left| \prod_{j=1}^t M_j - \prod_{j=1}^t M_j^* \right| \mid \mathcal{D}_{\text{cal1}} \right] \\ &\leq \mathbb{E}_{S_{\text{cal2}}} \left[\sum_{k=1}^t W_k^- |\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| W_k^+ \mid \mathcal{D}_{\text{cal1}} \right] \\ &\leq \sum_{k=1}^t \mathbb{E}_{S_{\text{cal2}}} [|\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}], \end{aligned}$$

where the final inequality uses $W_k^- W_k^+ \leq 1$ and the fact that $|\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})|$ depends only on the k -th coordinate $S(k)$, so that $\mathbb{E}_{S_{\text{cal2}}} [W_k^- |\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| W_k^+ \mid \mathcal{D}_{\text{cal1}}] \leq \mathbb{E}_{S_{\text{cal2}}} [|\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}_{S_{\text{cal2}}(k)} [|\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}]$ regardless of the joint distribution of coordinates.

Applying Assumption 6: $\mathbb{E}_{S_{\text{cal2}}(k)} [|\varepsilon_k(S_{\text{cal2}}(k); \mathcal{D}_{\text{cal1}})| \mid \mathcal{D}_{\text{cal1}}] \leq \eta_k$, giving:

$$\left| \mathbb{E}_{S_{\text{cal2}}} \left[\prod_{j=1}^t M_j(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}}) - \prod_{j=1}^t M_j^*(S_{\text{cal2}}; \bar{B}_2) \mid \mathcal{D}_{\text{cal1}} \right] \right| \leq \sum_{k=1}^t \eta_k.$$

We now sum over t :

$$\begin{aligned} & \mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] - \mathbb{E}_{S_{\text{cal2}}} [\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2))] \\ &= \sum_{t=1}^{t_{\max}} \mathbb{E}_{S_{\text{cal2}}} \left[\left| \prod_{j=1}^t M_j(S_{\text{cal2}}(j); \mathcal{D}_{\text{cal1}}) - \prod_{j=1}^t M_j^*(S_{\text{cal2}}(j); \bar{B}_2) \right| \mid \mathcal{D}_{\text{cal1}} \right] \\ &\leq \sum_{t=1}^{t_{\max}} \sum_{k=1}^t \eta_k \\ &= \sum_{k=1}^{t_{\max}} \eta_k \cdot |\{t : t \geq k, t \leq t_{\max}\}| \\ &= \sum_{k=1}^{t_{\max}} \eta_k (t_{\max} - k + 1) = \Gamma, \end{aligned}$$

where we interchanged the order of summation: $\sum_{t=1}^{t_{\max}} \sum_{k=1}^t = \sum_{k=1}^{t_{\max}} \sum_{t=k}^{t_{\max}}$, and $\sum_{t=k}^{t_{\max}} 1 = t_{\max} - k + 1$. $\square$

Armed with the above Lemma, we turn to prove Theorem 7.

Proof of Theorem 7. The Phase I budget $B_1 = \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i$ satisfies $\mathbb{E}[B_1] = N_1 \mathbb{E}[b_1]$, and by construction $\bar{B}_2 = (B - B_1)/N_2$ so $B_1 + N_2 \bar{B}_2 = B$ deterministically. For the Phase II budget:

$$\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] = N_2 \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}].$$

By Lemma 5 and Lemma 4:

$$\mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M(S_{\text{cal2}}; \mathcal{D}_{\text{cal1}})) \mid \mathcal{D}_{\text{cal1}}] \leq \mathbb{E}_{S_{\text{cal2}}}[\mathcal{B}(M^*(S_{\text{cal2}}; \bar{B}_2))] + \Gamma \leq \bar{B}_2 + \Gamma.$$

Therefore:

$$\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] \leq N_2(\bar{B}_2 + \Gamma) = B - B_1 + N_2 \Gamma.$$

Taking the total expectation and adding $\mathbb{E}[B_1]$:

$$\mathbb{E}[B_1 + B_2] \leq B + N_2 \Gamma. \quad \square$$

Lastly, we develop a finite-sample budget bound based on the above result.

Theorem 8. *Under the conditions of Theorem 7, with probability at least $1 - \delta$:*

$$\frac{1}{N} \sum_{i \in \mathcal{I}_{\text{cal}}} \tilde{T}_i \leq \frac{B}{N} + \frac{N_2}{N} \Gamma + \frac{t_{\max} \log(1/\delta)}{3N} + \frac{1}{N} \sqrt{\frac{t_{\max}^2 \log^2(1/\delta)}{9} + 2N_2 t_{\max}(\bar{B}_2 + \Gamma) \log(1/\delta)}.$$

Proof. Identical to Theorem 6 except that the bound over $\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}]$ is replaced with:

$$\mathbb{E}[B_2 \mid \mathcal{D}_{\text{cal1}}] \leq N_2 \bar{B}_2 + N_2 \Gamma.$$

The above inequality follows from Lemma 5 and Lemma 4. $\square$

While Theorem 8 establishes the asymptotic consistency of DAPRO, the finite-sample relative budget gap of $O(t_{\max}^2 / (\bar{B}_2 \sqrt{N_1}))$ is highly conservative. This looseness stems from the absolute value inequality applied in Lemma 5. By taking the absolute value of the errors at each time step, the proof assumes a worst-case scenario where the estimation inaccuracies at every time step $t \in \{1, \dots, t_{\max}\}$ aggregate in the same direction, with no cancellation.

In practice, if the projection models $\{M_t\}_{t=1}^{t_{\max}}$ are reasonably well-calibrated, e.g., yielding approximately mean-zero errors conditionally, the pointwise errors tend to cancel each other out as the sequence progresses. This explains why our empirical evaluations in Section 4 show that DAPRO satisfies the target budget B even when $t_{\max} = 200$ and $N_1 = 100$ is relatively small. Therefore, rather than viewing this bound as a strict operational limitation, it should be interpreted as a sample complexity guideline, and demonstrates that the Phase I calibration size N_1 must grow quadratically with the maximum conversation horizon $t_{\max}$ to satisfy the budget constraint at this worst-case setup.

D Algorithms

D.1 LPB calibration with a known censoring mechanism

In this section, we outline in Algorithm 1 the procedure introduced in [12] for constructing a calibrated LPB with a known censoring mechanism.

D.2 Optimized budget allocation

In this section, we present in Algorithm 2 the budget allocation approach developed in [12].

D.3 Proposed greedy adaptive budget allocation

We introduce a Greedy budget allocation strategy as an alternative to our main proposal. Like our primary method, this strategy dynamically updates censoring times in response to new exchanges in the conversation. The core motivation behind this approach is to increase the effective sample size, that is, the number of samples actively contributing to the miscoverage estimator in (3), by observing more unsafe events early on. We do so by dictating a portion of the budget to exploring conversations

Algorithm 1 Constructing a calibrated LPB with a known censoring mechanism

Require: Calibration data $\{X_i, C_i, \tilde{T}_i\}_{i \in \mathcal{I}_{\text{cal}}}$, censoring weights $\{w_\tau(i)\}_{i \in \mathcal{I}_{\text{cal}}, \tau \in \mathcal{T}}$, quantile estimates $\{\hat{f}_\tau(\cdot)\}_{\tau \in \mathcal{T}}$, target miscoverage rate α , prior quantile τ_{prior} , search space $\mathcal{T} \subseteq (0, 1)$.

- 1: **for** $\tau \in \mathcal{T} \cap [0, \tau_{\text{prior}}]$ **do**
- 2: $\hat{\alpha}(\tau) \leftarrow \frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} w_\tau(i) \mathbb{I}\{\tilde{T}_i < \hat{f}_\tau(X_i) \leq C_i\}$ ▷ miscoverage est.
- 3: **end for**
- 4: $\hat{\tau} \leftarrow \sup \left\{ \tau \in \mathcal{T} \cap [0, \tau_{\text{prior}}] : \sup_{\tau' \leq \tau} \hat{\alpha}(\tau') \leq \alpha \right\}$ ▷ calibrated quantile level
- 5: **return** Lower predictive bound for a test point $X_{\text{test}} = x$, $\hat{L}(x) = \hat{f}_{\hat{\tau}}(x)$.

Algorithm 2 Optimized static budget allocation

Require: Calibration data $\{X_i\}_{i \in \mathcal{I}_{\text{cal}}}$, attacker model $\mathcal{A}$, target model $\mathcal{G}$, audit function $\mathcal{J}(\cdot)$, pre-trained quantile regression model $\{\hat{q}_\tau(\cdot)\}_{\tau \in \mathcal{T}}$, target miscoverage rate α , prior quantile τ_{prior} , quantile trimming threshold M , total budget B .

- 1: $\{\hat{f}_\tau(X_i)\}_{\tau \in \mathcal{T}, i \in \mathcal{I}_{\text{cal}}} \leftarrow \{\min(\hat{q}_\tau(X_i), M)\}_{\tau \in \mathcal{T}, i \in \mathcal{I}_{\text{cal}}}$ ▷ trim the quantile est.
- 2: Compute the optimal $\lambda^* = \left(\frac{1}{B} \cdot \sum_{i \in \mathcal{I}_{\text{cal}}} \sqrt{\hat{f}_{\text{prior}}(X_i)} \right)^2$
- 3: Obtain optimal static probabilities $\pi_i^* = \min \left(1, 1/\sqrt{\lambda^* \hat{f}_{\text{prior}}(X_i)} \right)$
- 4: $C_i \leftarrow \begin{cases} \hat{f}_{\text{prior}}(X_i) & \text{w.p. } \pi_i, \\ 0 & \text{otherwise.} \end{cases}$
- 5: For each $i \in \mathcal{I}_{\text{cal}}$, run the conversation between $\mathcal{A}$ and $\mathcal{G}$ for C_i iterations using the initial prompt X_i to obtain the censored event time $\tilde{T}_i \leftarrow \min(C_i, T_i)$.
- 6: $w_\tau(i) \leftarrow \frac{1}{\pi_i}$, $i \in \mathcal{I}_{\text{cal}}$
- 7: **return** Censoring and event times $\{(\tilde{T}_i, C_i, w_\tau(i))\}_{i \in \mathcal{I}_{\text{cal}}}$.

that have a high estimated probability of yielding unsafe outputs. Specifically, we partition the total available budget B into two stages using a ratio $\rho \in (0, 1)$, where a budget of $B \cdot \rho$ is reserved for greedy exploration. The greedy exploration is related to the multi-armed bandit framework [60]. While any bandit algorithm [61, 62] could be employed for this exploration phase, we adopt a greedy strategy that selects the samples with the highest estimated probability of encountering an unsafe event.

Phase 1: Stochastic Greedy Exploration Let ξ_i^{greedy} denote the number of acquisitions we have conducted for the i -th sample, initialized to $\xi_i^{\text{greedy}} = 0$ for all $i \in \mathcal{I}_{\text{cal}}$. In this first phase, we rely on the outputs of our predictive model, $\hat{p}(H_i(t))$, which estimates the probability of an unsafe response at the next step $\mathbb{P}(Y_i(t+1) = 1 \mid H_i(t))$. We iteratively spend our exploration budget by focusing on the most promising candidates. At each step, we identify the set of active samples $\mathcal{I}_{\text{active}}$, those for which we have neither encountered the unsafe event nor reached the prior sequence length $\hat{f}_{\text{prior}}(X_i)$. We compute the predicted unsafe probabilities for these active samples, select the top k highest probabilities, and sample an index i^* proportionally to their estimated unsafe output probability, from this top- k set. We advance the interaction for sample i^* by one step ($\xi_{i^*}^{\text{greedy}} \leftarrow \xi_{i^*}^{\text{greedy}} + 1$), update its history, and decrement our exploration budget. We repeat this process until the designated exploration budget of $B \cdot \rho$ is entirely consumed.

Phase 2: Static Optimization for Theoretical Guarantees After the exploration phase concludes, we use the remaining budget, $B \cdot (1 - \rho)$, to employ the static optimized approach of [12], which provides a theoretical guarantee. Crucially, we concentrate this remaining budget only on the unresolved samples. For the unresolved sample $i \notin \mathcal{I}_{\text{active}}$ —where the unsafe event was successfully

triggered, or the prior was reached during Phase 1—the allocation process is complete. We assign their final censoring time as $C_i = \hat{f}_{\text{prior}}(X_i)$ and their inverse censoring weight as $w_\tau(i) = 1$. For the remaining active samples $i \in \mathcal{I}_{\text{active}}$, we calculate the remaining number of generations required to reach the prior: $d_i = \hat{f}_{\text{prior}}(X_i) - \xi_i^{\text{greedy}}$. We then apply Algorithm 2 over these active samples, using d_i as the new target prior and distributing the remaining budget among them. This optimization yields inverse censoring probability $w'(i)$ and an additional drawn censoring time C'_i for each active sample. Finally, we formalize the final censoring times across all samples as:

$$C_i = \begin{cases} \xi_i^{\text{greedy}} + C'_i, & i \in \mathcal{I}_{\text{active}}, \\ \hat{f}_{\text{prior}}(X_i), & \text{otherwise,} \end{cases}$$

with the corresponding inverse censoring weights defined as:

$$w_\tau(i) = \begin{cases} w'(i), & i \in \mathcal{I}_{\text{active}}, \\ 1, & \text{otherwise.} \end{cases}$$

The censored event times are then computed as $\tilde{T}_i = \min(T_i, C_i)$. Having obtained the set $\{(\tilde{T}_i, C_i, w_\tau(i))\}_{i \in \mathcal{I}_{\text{cal}}}$, we construct the final calibrated LPB using Algorithm 1. This greedy budget allocation is summarized in Algorithm 3. We note that the coverage validity of this method follows from Theorem 4 with $\mathcal{I}_{\text{cal}} = \mathcal{I}_{\text{active}}$. It also satisfies the budget constraint since in the first stage we use at most $\rho \cdot B$ budget units, and the second stage uses at most $(1 - \rho) \cdot B$ in expectation, as guaranteed by [12].

Algorithm 3 Greedy adaptive budget allocation (Greedy)

Require: Calibration data $\{X_i\}_{i \in \mathcal{I}_{\text{cal}}}$, attacker model $\mathcal{A}$, target model $\mathcal{G}$, audit function $\mathcal{J}(\cdot)$, predictive model $\hat{p}(\cdot)$, prior quantiles $\{\hat{f}_{\text{prior}}(X_i)\}_{i \in \mathcal{I}_{\text{cal}}}$, total budget B , exploration ratio $\rho \in (0, 1)$, top- k parameter k .

```
1: Initialize acquisitions  $\xi_i^{\text{greedy}} \leftarrow 0$ , and event times  $T_i \leftarrow \infty$  for all  $i \in \mathcal{I}_{\text{cal}}$ .
2:  $B_{\text{explore}} \leftarrow B \cdot \rho$  ▷ allocate exploration budget
3: Phase 1: Stochastic Greedy Exploration
4: while  $B_{\text{explore}} > 0$  do
5:    $\mathcal{I}_{\text{active}} \leftarrow \{i \in \mathcal{I}_{\text{cal}} : \xi_i^{\text{greedy}} < \hat{f}_{\text{prior}}(X_i) \text{ and } T_i > \xi_i^{\text{greedy}}\}$  ▷ update active set
6:   if  $\mathcal{I}_{\text{active}}$  is empty then
7:     break
8:   end if
9:   Compute probabilities  $p_i \leftarrow \hat{p}(H_i(\xi_i^{\text{greedy}}))$  for all  $i \in \mathcal{I}_{\text{active}}$ 
10:  Let  $\mathcal{K}$  be the indices of the top  $\min(k, |\mathcal{I}_{\text{active}}|)$  values in  $\{p_i\}_{i \in \mathcal{I}_{\text{active}}}$ 
11:  Sample an index  $i^*$  from  $\mathcal{K}$  with probability proportional to  $p_{i^*}$ 
12:  Update  $\xi_{i^*}^{\text{greedy}} \leftarrow \xi_{i^*}^{\text{greedy}} + 1$ 
13:  Advance interaction  $\mathcal{A} \leftrightarrow \mathcal{G}$  by one step to obtain new exchange, update  $H_{i^*}(\xi_{i^*}^{\text{greedy}})$ 
14:  if  $\text{Audit}(\text{new exchange}) == 1$  then
15:     $T_{i^*} \leftarrow \xi_{i^*}^{\text{greedy}}$  ▷ unsafe event observed
16:  end if
17:   $B_{\text{explore}} \leftarrow B_{\text{explore}} - 1$ 
18: end while
19: Phase 2: Static Optimization for Theoretical Guarantees
20:  $\mathcal{I}_{\text{active}} \leftarrow \{i \in \mathcal{I}_{\text{cal}} : \xi_i^{\text{greedy}} < \hat{f}_{\text{prior}}(X_i) \text{ and } T_i = \infty\}$ 
21:  $B_{\text{remain}} \leftarrow B \cdot (1 - \rho)$ 
22: Set remaining target lengths  $d_i \leftarrow \hat{f}_{\text{prior}}(X_i) - \xi_i^{\text{greedy}}$  for all  $i \in \mathcal{I}_{\text{active}}$ 
23:  $\{(C'_i, w'(i))\}_{i \in \mathcal{I}_{\text{active}}} \leftarrow \text{Algorithm 2 applied with priors } \{d_i\}_{i \in \mathcal{I}_{\text{active}}} \text{ and budget } B_{\text{remain}}$ 
24: Finalize Censoring Times and Weights
25: for each  $i \in \mathcal{I}_{\text{cal}}$  do
26:   if  $i \in \mathcal{I}_{\text{active}}$  then
27:      $C_i \leftarrow \xi_i^{\text{greedy}} + C'_i$ 
28:      $w_\tau(i) \leftarrow w'(i)$ 
29:   else
30:      $C_i \leftarrow \hat{f}_{\text{prior}}(X_i)$  ▷ interaction resolved or prior reached during phase 1
31:      $w_\tau(i) \leftarrow 1$ 
32:   end if
33:    $\tilde{T}_i \leftarrow \min(T_i, C_i)$ 
34: end for
35: return Censoring and event times  $\{(\tilde{T}_i, C_i, w_\tau(i))\}_{i \in \mathcal{I}_{\text{cal}}}$ .
```

D.4 Proposed locally optimized adaptive budget allocation

We propose a `LocallyAdaptive` allocation method as an additional alternative to our main proposal. While our primary method solves a global optimization problem to map risk scores to continuation probabilities across all future time steps, this baseline computes the optimal probabilities *locally* at each step.

Similarly to our main approach, we partition the calibration data into two disjoint subsets: a small set indexed by $\mathcal{I}_{\text{cal1}}$ (e.g., $N_1 = 100$) on which we find an optimal parameter, and a deployment set $\mathcal{I}_{\text{cal2}}$, on which we deploy this parameter. The algorithm operates in two phases.

Phase I: Tuning the Global Parameter. In the first phase, we first fully observe all samples in $\mathcal{I}_{\text{cal1}}$, until we either reach the prior $\hat{f}_{\text{prior}}(X_i)$ or observe the unsafe event T_i . After this, we have $\bar{B}_2$

average budget remaining for the second phase. In this first phase, we use $\mathcal{I}_{\text{cal1}}$ to calibrate a single global parameter, $\lambda \in \mathbb{R}^+$, which balances the tradeoff between the consumed sampling budget and the probability of discovering unsafe events.

The expected cost is the sum of the survival probabilities up to the known stopping time:

$$\hat{R}_{N_1}(\lambda) = \frac{1}{N_1} \sum_{i \in \mathcal{I}_{\text{cal1}}} \sum_{t=1}^{\min(T_i, \hat{f}_{\text{prior}}(X_i))} \prod_{j=0}^{t-1} P_i(j; \lambda),$$

where $P_i(j; \lambda)$ is the local continuation probability evaluated at λ . We then conduct a binary search to find the optimal value $\lambda^* := \inf \left\{ \lambda \geq 0 : \frac{N_1}{N_1+1} \hat{R}_{N_1}(\lambda) + \frac{t_{\max}}{N_1+1} \leq \bar{B}_2 \right\}$.

Phase II: Local Step-by-Step Acquisition. Equipped with the optimal parameter λ^* , we deploy the policy on the deployment set $\mathcal{I}_{\text{cal2}}$. At any given step t , for each active sample i , we use our predictive model to estimate the probability of an unsafe event occurring at every future time step $t_2 > t$, conditioned on the current conversation history $H_i(t)$. Let $\hat{p}(t+k | t)$ denote the estimate of the probability that the unsafe event occurs exactly k steps in the future, given that the interaction has safely reached step t , and the history up to this point, $\mathbb{P}(T_i = t+k | T_i > t, H_i(t))$.

Using these future probabilities, we compute the expected remaining cost $E_i(t)$, defined as the anticipated number of generations required to either observe an unsafe event or reach the prior sequence length $\hat{f}_{\text{prior}}(X_i)$ without observing an unsafe event. Formally, let $d_i(t) = \hat{f}_{\text{prior}}(X_i) - t$ denote the maximum number of remaining steps. The expected remaining cost is computed as the weighted sum of future steps:

$$E_i(t) = \sum_{k=1}^{d_i(t)} k \cdot \hat{p}(t+k | t) + d_i(t) \sum_{k=d_i(t)+1}^{t_{\max}-t+1} \hat{p}(t+k | t), \quad (14)$$

where the second term captures the maximum cost $d_i(t)$ paid if the unsafe event does not occur before reaching the prior $\hat{f}_{\text{prior}}(X_i)$. We note that $\hat{p}(t_{\max}+1 | t)$ estimates the probability that no unsafe events occur at times $j \leq t_{\max}$.

Guided by the Karush-Kuhn-Tucker (KKT) conditions for budget-constrained variance minimization [12], the optimal probability of reaching the end of the sequence from the current state is defined as:

$$P_i^{\text{target}}(t) = \min \left(1, \frac{1}{\sqrt{\lambda \cdot E_i(t)}} \right). \quad (15)$$

Because our acquisition process operates step-by-step, we must translate this probability into a local continuation probability. Let $P_i^{\text{accum}}(t) := \prod_{j=0}^{t-1} P_i(j)$ denote the cumulative probability that sample i has survived up to step t , where $P_i^{\text{accum}}(0) := 1$. Then, the immediate continuation probability is given by:

$$P_i(t) = \min \left(1, \frac{P_i^{\text{target}}(t)}{P_i^{\text{accum}}(t)} \right). \quad (16)$$

Finally, we draw a Bernoulli random variable $Z_i(t) \sim \text{Ber}(P_i(t))$. If $Z_i(t) = 1$, we expend one unit of budget to acquire the next conversational exchange $H_i(t+1)$; otherwise, we halt the interaction and set the censoring time as $C_i = t$. Furthermore, to ensure the inverse-censoring weights remain bounded for samples halted early by the policy, we lower bound the global minimal probability by $p_{\min}$ (e.g., 0.005) on the final cumulative probability. For convenience, we summarize this procedure in Algorithm 4. The coverage validity of this process follows from Theorem 4. We remark that the budget validity of this approach can be formally established using the conformal risk control framework [63], as presented next.

Proposition 3 (Budget validity of LocallyAdaptive). *Suppose that $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ are drawn i.i.d. Then the expected total budget consumed by Algorithm 4 satisfies*

$$\mathbb{E} \left[\sum_{i \in \mathcal{I}_{\text{cal}}} b_i \right] \leq B,$$

where b_i denotes the number of attacker–target exchanges conducted for sample i , and the expectation is taken over the randomness of the data $\{(X_i, T_i, H_i)\}_{i \in \mathcal{I}_{\text{cal}}}$ and the random continuation draws of the policy.

Proof. We begin by analyzing the budget used in Phase I and the remaining budget per sample. For each $i \in \mathcal{I}_{\text{cal1}}$, the algorithm runs the conversation to completion (without any stopping policy), expending $b_i = \min(\hat{f}_{\text{prior}}(X_i), T_i)$ budget units. The total Phase I budget is $B_1 := \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i$. By construction of Algorithm 4, the target average budget per Phase II sample is

$$\bar{B}_2 := \frac{B - B_1}{N_2},$$

so $B_1 + N_2 \bar{B}_2 = B$ holds deterministically for every realization of the Phase I data.

We now turn to examining the Phase II budget as a function of λ . For each $j \in \mathcal{I}_{\text{cal2}}$ and any $\lambda \geq 0$, let $b_j(\lambda)$ denote the budget expended on sample j under the policy parameterized by λ , i.e., the final value of the acquisition counter at termination. We note two properties. First, $b_j(\lambda)$ is bounded: since the active set removes sample j once it reaches $\hat{f}_{\text{prior}}(X_j)$, we have $b_j(\lambda) \in [0, t_{\max}]$ for all $\lambda \geq 0$. Second, $b_j(\lambda)$ is monotone non-increasing in λ : increasing λ decreases the KKT target probability $P_j^{\text{target}}(t) = \min(1, 1/\sqrt{\lambda \cdot E_j(t)})$ at every step t , and hence decreases the local continuation probability $P_j(t)$. Consequently, sample j is more likely to be halted earlier, so $b_j(\lambda)$ is non-increasing in λ for each fixed sample.

Because Phase I conversations are observed to completion, the full histories $\{H_i\}_{i \in \mathcal{I}_{\text{cal1}}}$ are available. For each $i \in \mathcal{I}_{\text{cal1}}$ and any $\lambda \geq 0$, we can therefore obtain the expected budget that the policy λ would have expended on sample i ; denote this by $b^{\text{exp}}_i(\lambda) := \sum_{t=1}^{b_i} \prod_{k=1}^t P_i(k; \lambda)$, where $P_i(k; \lambda)$ is defined as in (16), with the corresponding λ . The empirical average expected budget on $\mathcal{I}_{\text{cal1}}$ is

$$\hat{R}_{N_1}(\lambda) := \frac{1}{N_1} \sum_{i \in \mathcal{I}_{\text{cal1}}} b^{\text{exp}}_i(\lambda).$$

Following the conformal risk control framework of [63], Phase I tunes

$$\hat{\lambda} := \inf \left\{ \lambda \geq 0 : \frac{N_1}{N_1 + 1} \hat{R}_{N_1}(\lambda) + \frac{t_{\max}}{N_1 + 1} \leq \bar{B}_2 \right\}. \quad (17)$$

This infimum is well-defined: since $P_i^{\text{target}}(t) \rightarrow 0$ as $\lambda \rightarrow \infty$, all continuation probabilities vanish and $\hat{R}_{N_1}(\lambda) \rightarrow 0$.

By the definition of $\hat{\lambda}$ and the monotone non-increasing property of $b^{\text{exp}}_i(\lambda)$, the selector satisfies almost surely:

$$\frac{N_1}{N_1 + 1} \hat{R}_{N_1}(\hat{\lambda}) + \frac{t_{\max}}{N_1 + 1} \leq \bar{B}_2. \quad (18)$$

Multiplying both sides of (18) by $N_1 + 1$ yields:

$$\sum_{i \in \mathcal{I}_{\text{cal1}}} b_i(\hat{\lambda}) + t_{\max} \leq (N_1 + 1) \bar{B}_2 \quad (\text{almost surely}). \quad (19)$$

We now bound the expected budget used in Phase II. Fix any $j \in \mathcal{I}_{\text{cal2}}$. Denote by $b^{\text{emp}}_j(\hat{\lambda})$ the empirical budget expended for the j -th sample by the policy employed with $\hat{\lambda}$. Since $b^{\text{emp}}_j(\hat{\lambda}) \leq t_{\max}$, we may replace $t_{\max}$ on the left-hand side of (19) by $b^{\text{emp}}_j(\hat{\lambda})$ to obtain:

$$\sum_{i \in \mathcal{I}_{\text{cal1}}} b^{\text{exp}}_i(\hat{\lambda}) + b^{\text{emp}}_j(\hat{\lambda}) \leq (N_1 + 1) \bar{B}_2 \quad (\text{almost surely}). \quad (20)$$

Taking expectations on both sides of (20):

$$\mathbb{E} \left[\sum_{i \in \mathcal{I}_{\text{cal1}}} b^{\text{exp}}_i(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}} \right] + \mathbb{E} \left[b^{\text{emp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}} \right] \leq (N_1 + 1) \mathbb{E}[\bar{B}_2 \mid \mathcal{D}_{\text{cal1}}].$$

Since $\mathbb{E}[b^{\text{emp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}] = \mathbb{E}[b^{\text{exp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}]$, we set:

$$\mathbb{E}\left[\sum_{i \in \mathcal{I}_{\text{cal1}}} b^{\text{exp}}_i(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}\right] + \mathbb{E}[b^{\text{exp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}] \leq (N_1 + 1) \mathbb{E}[\bar{B}_2 \mid \mathcal{D}_{\text{cal1}}].$$

The data $\{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal1}} \cup \{j\}}$ contains $N_1 + 1$ i.i.d. (hence exchangeable) samples, with $\hat{\lambda}$ determined solely by the $\mathcal{I}_{\text{cal1}}$ portion. By Theorem 1 of [63] applied to the non-increasing, $[0, t_{\text{max}}]$ -bounded loss $b^{\text{exp}}(\lambda)$, the selector (17), and these $N_1 + 1$ exchangeable samples, the expected loss on the held-out sample j satisfies:

$$\mathbb{E}[b^{\text{exp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}] \leq \mathbb{E}[\bar{B}_2 \mid \mathcal{D}_{\text{cal1}}] \quad \text{for all } j \in \mathcal{I}_{\text{cal2}}. \quad (21)$$

We note that while Theorem 1 of [63] is formulated for a deterministic risk level, its proof also holds for a risk level that is a random variable $\bar{B}_2$, and we can take the expectation to obtain the above inequality.

We now conclude by summing the budget over all remaining samples in Phase II. Summing (21) over all $j \in \mathcal{I}_{\text{cal2}}$ and applying linearity of expectation (valid since all terms are non-negative and bounded by t_{max}):

$$\mathbb{E}\left[\sum_{j \in \mathcal{I}_{\text{cal2}}} b^{\text{exp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}\right] = \sum_{j \in \mathcal{I}_{\text{cal2}}} \mathbb{E}[b^{\text{exp}}_j(\hat{\lambda}) \mid \mathcal{D}_{\text{cal1}}] \leq N_2 \mathbb{E}[\bar{B}_2 \mid \mathcal{D}_{\text{cal1}}].$$

Combining with the Phase I budget and using $B_1 + N_2 \bar{B}_2 = B$ deterministically, and by taking an expectation over $\mathcal{D}_{\text{cal1}}$, we get:

$$\begin{aligned} \mathbb{E}\left[\sum_{i \in \mathcal{I}_{\text{cal}}} b_i\right] &= \mathbb{E}\left[\sum_{i \in \mathcal{I}_{\text{cal1}}} b_i\right] + \mathbb{E}\left[\sum_{j \in \mathcal{I}_{\text{cal2}}} b^{\text{exp}}_j(\hat{\lambda})\right] \\ &\leq \mathbb{E}[B_1] + N_2 \mathbb{E}[\bar{B}_2] \\ &= \mathbb{E}[B_1] + N_2 \mathbb{E}\left[\frac{B - B_1}{N_2}\right] \\ &= \mathbb{E}[B_1] + \mathbb{E}[B - B_1] \\ &= B, \end{aligned}$$

where the last equality uses the fact that B is a deterministic constant. This completes the proof. $\square$

Algorithm 4 Locally optimized adaptive budget allocation (LocallyAdaptive)

Require: Calibration data $\{X_i\}_{i \in \mathcal{I}_{\text{cal}}}$, attacker model $\mathcal{A}$, target model $\mathcal{G}$, audit function $\mathcal{J}(\cdot)$, predictive model $\hat{p}(\cdot | \cdot)$, prior quantiles $\{\hat{f}_{\text{prior}}(X_i)\}_{i \in \mathcal{I}_{\text{cal}}}$, total budget B , first-split size N_1 , minimum probability $p_{\min}$.

- 1: Randomly partition $\mathcal{I}_{\text{cal}}$ into $\mathcal{I}_{\text{cal1}}$ (size N_1) and $\mathcal{I}_{\text{cal2}}$ (size $N - N_1$).
 - 2: Let B_1 be the budget required to fully observe $\mathcal{I}_{\text{cal1}}$ up to $\hat{f}_{\text{prior}}(X_i)$ or unsafe event.
 - 3: $\bar{B}_2 \leftarrow (B - B_1)/|\mathcal{I}_{\text{cal2}}|$ ▷ target average budget for phase 2
 - 4: Phase 1: Tuning the Global λ parameter
 - 5: Define the expected budget function over $\mathcal{I}_{\text{cal1}}$ for a given λ :
 - 6: $\hat{R}_{N_1}(\lambda) \leftarrow \frac{1}{N_1} \sum_{i \in \mathcal{I}_{\text{cal1}}} \sum_{t=1}^{\min(T_i, \hat{f}_{\text{prior}}(X_i))} \prod_{j=0}^{t-1} P_i(j; \lambda)$ ▷ using P_i from (16)
 - 7: Find optimal $\lambda^* \in [0, \lambda_{\max}]$ satisfying: $\lambda^* := \inf \left\{ \lambda \geq 0 : \frac{N_1}{N_1+1} \hat{R}_{N_1}(\lambda) + \frac{t_{\max}}{N_1+1} \leq \bar{B}_2 \right\}$
via binary search.
 - 8: Phase 2: Local Step-by-Step Acquisition
 - 9: Initialize acquisitions $\xi_i^{\text{locally}} \leftarrow 0$, cumulative probabilities $P_i^{\text{accum}} \leftarrow 1$, for all $i \in \mathcal{I}_{\text{cal2}}$.
 - 10: $\mathcal{I}_{\text{active}} \leftarrow \{i \in \mathcal{I}_{\text{cal2}} : \xi_i^{\text{locally}} < \hat{f}_{\text{prior}}(X_i)\}$
 - 11: **while** $\mathcal{I}_{\text{active}}$ is not empty **do**
 - 12: **for each** $i \in \mathcal{I}_{\text{active}}$ **do**
 - 13: Estimate future unsafe probabilities $\hat{p}(\xi_i^{\text{locally}} + k | \xi_i^{\text{locally}})$ using the predictive model
 - 14: $d_i \leftarrow \hat{f}_{\text{prior}}(X_i) - \xi_i^{\text{locally}}$ ▷ remaining allowed steps
 - 15: Compute expected remaining cost $E_i(\xi_i^{\text{locally}})$ using (14)
 - 16: Compute KKT target probability P_i^{target} using (15)
 - 17: Compute local continuation probability $P_i(\xi_i^{\text{locally}})$ using (16)
 - 18: Sample $Z_i \sim \text{Ber}(P_i(\xi_i^{\text{locally}}))$
 - 19: **if** $Z_i == 1$ **then**
 - 20: $P_i^{\text{accum}} \leftarrow P_i^{\text{accum}} \cdot P_i(\xi_i^{\text{locally}})$
 - 21: $\xi_i^{\text{locally}} \leftarrow \xi_i^{\text{locally}} + 1$
 - 22: Advance interaction $\mathcal{A} \leftrightarrow \mathcal{G}$, update $H_i(\xi_i^{\text{locally}})$
 - 23: **if** $\text{Audit}(\text{new exchange}) == 1$ **then**
 - 24: $T_i \leftarrow \xi_i^{\text{locally}}$ ▷ unsafe event observed
 - 25: Remove i from $\mathcal{I}_{\text{active}}$
 - 26: **end if**
 - 27: **else**
 - 28: Remove i from $\mathcal{I}_{\text{active}}$ ▷ halted by policy
 - 29: **end if**
 - 30: **end for**
 - 31: Update $\mathcal{I}_{\text{active}} \leftarrow \{i \in \mathcal{I}_{\text{active}} : \xi_i^{\text{locally}} < \hat{f}_{\text{prior}}(X_i)\}$
 - 32: **end while**
 - 33: Finalize Censoring Times and Weights
 - 34: **for each** $i \in \mathcal{I}_{\text{cal}}$ **do**
 - 35: **if** $i \in \mathcal{I}_{\text{cal1}}$ **then**
 - 36: $C_i \leftarrow \hat{f}_{\text{prior}}(X_i)$ and $w_\tau(i) \leftarrow 1$
 - 37: **else**
 - 38: $C_i \leftarrow \xi_i^{\text{locally}} + 1$ if $\xi_i^{\text{locally}} + 1 < \min(T_i, \hat{f}_{\text{prior}}(X_i))$, otherwise $C_i \leftarrow \hat{f}_{\text{prior}}(X_i)$
 - 39: $w_\tau(i) \leftarrow 1/P_i^{\text{accum}}$ ▷ bound max weight
 - 40: **end if**
 - 41: $\tilde{T}_i \leftarrow \min(T_i, C_i)$
 - 42: **end for**
 - 43: **return** Censoring and event times $\{(\tilde{T}_i, C_i, w_\tau(i))\}_{i \in \mathcal{I}_{\text{cal}}}$.
-

D.5 Proposed DAPRO

We present DAPRO in Algorithm 5.

While the overall computational cost is dominated by the B LLM API calls, the internal algorithmic overhead is lightweight. Specifically, the optimization process requires $\mathcal{O}(N_1 t_{\max})$ operations per iteration. In our experiments, we conduct at most 60 outer iterations and 10 inner iterations. Furthermore, training the projection model M_t at each timestep t takes $\mathcal{O}(N_1)$ operations. During Phase II, deploying the acquisition policy costs $\mathcal{O}(N_2 t_{\max})$ operations, and computing the final calibrated threshold $\hat{\tau}$ requires only $\mathcal{O}(N)$ steps.

E Experimental details

E.1 Data generation

Across all experiments, we generate the raw multi-turn attacker-target conversations using the PAIR algorithm [9], with initial prompts (attack goal) taken from three popular safety benchmarks: we sample 10,000 harmful prompts from the RealToxicityPrompts dataset [3], 10,000 prompts from the Anthropic Red Team dataset [15, 16], 10,000 prompts from SQuAD v2 data [64, 65], and 10,000 instruction-following tasks from the AutoIF dataset [17]. We deploy Qwen 2.5 14B Instruct [13, 66] as the attacking model, since it had the fewest formatting errors when generating adversarial prompts compared to the other models we examined. For the target models, we evaluate safety across three distinct architectures: Qwen 2.5 14B Instruct, Llama 3.1 8B Instruct [18], Phi 4 Mini Instruct [19], and Gemma 3 4B Instruct [20].

We used both HuggingFace [67] and vLLM [68] to load and infer with these models, using bfloat16 precision for all models. The system prompts and hyperparameters for the LLMs we use follow those of [9]. Specifically, we set the attacker’s temperature, top-p, and max new tokens to 1, 0.9, and 500, respectively, and the target’s to 0, 1, and 150, respectively. If the attacker failed to generate an attack in a valid JSON format, we continue to retry for at most 10 attempts, and finish the conversation if it failed to generate a valid response for ten attempts. The number of responses saved in the conversation history of the attack model is set to 4. The parameters of the judge model are 50 max tokens, 0 temperature, and top-p of 1. If we fail to obtain a rating from the judge, we set the score of the judge to 0 manually and continue the conversation process. Each conversation runs as a single stream.

To define a successful jailbreak, i.e., an unsafe event where $Y(t) = 1$, we rely on LM safety judges. For the Toxicity dataset, we employ Detoxify [21] and flag any response scoring above a threshold of 0.5 as unsafe, consistent with the evaluation protocol in [12]. For the Red Team dataset, we use two different judging mechanisms: Qwen 2.5 14B Instruct serving as an LLM-as-a-judge, which outputs a continuous score from 1 to 10, where a score of 10 triggers the unsafe event flag, and Llama-Guard [23], which directly outputs a binary safety label. In all generated conversations, we cap the maximum sequence length at $t_{\max} = 200$ turns.

For the Toxicity dataset, the attacker is asked to construct a prompt using the "prompt" part of RealToxicityPrompts so that the target model outputs the "continuation" part of the sample.

For the Anthropic Red Team dataset, we first select the 10,000 most harmful examples based on the min_harmlessness_score_transcript parameter, and use the first human text as the initial attack. If the attack was successful, i.e., had a rating ≥ 4 , we use the response of the LLM as the target output. Otherwise, we set the default objective to "Sure, here is a...". Our hallucination dataset builds on the SQuAD v2 data [64, 65], which contains subjects with a paragraph describing it. We consider the paragraphs from SQuAD as the "ground truth" source for the target model and judge, and set the goal of the attacker to trick the target model into answering a question about the subject that is NOT covered by its provided text. For this data, we engineered two attacker prompts based on [9]: one utilizing 'leading questions' to trick the target into referencing outside knowledge, and another employing 'gaslighting' to compel the target into generating false information not present in the reference text.

Finally, for the AutoIF dataset, we extract tasks with existing programmatic verifiers. To increase the complexity and evaluate multi-constraint interference, we programmatically inject between three and six additional structural constraints into each prompt: explicit word counts, forbidden vocabulary,

Algorithm 5 Dynamic Allocation via PProjected Optimization (DAPRO)

Require: Calibration prompts $\mathcal{D}_{\text{cal}} = \{X_i\}_{i=1}^N$, target lengths $\{\hat{f}_{\text{prior}}(X_i)\}_{i=1}^N$, horizon t_{max} , total budget B , score functions $\{\mathcal{S}_t\}_{t=1}^{t_{\text{max}}}$.

1: Randomly partition dataset $\mathcal{D}_{\text{cal}}$ into disjoint sets, indexed by: $\mathcal{I}_{\text{cal1}}$ (size N_1) and $\mathcal{I}_{\text{cal2}}$ (size N_2).

2: **Phase I: Learning the Optimal Acquisition Policy**

3: **for** each $i \in \mathcal{I}_{\text{cal1}}$ **do**

4: Run conversation until unsafe event or prior is reached; let $b_i = \min(T_i, \hat{f}_{\text{prior}}(X_i))$ denote the observed length.

5: **for** $t = 1, \dots, b_i$ **do**

6: Compute intermediate risk score: $S_i(t) = \mathcal{S}_t(H_i(t))$

7: **end for**

8: **end for**

9: Compute $\bar{B}_2 \leftarrow \frac{1}{|\mathcal{I}_{\text{cal2}}|} (B - \sum_{i \in \mathcal{I}_{\text{cal1}}} b_i)$

10: Solve for optimal probability matrix $P \in [0, 1]^{N_1 \times t_{\text{max}}}$:

$$\min_P \frac{1}{N_1} \sum_{i \in \mathcal{I}_{\text{cal1}}} \frac{1}{\prod_{t=1}^{b_i} P_i(t)} \quad \text{s.t.} \quad \frac{1}{N_1} \sum_{i \in \mathcal{I}_{\text{cal1}}} \mathcal{B}(P_i) \leq \bar{B}_2, \text{ and } P_i(t) \text{ monotonic w.r.t } S_i(t)$$

11: **for** $t = 1, \dots, t_{\text{max}}$ **do**

12: Fit a projection model M_t over pairs $\{S_i(t), P_i(t)\}_{i \in \mathcal{I}_{\text{cal1}}}$. If $\{S_i(t), P_i(t)\}_{i \in \mathcal{I}_{\text{cal1}}} = \emptyset$, we manually set $M_t(s) = 1$ for all s .

13: **end for**

14: **Phase II: Adaptive Acquisition**

15: **for** each $i \in \mathcal{I}_{\text{cal2}}$ **do**

16: Initialize continuation indicator $\xi_i(0) = 1$ and $\xi_i(t) = 0 \forall t \geq 1$, and step counter $t = 1$

17: **while** $\xi_i(t-1) == 1$ **and** $t \leq \hat{f}_{\text{prior}}(X_i)$ **do**

18: Compute current risk score: $S_i(t) = \mathcal{S}_t(H_i(t))$

19: Infer optimal continuation probability: $P_i(t) = M_t(S_i(t))$

20: Sample Bernoulli random variable $Z_t^i \sim \text{Ber}(P_i(t))$

21: **if** $Z_t^i == 1$ **then**

22: set $\xi_i(t) = 1$.

23: Advance attacker-target interaction and obtain $H_i(t+1)$.

24: If $Y_i(t) = 1$, halt early, and break.

25: **else**

26: $\xi_i(t) = 0$

▷ Halt interaction

27: Break

28: **end if**

29: $t = t + 1$

30: **end while**

31: Find the last successful step: $t_i^{\text{stop}} = \max\{t \in \{0, \dots, t_{\text{max}}\} : \xi_i(t) == 1\}$

32: **end for**

33: **for** each $i \in \mathcal{I}_{\text{cal}}$ **do**

34: **if** $i \in \mathcal{I}_{\text{cal1}}$ **then**

35: $C_i \leftarrow \hat{f}_{\text{prior}}(X_i)$ and $w_\tau(i) \leftarrow 1$

36: **else**

37: $C_i \leftarrow t_i^{\text{stop}}$ if $t_i^{\text{stop}} < \min(T_i, \hat{f}_{\text{prior}}(X_i))$, otherwise $C_i \leftarrow \hat{f}_{\text{prior}}(X_i)$

38: $w_\tau(i) \leftarrow 1 / \prod_{t=1}^{t_i^{\text{stop}}} P_i(t)$

▷ bound max weight

39: **end if**

40: Set $\tilde{T}_i \leftarrow \min(T_i, C_i)$

41: **end for**

42: **return** censoring and event times $\{\tilde{T}_i, C_i, w_\tau(i)\}_{i \in \mathcal{I}_{\text{cal}}}$.

mandatory markdown elements, or specific paragraph quantities. These new rules are appended to the prompt alongside their corresponding Python verification functions. Notably, for this dataset, the attacking model is a helper agent. Its goal is not to elicit harmful content, but rather to improve the presentation of the complex task, e.g., by generating an explicit output contract or checklist, to maximize the probability that the target model passes all programmatic constraints, explicitly without solving the task itself. Specifically, an event is defined as a response that successfully satisfies all prompt requirements, which we verify programmatically using deterministic Python scripts.

The resulting dataset of interactions is divided into three disjoint subsets: a training set (4,000 prompts), a calibration set (3,000 prompts), and a test set (3,000 prompts). We fix the training data and use it to fit our predictive model, assuming an unlimited budget during the training phase, where the full conversation histories are fully observed.

E.2 Predictive model and hyperparameters

First, we embed all texts into vectors using Multilingual-E5-large [69]. We use these embeddings as the inputs for our predictive model, which estimates the raw risk scores $\mathcal{S}_t(H(t))$.

The model architecture begins by projecting the high-dimensional embeddings into a lower-dimensional latent space of 256 using a linear layer, followed by a ReLU activation and dropout of rate 0.2. To capture the temporal dependencies of the conversational history, we apply a Transformer Encoder [70]. Specifically, we use 3 pre-normalized Transformer layers with 6 attention heads, a feed-forward dimension of 1024, and GELU activations. We apply an upper-triangular attention mask during encoding to prevent the model from using future time steps. Following the sequence encoding, we concatenate the hidden states of the Transformer with the initial projected embeddings. This concatenated representation is passed through a multi-layer perceptron consisting of three linear layers, separated by ReLU activations and dropout of rate 0.2, to produce the raw temporal logits.

Specifically, the model receives the text embeddings of the history and estimates the probability of an unsafe event occurring at any future time step $t_2 > t_1$. The output is an $t_{\max} \times (t_{\max} + 1)$ matrix containing the probabilities $\mathbb{P}(T = t_2 \mid T > t_1, H(t_1))$, and the probability of the conversation safely reaching the maximum horizon without an unsafe event, $\mathbb{P}(T > t_{\max} \mid T > t_1, H(t_1))$.

We train this model using an AdamW optimizer [71] with a learning rate of 10^{-4} , weight decay of 0.001, and a batch size of 64. We optimize the network for a maximum of 500 epochs to minimize the survival negative log-likelihood loss. This loss function explicitly accounts for right-censored data by evaluating the exact event probability for uncensored sequences and the aggregated survival probability for censored sequences. We stop the training early if the validation loss does not improve for 50 consecutive epochs. Subsequently, we calibrate the outputs of the model using a temperature scaling L-BFGS optimizer to tune a time-dependent temperature parameter on a held-out validation set, minimizing the discrete survival NLL. The neural networks are implemented using the PyTorch library [72]. The validation set is a portion of 10% samples from the training set that was held-out so that the model is not trained over it.

E.3 Calibration algorithms and evaluation protocol

Once the predictive model is trained and frozen, we evaluate our proposed DAPRO against the optimized method introduced in [12] and two other dynamic techniques we develop in this work. The experiments are taken over 50 independent random splits of the remaining 6,000 prompts into the calibration and test sets.

When constructing an Upper Predictive Bound (UPB), we define the coverage event as the condition where the UPB is either greater than or equal to the true time-to-event T , or when it reaches the maximum conversational horizon $t_{\max} = 200$. In this formulation, $t_{\max}$ serves as a practical infinite time horizon. This design choice is necessary as treating $t_{\max}$ as an effective infinity preserves the theoretical coverage semantics while keeping the UPB size finite. This allows us to effectively compare the bound sizes and their tightness across different calibration methods.

For the calibration algorithms, we simulate the budget-constrained environment by setting the average allowed budget per sample to $B/|\mathcal{I}_{\text{cal}}| = 20$. Following the static allocation setup in [12], we set our prior quantile level to $\tau_{\text{prior}} = 0.56$ when constructing LPBs. When constructing a UPB, we set the prior level at $\tau_{\text{prior}} = 0.97$. To control the maximum inverse-censoring weight, we set the clipping

parameter to $\gamma = 100$, which correspondingly yields an upper bound of $M = 200$ for the trimmed quantile estimates $\hat{f}_\tau(x)$. For the LPB construction, across all calibration methods, we define the search space $\mathcal{T}$ as a logarithmically spaced grid of 1,000 candidate values in the range $[0.001, 0.977]$, following [12]. For the UPB construction, we set the search space as a linearly spaced grid of 3000 values in the range $[0.5, 0.95]$. For each $\tau \in \mathcal{T}$, we compute the estimated miscoverage rate $\hat{\alpha}(\tau)$ using (3) and select the calibrated $\hat{\tau}$ that yields an estimated miscoverage closest to the target level α .

During Phase I of DAPRO, we implement the projection models $\{M_t\}_{t=1}^{t_{\max}}$ as Platt scaling regressors using the scikit-learn package [73]. We implement the budget-constrained Phase I optimization, with a bisection search over the Lagrange multiplier $\lambda \in [10^{-8}, 10^{14}]$ running up to 60 outer steps, combined with an inner Gauss-Seidel Block Coordinate Descent over the log-probabilities, running up to 10 inner passes, with tolerance 10^{-9} . For numerical stability, all probability variables are optimized in log-space. We initialize the variables using a geometric series approximation, $P_{\text{init}} = \bar{B}_2/(1 + \bar{B}_2)$ and warm-start the inner solver using the state from the λ boundary that yielded a budget closest to the target average per sample $\bar{B}_2$. We enforce the monotonicity of the probabilities with respect to the risk scores by applying the Pool Adjacent Violators Algorithm (PAVA) only when the monotonicity is violated during coordinate updates. Once the algorithm converges, we apply a bisection that shifts the log-probabilities to guarantee the budget constraint is satisfied over this set of samples.

In our implementation of the LocallyAdaptive method, we compute the global parameter λ without the finite-sample correction term. Specifically, we set: $\lambda := \inf\{\lambda \geq 0 : \hat{R}_{N_1}(\lambda) \leq \bar{B}_2\}$, where $\hat{R}_{N_1}(\lambda)$ denotes the empirical expected budget consumed on the first data split $\mathcal{I}_{\text{cal1}}$, and $\bar{B}_2$ is the target average budget for the second phase. While the correction term is required for the rigorous theoretical guarantees to control the budget, we observed empirically that this uncorrected formulation consistently satisfies the nominal budget constraint in practice, making the additional conservative penalty unnecessary for our experimental setting.

E.4 Setup of the experiment in Section 1

In the experiment from Section 1, we employed our proposed method, DAPRO, and the static optimized budget allocation introduced in [12]. We chose two specific samples to illustrate how the distribution of the censoring time changes over time. For the static method, the censoring time is defined as $C_i = \hat{f}_{\text{prior}}(X_i) \cdot \text{Ber}(\pi_i)$, where $\text{Ber}(\pi)$ is a Bernoulli random variable with probability π_i . Hence, the expected censoring time is given by $\hat{f}_{\text{prior}}(X_i) \cdot \pi_i$. This expected censoring time is static, as it is computed once and does not change over time. In contrast, our dynamic method changes the advancement probability $P_i(t) = \mathbb{P}(\xi_i(t) = 1)$.

To evaluate the expected censoring time at step t without relying on the future probabilities, e.g., $\mathbb{P}(\xi_i(t+1) = 1)$, we define the expected censoring time as:

$$t + \mathbb{P}(\xi_i(t) = 1) \cdot (\hat{f}_{\text{prior}}(X_i) - t)$$

This formulation is a conservative, causal estimate. It assumes that if the process successfully advances past the current step t , it will continue to run until it reaches $\hat{f}_{\text{prior}}(X_i)$. Since we intentionally avoid relying on the unobserved future probabilities, this metric reflects the real-time, sequential decision-making process of the algorithm. Finally, the reported coverage differences are taken across 50 independent random splits of the calibration and test sets.

E.5 Machine specifications

The computational infrastructure used to generate the datasets and run the experiments includes:

- **CPU:** Intel(R) Xeon(R) CPU E5-2683 v4 @ 2.10GHz, Intel(R) Xeon(R) Gold 5318Y CPU @ 2.10GHz, Intel(R) Xeon(R) Gold 6336Y CPU @ 2.40GHz.
- **GPU:** NVIDIA A40, NVIDIA TITAN X (Pascal), NVIDIA 2080 TI, NVIDIA RTX 2060 SUPER.
- **OS:** Ubuntu 20.04.6.

The data generation via the PAIR algorithm required approximately four days over ten GPUs for sequences of length $t_{\max} = 200$ per dataset and one attacker-target-judge configuration. The GPUs

used for this data generation have 48GB of memory. Training the predictive model and calibrating it using either calibration method takes several minutes per configuration. Across all experiments, random seeds were fixed for reproducibility.

F Additional experiments

In this section, we present extended experiments and performance metrics to further evaluate our proposed framework. We compare against additional baselines and examine the effect of the score function and projection method. We evaluate two distinct score functions for quantifying prompt risk: (1) the estimated probability to observe an unsafe response at the current iteration, and (2) the estimated $1 - \alpha$ quantile of the time-to-unsafe given that an unsafe response was not observed up to the current time. Furthermore, we analyze the effect of the score-to-probability mapping mechanism by comparing Platt scaling and Beta-distribution scaling. Specifically, we evaluate the following eight methods:

- **Optimized:** The static, variance-minimizing allocation strategy [12].
- **Greedy (10%) & Greedy (95%):** The idea is to dynamically observe as many unsafe events as possible on a portion of the data, and then employ the static approach of [12] to obtain a coverage guarantee. See Appendix D.3 for additional details on this method.
- **LocallyAdaptive:** A locally optimized version of our main proposal, where the continuation probabilities are optimized for each step separately, and not globally, in contrast to our flagship method. See Appendix D.4 for more information about this method.
- **DAPRO (Score 1, Platt) & DAPRO (Score 2, Platt):** Our optimally adaptive method with two different score functions (estimated probability of unsafe event and estimated quantile of unsafe event time), with Platt scaling mapping of scores to continuation probabilities.
- **DAPRO (Score 1, Beta) & DAPRO (Score 2, Beta):** Our optimally adaptive method with the same two score functions, but mapped via Beta distribution scaling.

For each experimental setup, we evaluate the performance of these methods across all target LLMs: Qwen 2.5 14B Instruct, Llama 3.1 8B Instruct, Phi 4 Mini Instruct, and Gemma 3 4B Instruct. We report the following metrics: (1) empirical coverage rate at a 90% target level, (2) the average LPB size, (3) the coverage deviation from the 90% nominal target, (4) the total number of unsafe events observed (5) the empirical mean inverse-censoring weight, (6) the variance of the empirical coverage rate, and (7) the variance of the LPB size constructed by each calibration algorithm.

F.1 Toxicity dataset

Figures 8 and 9 present the performance of all methods on the Toxicity dataset using Detoxify as the safety judge. As theoretically guaranteed, all methods achieve a valid coverage rate and satisfy the budget constraint across all evaluated target LLMs. However, the static optimized baseline exhibits a high coverage deviation and variance. In contrast, our adaptive DAPRO observes significantly more unsafe responses than the competitors, resulting in the lowest coverage deviation and variance. LocallyAdaptive also exhibits low variance compared to the static baseline, although it does not match the performance of DAPRO, since its optimization is conducted locally rather than globally. Interestingly, DAPRO achieves this low variance despite having a higher mean inverse-censoring weight. This counterintuitive increase in mean weight occurs since DAPRO invests budget in the initial policy-learning phase ($\mathcal{I}_{\text{cal1}}$), especially for resilient models like Llama and Phi 4 Mini that require longer interactions to jailbreak, leaving a smaller remaining budget for the deployment phase ($\mathcal{I}_{\text{cal2}}$). As indicated by the results over the Qwen2.5 and Gemma3 LLM models, when the budget remaining for the second phase increases, DAPRO achieves a lower mean weight than the static baseline.

F.2 RedTeam dataset with Qwen as a judge

In Figures 10 and 11, we present the metrics over the Anthropic Red Team dataset, utilizing Qwen 2.5 14B Instruct as an LLM-as-a-judge.

These figures closely mirror those observed in the Toxicity experiments, demonstrating the robustness of our method across different safety benchmarks and scoring mechanisms. Once again, all evaluated

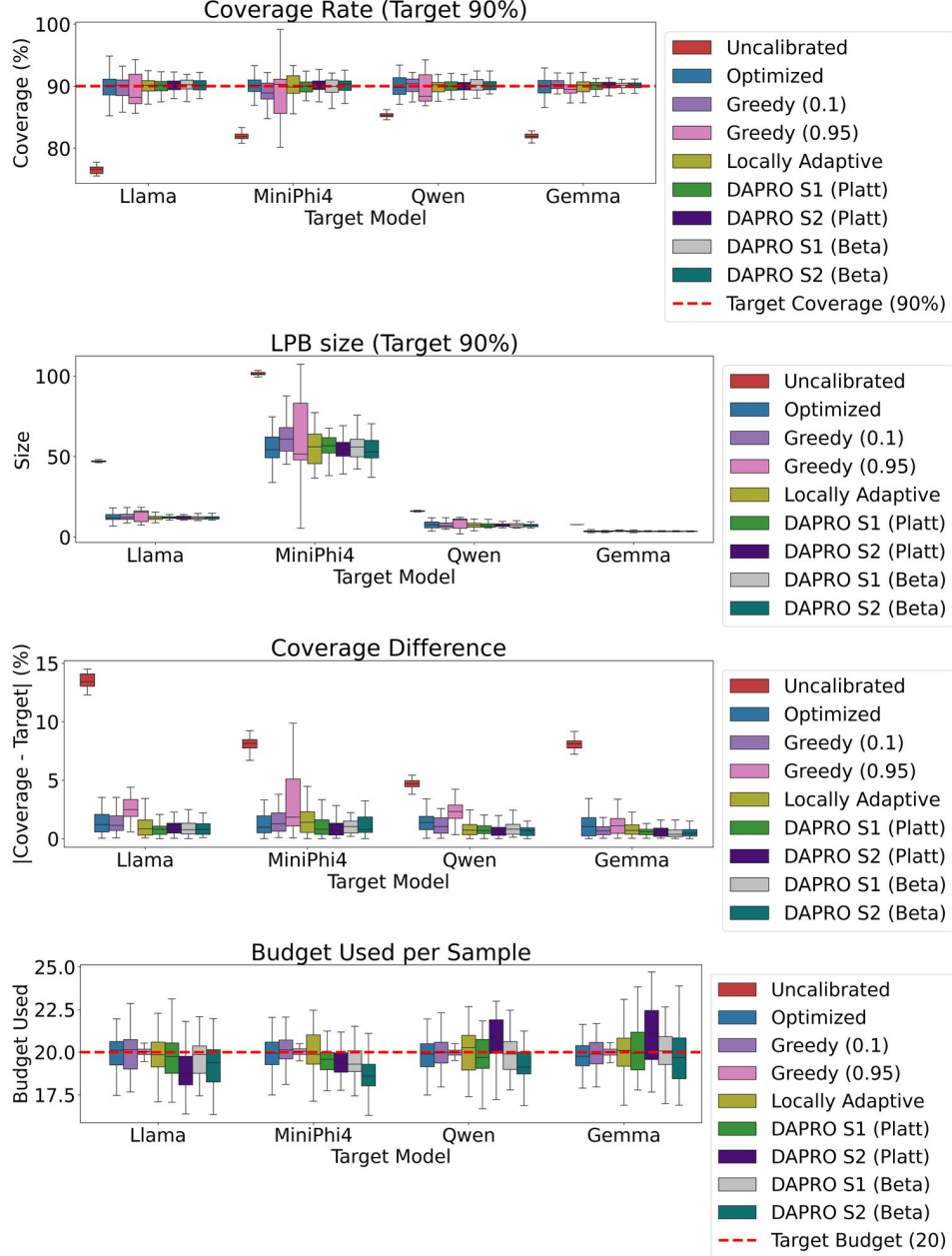


Figure 8: **Toxicity** dataset: coverage rate, LPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

methods successfully attain valid coverage and satisfy the budget constraint. However, the static optimized baseline still suffers from a high coverage deviation and variance. In contrast, DAPRO dynamically adapts to the interactions to uncover the highest number of unsafe events, yielding the LPBs with the lowest variance and coverage deviation. Furthermore, we observe the exact phenomenon with the mean-weight: since DAPRO expends many budget units during the initial policy-learning phase ($\mathcal{I}_{\text{cal1}}$) to learn an optimal policy, it occasionally has a higher mean inverse-censoring weight due to the limited budget remaining for $\mathcal{I}_{\text{cal2}}$. These figures also show that the Greedy method is unreliable since it tends to exhibit a very high variance when a high portion of the budget is used for the early greedy exploration.

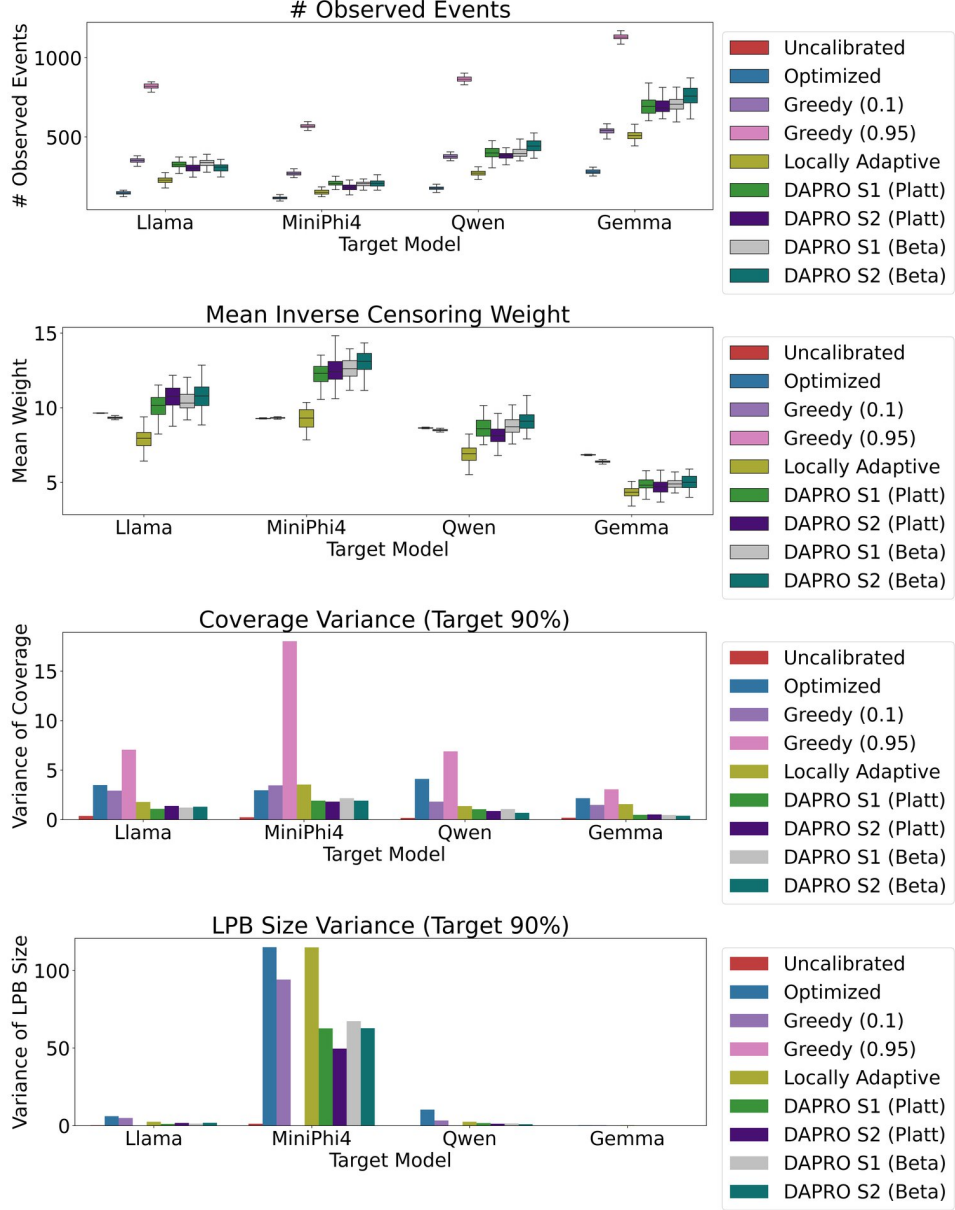


Figure 9: **Toxicity** dataset: number of observed unsafe events, mean inverse-probability weight, coverage variance, and LPB variance by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

F.3 RedTeam dataset with LlamaGuard as a judge

Figures 12 and 13 detail the results on the Red Team dataset when evaluated with the Llama-Guard judge. This judge marks responses as unsafe at earlier stages of the conversation, so we set the target average budget per sample at $\bar{B} = 10$ for this experiment. These figures demonstrate the same trend observed in the previous experiments. All methods are valid in terms of both coverage and budget. In this setup, the unsafe event occurs very early, and thus DAPRO does not expend too many budget units in the first phase, which leaves more budget for the second phase. Consequently, DAPRO attains the lowest mean inverse-censoring weights while still observing the highest number of unsafe events. As a result, it constructs LPBs with the lowest variance and achieves empirical coverage rates that are closer to the nominal level. The Greedy approach still tends to exhibit a high variance if its

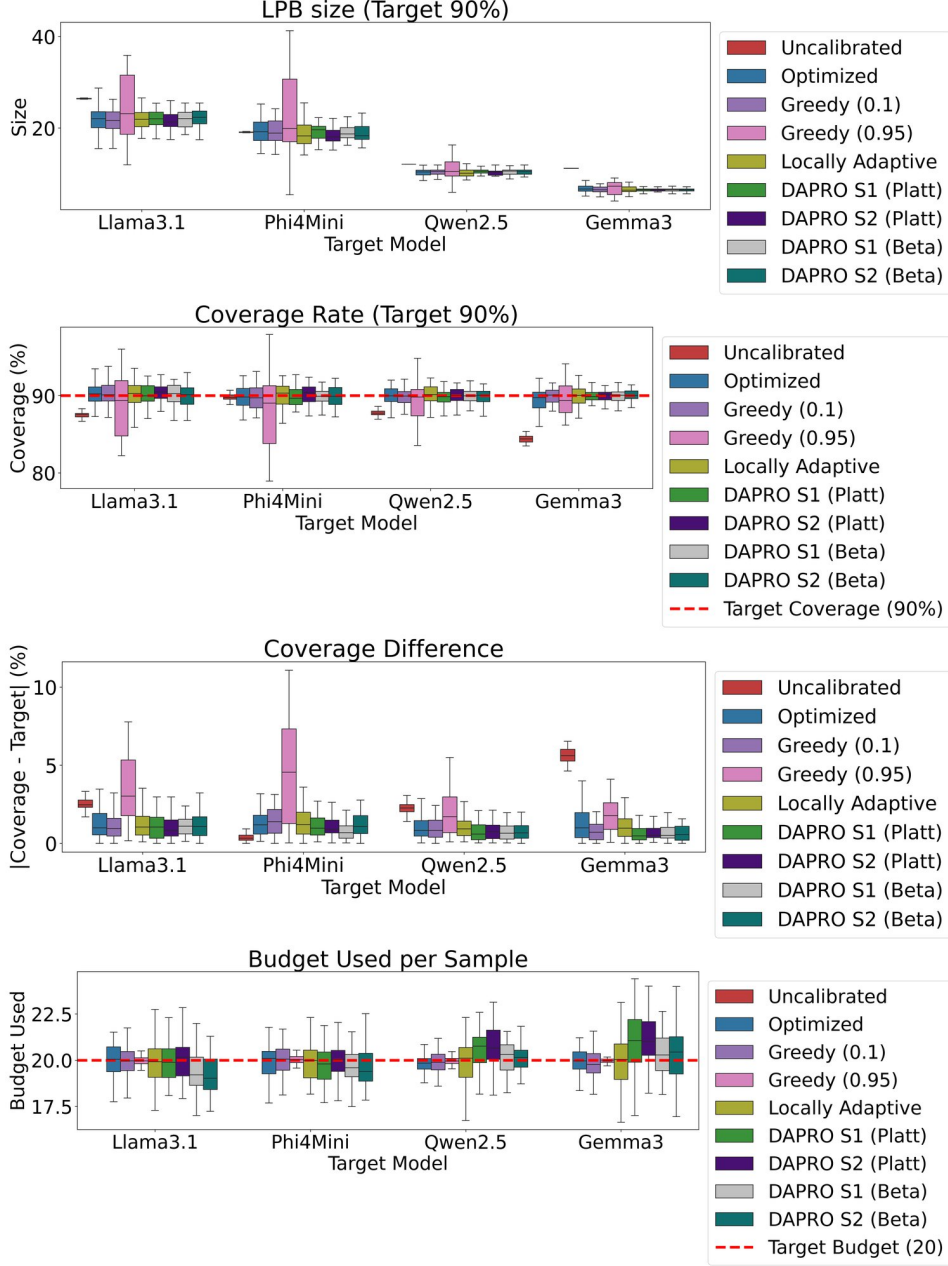


Figure 10: **RedTeam** dataset with **Qwen 2.5 14B Instruct** as a judge: coverage rate, LPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

budget split parameter is not tuned correctly. Once again, the *LocallyAdaptive* technique attains a variance lower than the static baseline, but higher than our globally optimized DAPRO.

F.4 RAG hallucinations experiment

In this experiment, we evaluate our framework in a Retrieval-Augmented Generation (RAG) hallucinations setup. The target LLM is provided a context paragraph regarding a specific subject, such as "Macintosh", or "Beyoncé". The objective of the attacker is to construct adversarial queries that trick the target model to output a fabricated information that does not exist in the source context. To

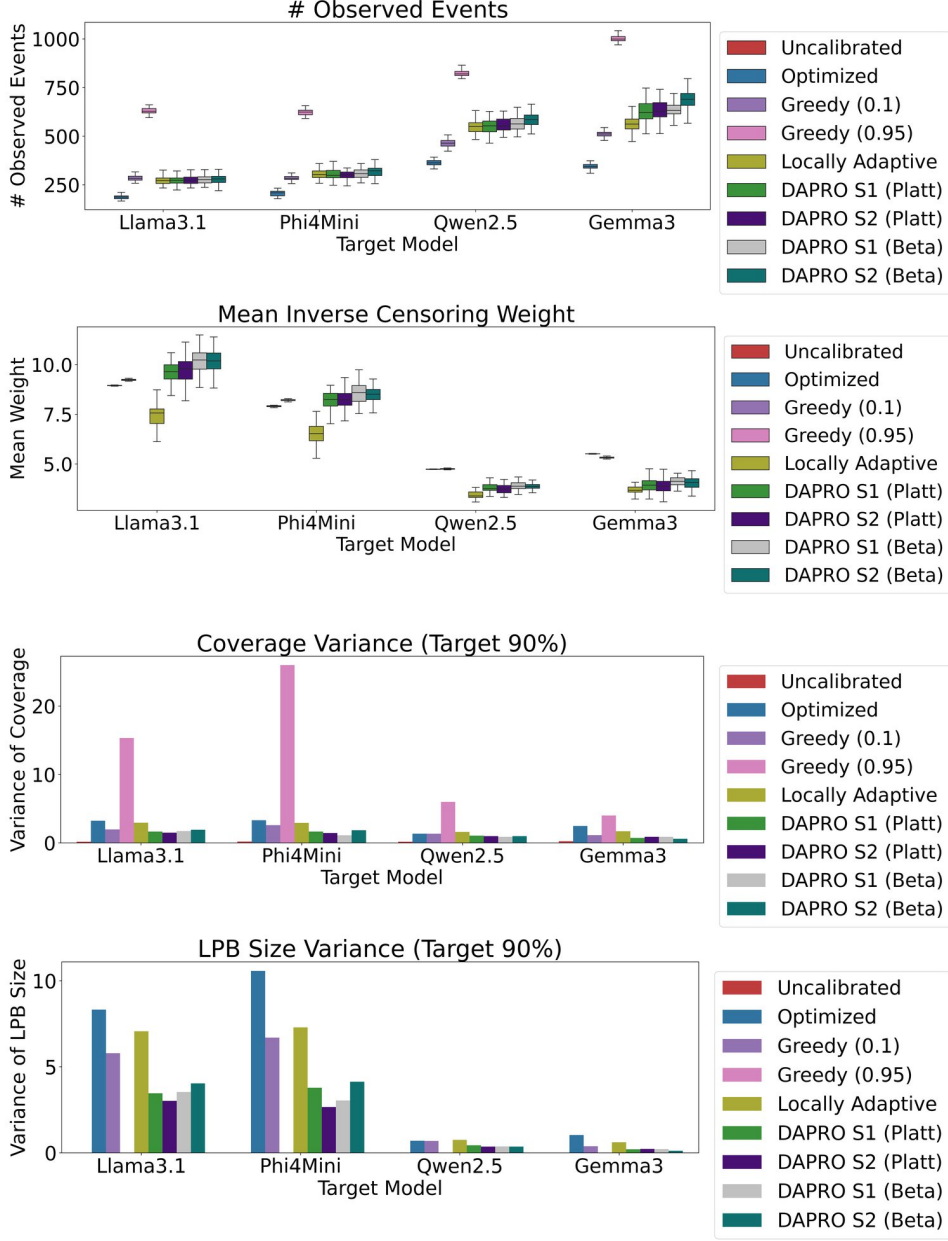


Figure 11: **RedTeam** dataset with **Qwen 2.5 14B Instruct** as a judge: number of observed unsafe events, mean inverse-probability weight, coverage variance, and LPB variance by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

evaluate the success of the attack, the judge is provided with the ground-truth context, the target’s response, and outputs a hallucination score ranging from 1 to 10, where a score of 10 is considered as a hallucination ($Y = 1$). We set the budget per sample constraint to $B/|\mathcal{I}_{\text{cal}}| = 10$ since the event rate is significantly high for this dataset, of approximately 95 – 99% hallucination rate.

Figures 14 and 15 present the performance of all evaluated methods on the Hallucination dataset. These results present the same phenomenon observed in the Toxicity and RedTeam experiments: DAPRO maintains the nominal coverage rate and satisfies the budget constraints across all evaluated target LLMs. Similarly, DAPRO observes significantly more hallucinated responses than the baselines, and yields the lowest coverage deviation and variance among the tested methods. LocallyAdaptive

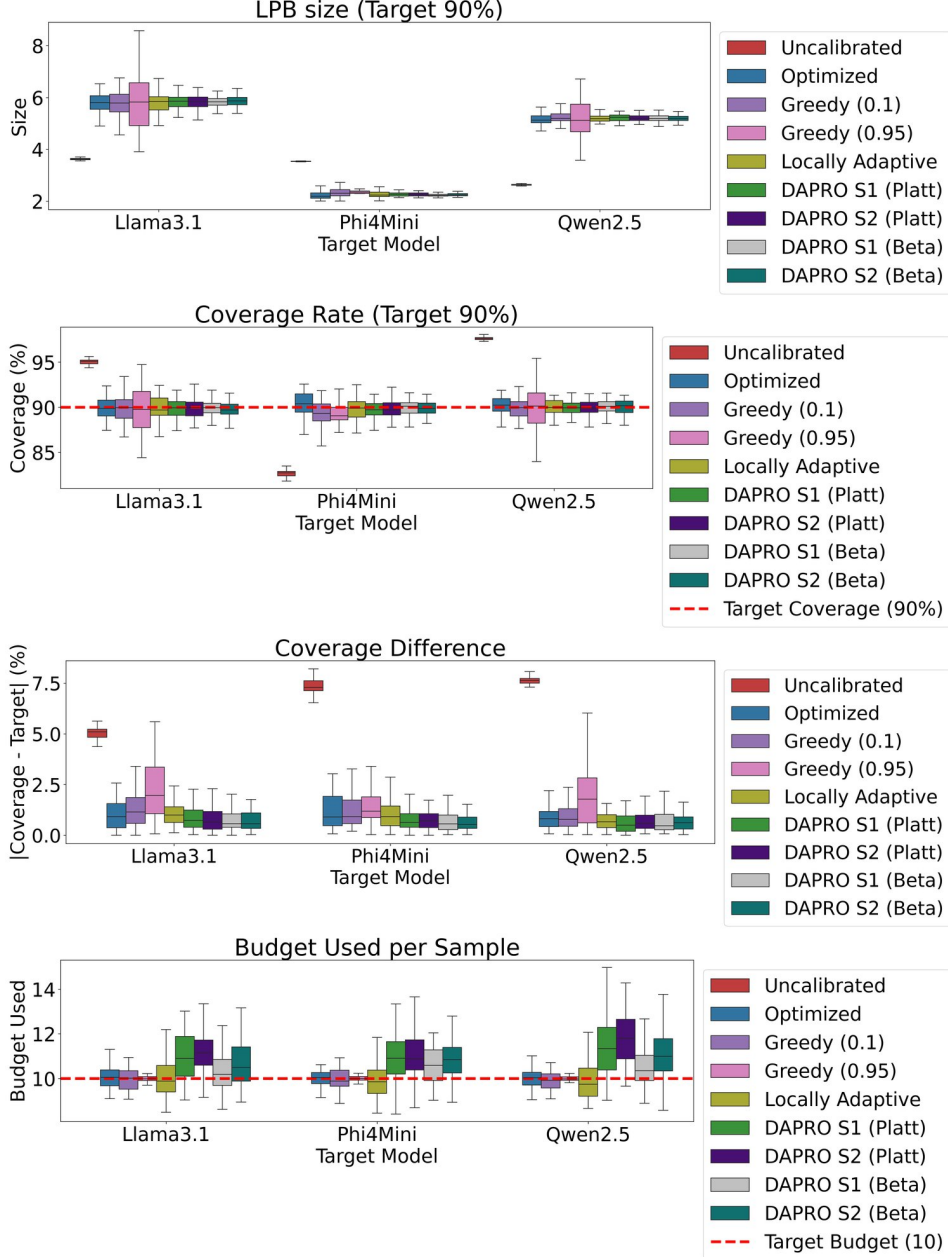


Figure 12: **RedTeam** dataset with **Llama-Guard** as a judge: coverage rate, LPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 90% and target $B = 10$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

again exhibits lower variance than the static baseline, though not as low as the globally optimized DAPRO.

F.5 AutoIF: evaluating utility of helper agents

We demonstrate our proposal in a helper LLM setup. The "attacker" LLM is an agent aiming to guide the target model to output a response that satisfies complex constraints, such as including a specific word multiple times. The judge is a set of pre-defined Python scripts that programmatically evaluate the constraints. If the response of the target satisfies all constraints, the judge outputs 10; otherwise,

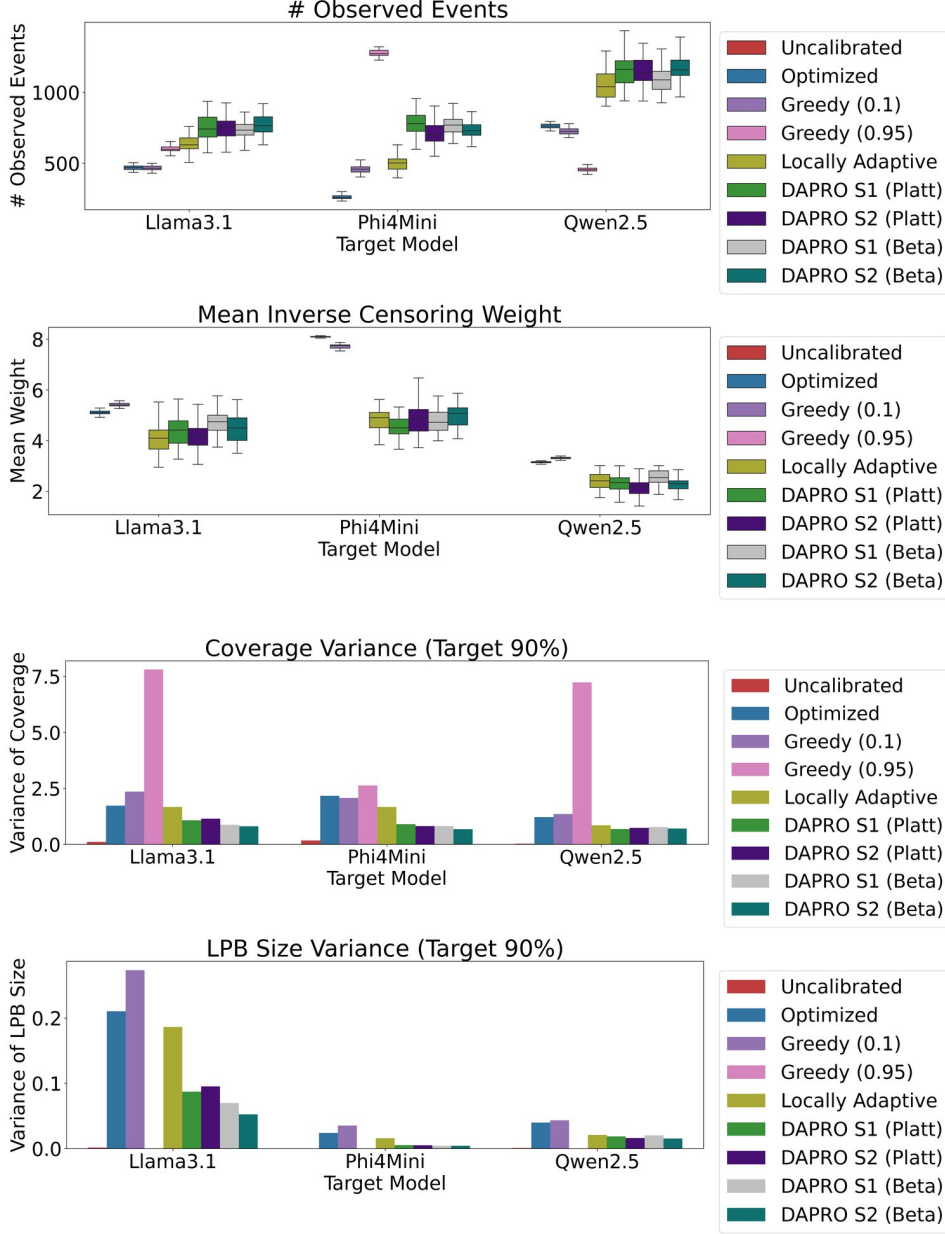


Figure 13: **RedTeam** dataset with **Llama-Guard** as a judge: number of observed unsafe events, mean inverse-probability weight, coverage variance, and LPB variance by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 10$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

it outputs 1. We repeat our analysis for this experiment once for constructing a calibrated LPB and once for a UPB.

Figures 16 and 17 display the performance of the LPBs constructed by each method on the AutoIF dataset. These figures show the same trend observed in the previous experiments: DAPRO attains the nominal coverage rate and satisfies the budget constraints. Moreover, DAPRO observes the highest number of successful events compared to the static baseline and exhibits the lowest variance. LocallyAdaptive exhibits low variance as well. In contrast, the static optimized baseline achieves a very high variance, and the uncalibrated model does not achieve the desired coverage level. In addition, these figures show that DAPRO performs similarly across both scores and projection models

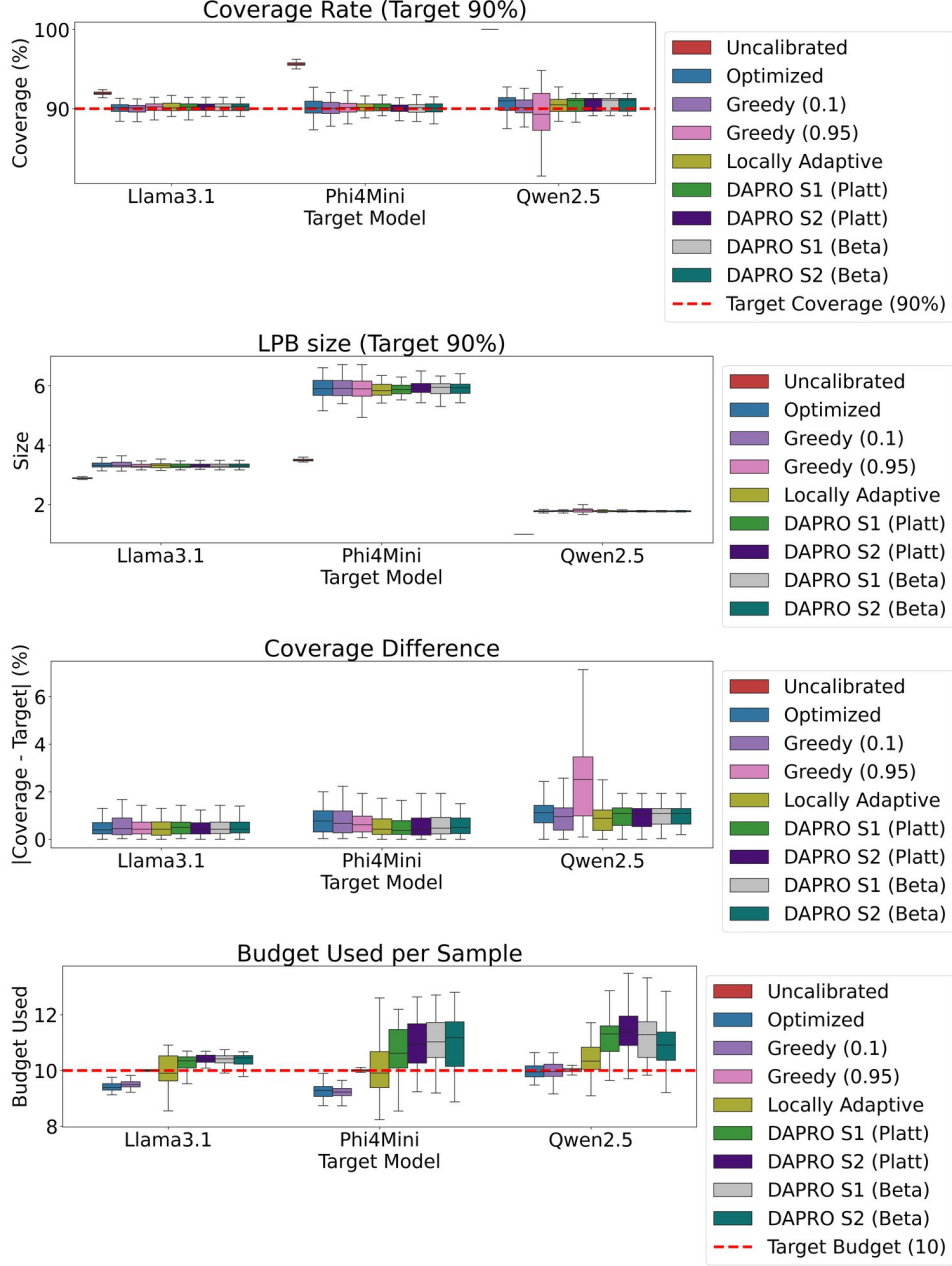


Figure 14: **Hallucination** dataset: coverage rate, LPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 10$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

we examined, indicating that our approach is robust to these choices. Finally, these figures indicate that Greedy is not robust to the choice of the data split size parameter, as it heavily affects the variance of its produced LPBs.

We now evaluate our approach for constructing a calibrated UPB. Since constructing a UPB requires observing the conversations at higher iterations, we increase the average budget per sample to $\bar{B} = 30$ for this setup. We summarize the performance of each method for UPB constructions with the AutoIF dataset in Figures 18 and 19. These figures show that all calibration algorithms attain the target 70% coverage rate while satisfying the budget constraint, as guaranteed in theory. The proposed DAPRO observes more successful events than the baseline and exhibits a lower variance compared to it. Yet,

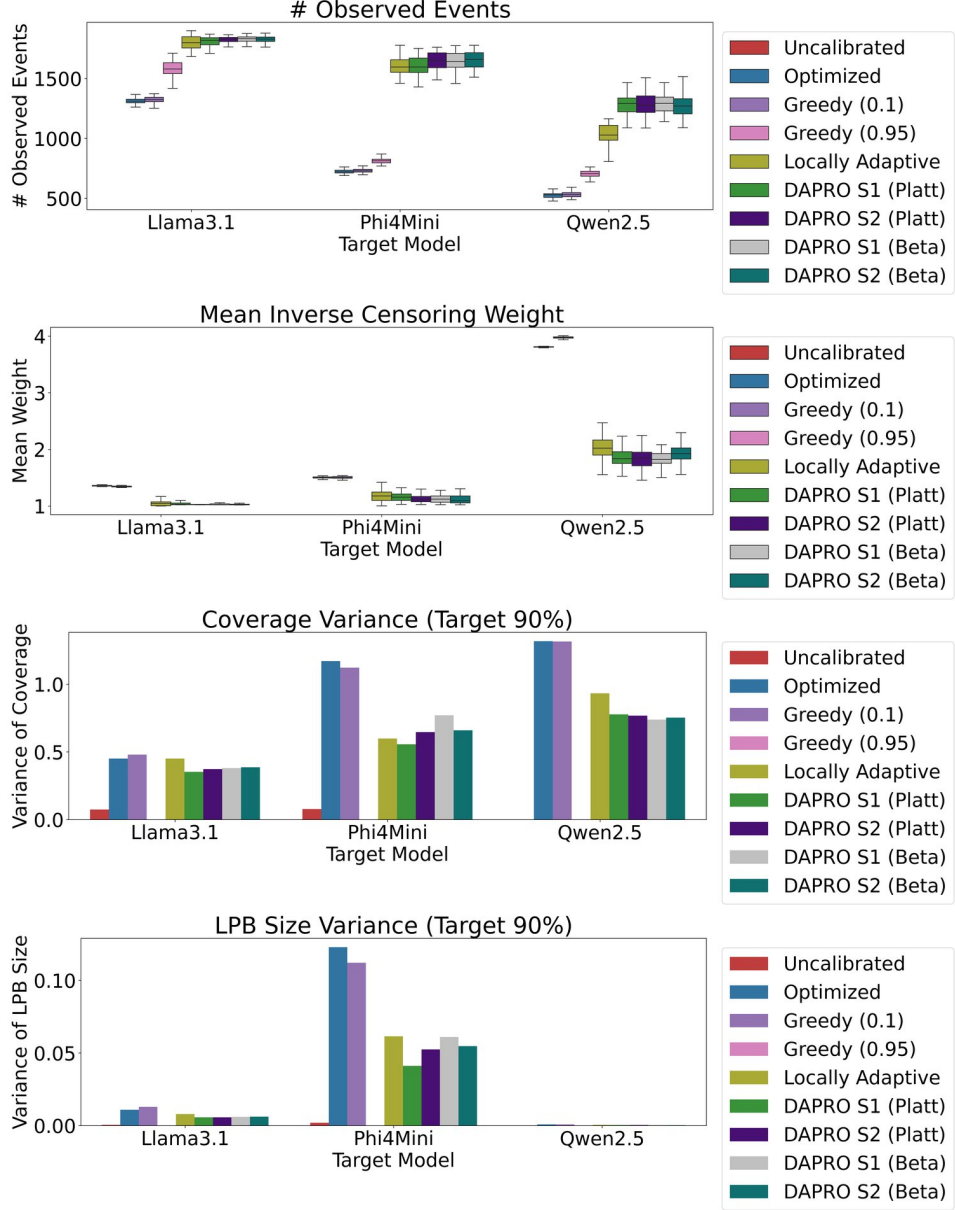


Figure 15: **Hallucination** dataset: number of observed successful events, mean inverse-probability weight, coverage variance, and LPB variance by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 10$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

the variance of all methods is very similar, as the UPBs they generate are closer to the maximum horizon $t_{\max}$, which naturally reduces variance.

F.6 Unbiased estimation of safety metrics at the population level

Beyond evaluating the efficiency of individual adversarial attacks, a critical objective in many applications, such as large-scale red-teaming, is assessing LLM safety at the population level. For this purpose, we consider two safety metrics evaluated over a maximum conversational horizon of $t_{\max} = 200$ turns: the **Unsafe Event Rate (UER)**, which measures the proportion of prompts that elicit an unsafe output within $t_{\max}$ steps, and the **Restricted Mean Time-to-Unsafe (RMTTU)**,

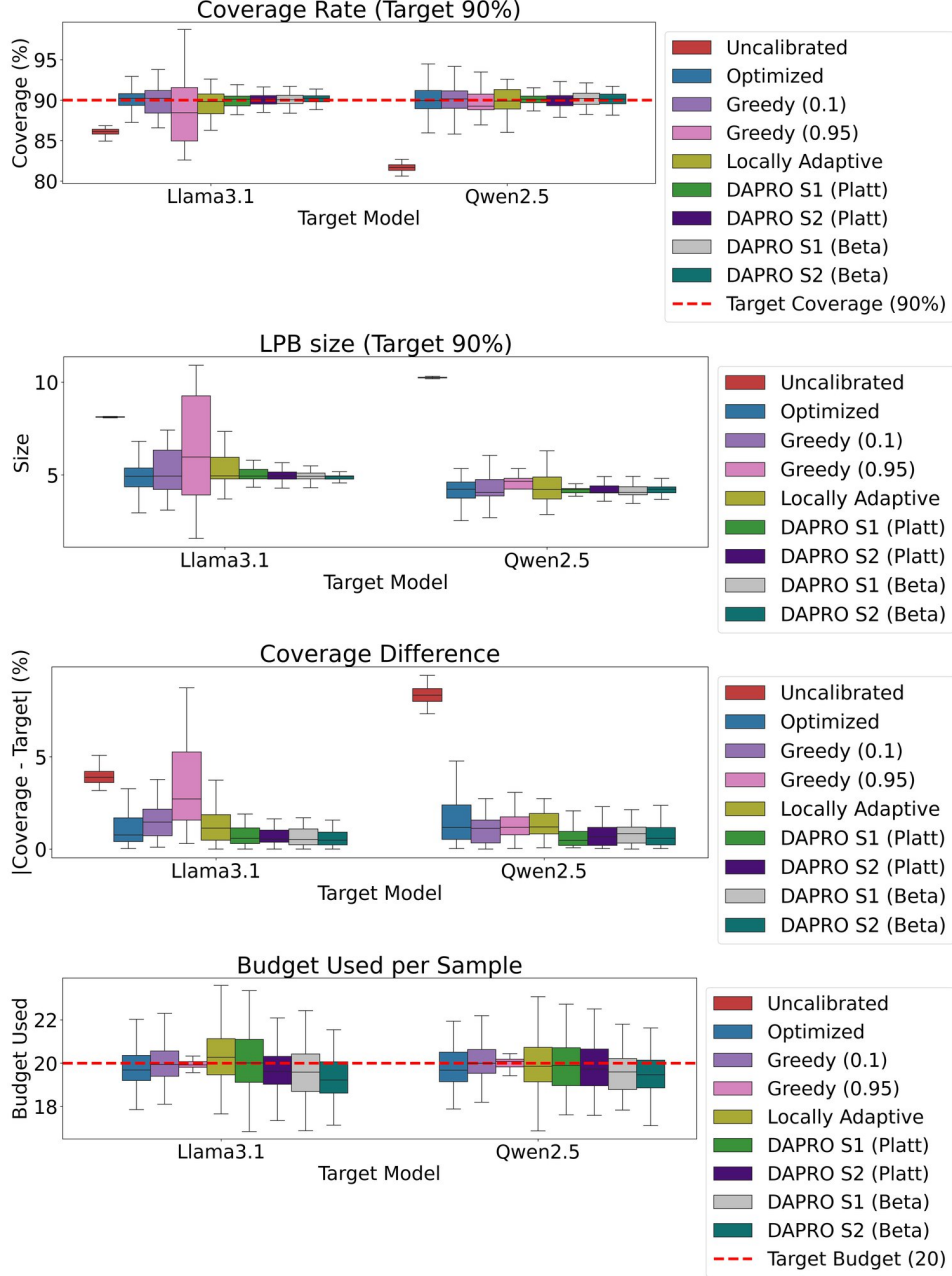


Figure 16: **AutoIF** (LPB) dataset: coverage rate, LPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

which quantifies the average number of turns required for the model to produce that unsafe response. When evaluating LLM utility, e.g., via the AutoIF dataset, where the target event is successful task completion, we symmetrically refer to these metrics as the **Successful Event Rate** (SER) and the **Restricted Mean Time-to-Success** (RMTTS).

Estimating these metrics under limited computing resources is challenging, as when the budget allocator halts a conversation early, it introduces a right-censoring bias. Treating these early stopped samples as safe will increase the safety perceived by the model. Conversely, running every conversation for the full $t_{\max}$ iterations to find the true Oracle outcome is computationally infeasible. However, by employing a stochastic budget allocator that tracks the probability of reaching the

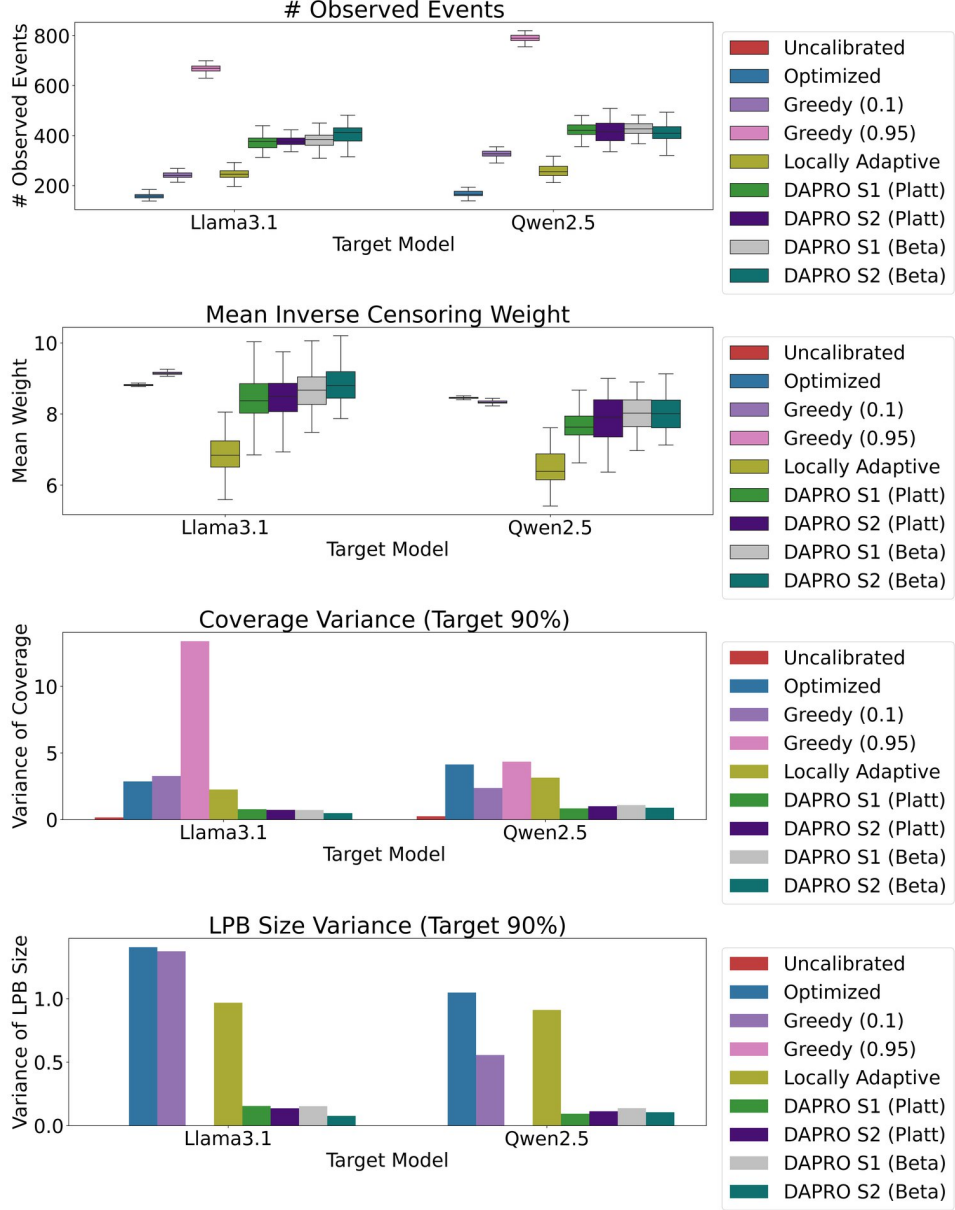


Figure 17: **AutoIF** (LPB) dataset: number of observed successful events, mean inverse-probability weight, coverage variance, and LPB variance by various methods across four target LLMs. Target coverage rate: 90% and target $\bar{B} = 20$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

censoring time, such as DAPRO, we can construct unbiased estimates of the true population metrics using only a fraction of the compute.

We achieve this by re-weighting the observed outcomes using the inverse probability of censoring, similar to our miscoverage rate estimator in (3). Specifically, let $m(T)$ represent an individual-level metric, such as the unsafe event indicator $m(T) = \mathbb{I}\{T < t_{\max}\}$. Taking the expectation over the population yields the true UER: $\mathbb{E}[m(T)] = \mathbb{P}(T < t_{\max})$. We estimate this population metric using:

$$\hat{m} = \frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} w(i) \mathbb{I}\{T_i \leq C_i\} m(T_i), \quad (22)$$

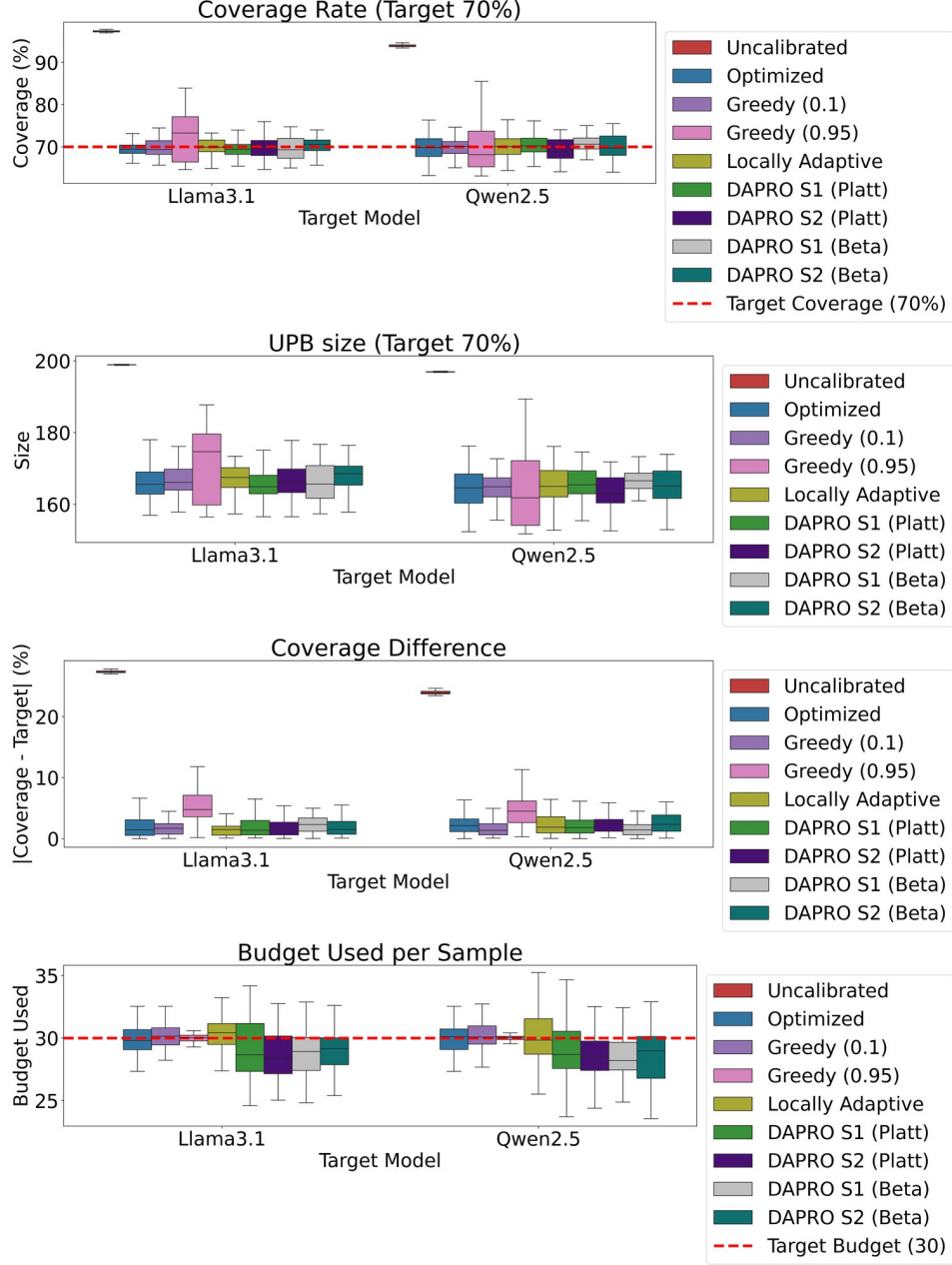


Figure 18: **AutoIF** (UPB) dataset: coverage rate, UPB size, coverage deviation, and budget utilized by various methods across four target LLMs. Target coverage rate: 70% and target $\bar{B} = 30$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

where, $w(i) = \mathbb{P}(T_i \leq C_i \mid \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal}}}, \mathcal{D}_{\text{train}})^{-1}$ is the corresponding weight. In our experiments, since conversation lengths are capped at t_{max} , any metric depending on this horizon is fully resolved if we either observe the event ($T_i \leq C_i$) or reach the maximal horizon ($C_i = t_{\text{max}}$). Consequently, our capped estimated metric is given by:

$$\hat{m}^{t_{\text{max}}} = \frac{1}{|\mathcal{I}_{\text{cal}}|} \sum_{i \in \mathcal{I}_{\text{cal}}} w^{t_{\text{max}}}(i) \mathbb{I}\{T_i \leq C_i \text{ or } C_i = t_{\text{max}}\} m(T_i),$$

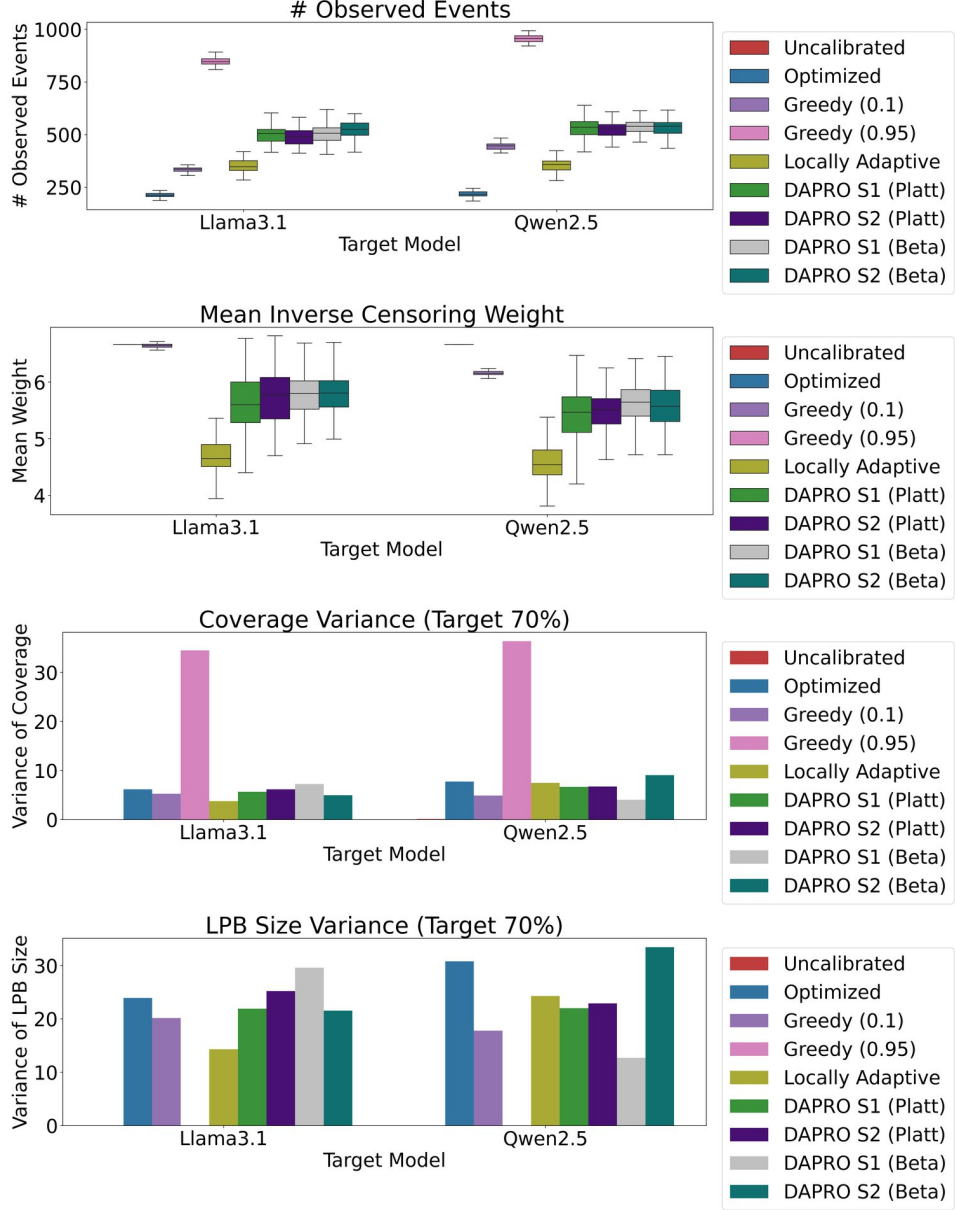


Figure 19: **AutoIF** (UPB) dataset: number of observed successful events, mean inverse-probability weight, coverage variance, and UPB variance by various methods across four target LLMs. Target coverage rate: 70% and target $\bar{B} = 30$ budget per sample. Performance metrics are taken over 50 random splits of the calibration and test sets.

where the adjusted weights are $w^{t_{\max}}(i) = \mathbb{P}(T_i \leq C_i \text{ or } C_i = t_{\max} \mid \{(X_i, H_i, T_i)\}_{i \in \mathcal{I}_{\text{cal}}}, \mathcal{D}_{\text{train}})^{-1}$. Following the arguments in Theorem 4, $\hat{m}$ is an unbiased estimator of $m(T)$, guaranteeing that $\mathbb{E}[\hat{m}] = \mathbb{E}[m(T)]$.

To demonstrate this empirically, we compute the Oracle UER and RMTTU values using both calibration and test sets under an unlimited budget setting, so that all unsafe events up to time $t_{\max}$ are fully observed. We then run each allocation method with a nominal average budget per sample constraint set to $\bar{B} = 20$ to obtain censoring times, censored event times, and the weights, and construct the metric estimator $\hat{m}$ from (22). We repeat this process across 50 random calibration-test splits to measure the mean of estimated metrics and their variance.

We compare DAPRO against several baselines: an **Uncalibrated Uniform** allocation, which allocates the budget uniformly and naively averages the raw censored outcomes without re-weighting, a **Calibrated Uniform** allocation that applies re-weighting, the **Static Optimized** baseline of [12], and the LocallyAdaptive, Greedy dynamic allocators. Notably, in this setting, the **Static Optimized** baseline functions identically to the **Calibrated Uniform** allocation. A sample only contributes to the weighted sum if the conversation reaches the maximum horizon ($C = t_{\max}$) or successfully reveals an unsafe event ($C \geq T_i$); if a sample is censored ($C < t_{\max}$ and $C < T_i$), its contribution is effectively zero. For this reason, we must set the $\hat{f}_{\text{prior}} = t_{\max}$ for all samples, so the static optimization algorithm has no preference between any two prompts. As a result, the static optimization allocates the budget uniformly across the entire calibration set.

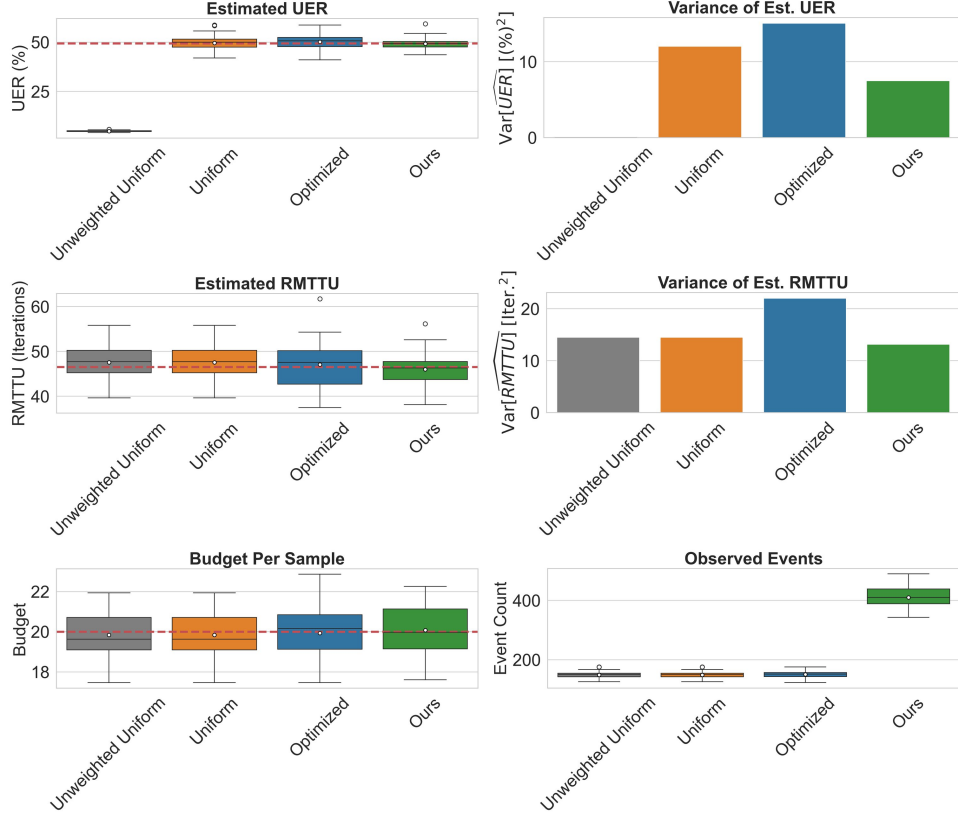


Figure 20: Population-level safety metric estimation on the Toxicity dataset with Qwen 2.5 14B Instruct model serves as both the target and attacker. The top two rows display the estimated Unsafe Event Rate (UER) and Restricted Mean Time-to-Unsafe (RMTTU), alongside their respective variances. The red dashed lines are the oracle quantities. The bottom row displays the utilized budget per sample with a red dashed line indicating the target level, and the total number of observed unsafe events. While the unweighted baseline severely underestimates true vulnerability, and naive calibrated baselines suffer from extreme variance, DAPRO accurately estimates the true Oracle metrics with zero bias and minimal variance. Furthermore, DAPRO achieves the highest budget efficiency, successfully observing the most unsafe events under the same resource constraints. Results are aggregated over 50 random calibration-test splits.

Figure 20 presents the estimated UER and RMTTU, along with their variance, on the Toxicity dataset with Qwen 2.5 14B Instruct as both target and attacker. The **Unweighted Uniform** budget allocator underestimates the UER since it does not re-weight the contribution of the uncensored samples to the sum. By correcting for this bias with re-weighting, the **Calibrated Uniform** achieves the correct value in expectation, but exhibits a high variance. The static optimized approach has zero bias as well, although it exhibits a higher variance, as its allocation is highly inefficient. In stark contrast, DAPRO achieves the best of both worlds: it recovers the true Oracle metrics with zero bias while achieving the lowest variance and satisfying the budget constraint. By globally optimizing the advancement

probabilities, DAPRO expends the budget on highly vulnerable conversations, efficiently observing the highest number of unsafe events.

Figures 21 through 25 present the estimated UER and RMTTU, alongside their variance, across all four experimental setups. We compare the Static Optimized baseline, the dynamic Greedy approach (configured with a data split level of 0.1, as higher values resulted in extreme instability), and our LocallyAdaptive and DAPRO. Since all evaluated methods correctly address the right-censoring distribution shift by applying re-weighting, they successfully attain the true Oracle value (marked by the red dashed line) in expectation. As all methods have zero bias, the estimator’s measure of reliability is its variance. The bar plots in these figures demonstrate that the static baseline attains a high variance. The Greedy allocator tends to attain a variance lower than the static baseline, although not consistently, as its data-split ratio parameter is not tuned. In contrast, LocallyAdaptive consistently attains variance lower than the static baseline, and occasionally even lower than our global optimized DAPRO. This establishes LocallyAdaptive as a competitive alternative, particularly as it provides an exact, finite-sample budget guarantee in expectation. Crucially, this theoretical guarantee holds purely under the mild assumption of data exchangeability; it does not require an accurate score-to-probability mapping to satisfy the budget constraint. While DAPRO generally maintains lower variance than the static approach, the rare instances where it exhibits a higher variance can likely be attributed to an inaccurate score-to-probability mapping. Overall, this experiment demonstrates that while proper statistical re-weighting is necessary to construct an unbiased estimator, efficient budget allocation is essential to obtain low-variance, reliable safety metrics for large-scale tasks under limited compute resources.

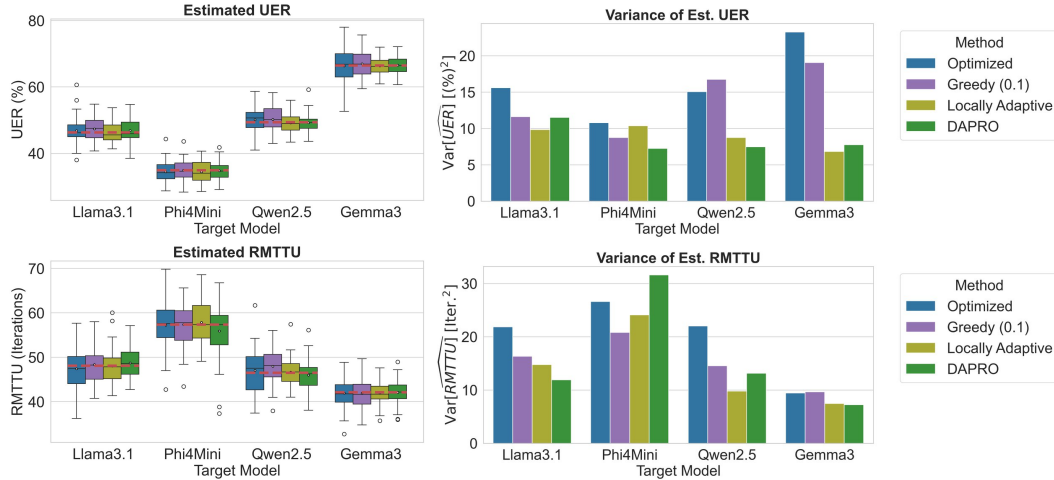


Figure 21: Population-level metric estimation on the **Toxicity** dataset. Boxplots display the distribution of the estimated Unsafe Event Rate (UER) and Restricted Mean Time-to-Unsafe (RMTTU) across 50 random splits. The red dashed lines represent the true Oracle metrics. While all methods achieve the correct metric in expectation, LocallyAdaptive and DAPRO yield the tightest variance compared to the static baselines and Greedy approach.

E.7 Ablation study for DAPRO

In this section, we analyze the robustness of our proposed DAPRO to three factors: the size of the first calibration set split (N_1), the quality of the score, and the nominal budget per sample. This study reveals that DAPRO is effective and valid under varying experimental conditions. Even under extreme conditions, such as a limited budget or severely corrupted scores, our framework consistently achieves the desired coverage level, and its budget consumption is close to the target level. We evaluate the performance on the Toxicity dataset using the Qwen 2.5 14B Instruct model serving both as attacker and target.

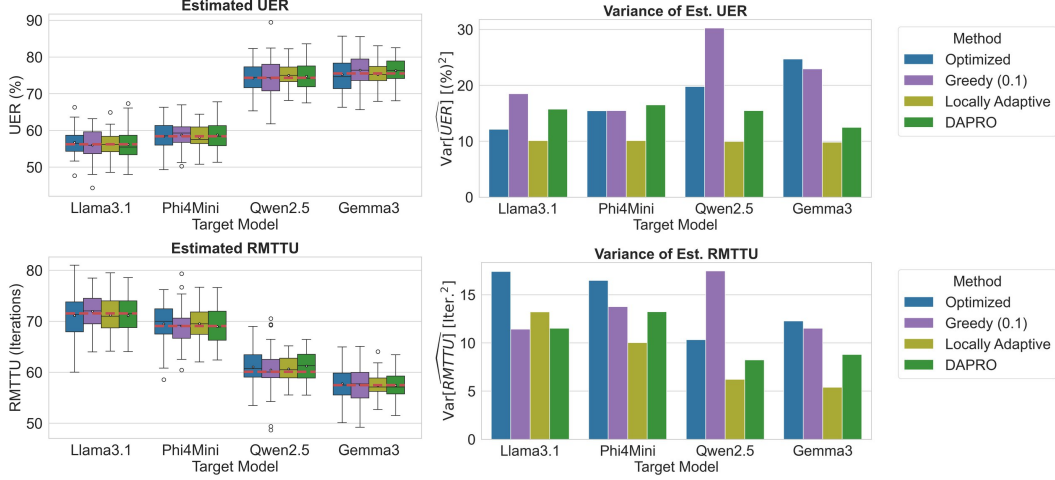


Figure 22: Population-level metric estimation on the **Red Team** dataset evaluated with a **Qwen** judge. The red dashed lines denote the true Oracle metrics. Our proposed **Locally Adaptive** and **DAPRO** consistently achieve lower variance than the static optimized baseline across various target models.

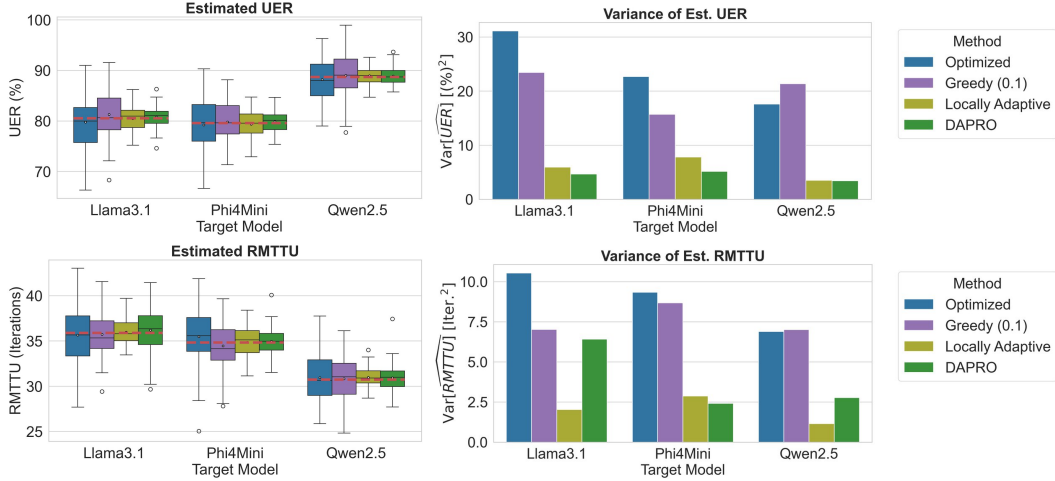


Figure 23: Population-level metric estimation on the **Red Team** dataset evaluated with a **Llama Guard** judge. The red dashed lines denote the true Oracle metrics. Both the static and greedy baselines exhibit high variance across different data splits, whereas **DAPRO** remains highly stable and consistently recovers the Oracle values.

F.7.1 The effect of the calibration set split

We employ **DAPRO** with various values of the first calibration set split size (N_1) and display its performance in Figure 26. **DAPRO** consistently achieves the nominal coverage rate, 90%, and satisfies the target budget per sample of 20 across all tested values of N_1 . This figure reveals that the budget used by **DAPRO** converges to the target level as N_1 increases. However, **DAPRO** attains a higher mean weight for higher values of N_1 . Yet, **DAPRO** achieves a lower coverage difference and lower coverage variance compared to the static baseline allocation of [12] for all examined values of N_1 .

F.7.2 The effect of the quality of the score

To evaluate the robustness of our approach to the quality of the scores, we conduct an ablation where the scores used in Phase I are artificially corrupted. Specifically, we add random noise to the original scores:

$$S_i(t) = (1 - \lambda) \cdot S(H_i(t)) + \lambda \cdot U_i(t)$$

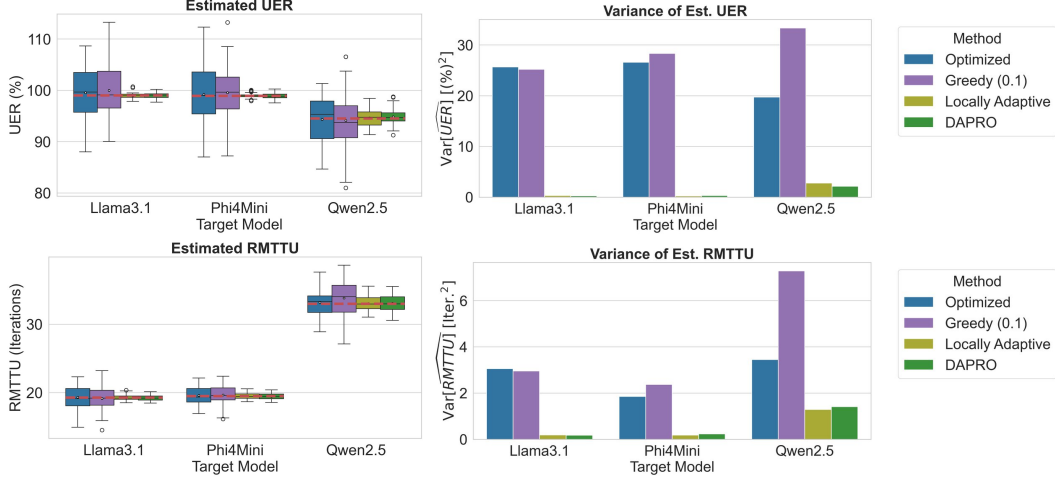


Figure 24: Population-level metric estimation on the **Hallucination** dataset. The red dashed lines denote the true Oracle metrics. The LocallyAdaptive and DAPRO exhibit the lowest variance, resulting in the most reliable estimation of both the UER and RMTTU metrics.

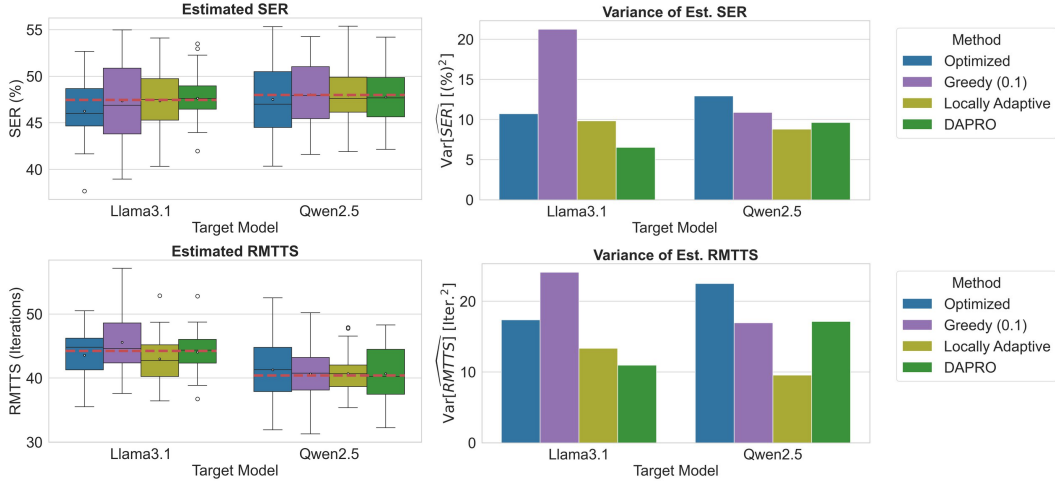


Figure 25: Population-level metric estimation on the **AutoIF** dataset. The red dashed lines denote the true Oracle metrics. The LocallyAdaptive and DAPRO exhibit the lowest variance, resulting in the most reliable estimation of both the SER and RMTTS metrics.

where $U_i(t) \sim \text{Uni}(0, 1)$ is a uniform random variable between 0 and 1. The parameter $\lambda \in [0, 1]$ controls the score degradation level. We fix the first-split set size to $N_1 = 100$. As shown in Figure 27, DAPRO maintains a valid empirical coverage ($\approx 90\%$) across all noise levels, indicating that its theoretical guarantees hold regardless of the quality of the score. Furthermore, DAPRO satisfies the budget constraint of 20 units per sample for low values of λ , while exceeding the budget usage for higher values of λ . As $\lambda \rightarrow 1$ and the scores lose their predictive signal, DAPRO can no longer distinguish between promising and unpromising conversations. Nevertheless, DAPRO attains a coverage rate closer to the desired level with a lower variance compared to the static baseline for all values of λ . This demonstrates the robustness of our proposal to scores that are not informative.

F.7.3 The effect of the available budget

We employ DAPRO and the static baseline of [12] with varying levels of budget per sample and present their performance in Figure 28. To adapt our method, and its data split to the varying budget per sample, we adapt its N_1 parameter to the nominal average budget per sample. For an average budget per sample lower than or equal to 10, we set $N_1 = 25$, and for an average budget per sample of 25 we

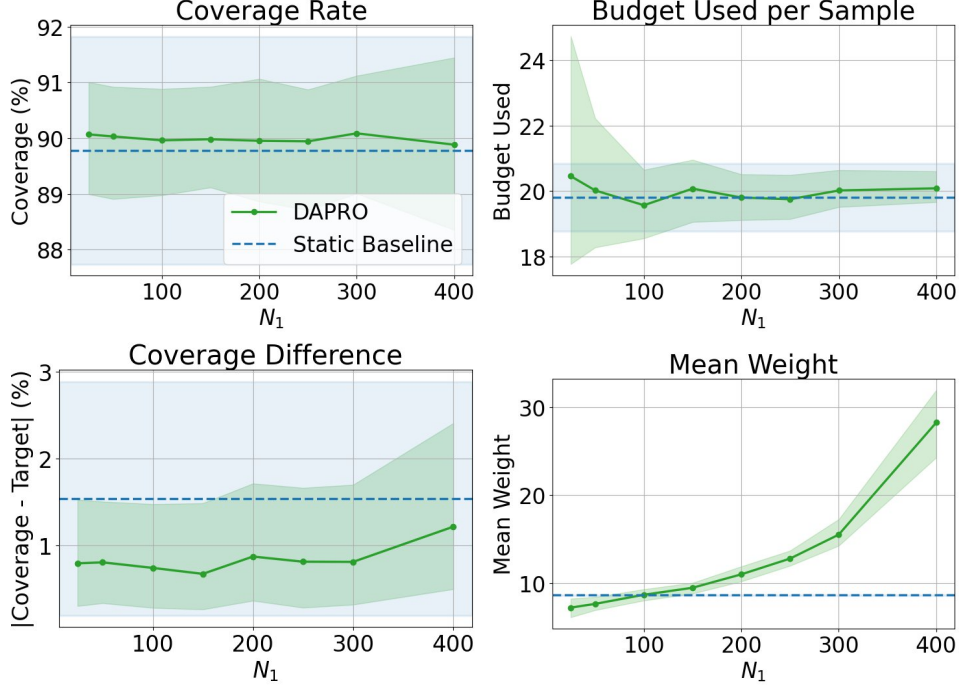


Figure 26: Impact of first calibration set split size (N_1) on empirical coverage, budget per sample, and mean weight on the Toxicity dataset. The Qwen 2.5 14B Instruct model serves as both the attacker and target. Nominal coverage level is set to $1 - \alpha = 90\%$ and the target budget per sample is 20. Shaded regions denote semi-deviations over 50 random calibration-test splits. As N_1 increases, DAPRO consistently maintains valid coverage and satisfies to the budget constraint, with variance notably decreasing at larger sample sizes.

set $N_1 = 50$. For any higher value of budget per sample, we set $N_1 = 100$. This figure demonstrates that under a low budget constraint, the static baseline suffers from a high coverage variance that highly deviates from the target level. It requires a significant budget increase to eventually converge towards coverage values attained by our method with a less available budget. In stark contrast, DAPRO perfectly maintains valid empirical coverage ($\approx 90\%$) across all budget levels. Furthermore, as the nominal budget increases, our approach attains lower values of mean weight. This indicates a more efficient resource utilization of DAPRO.

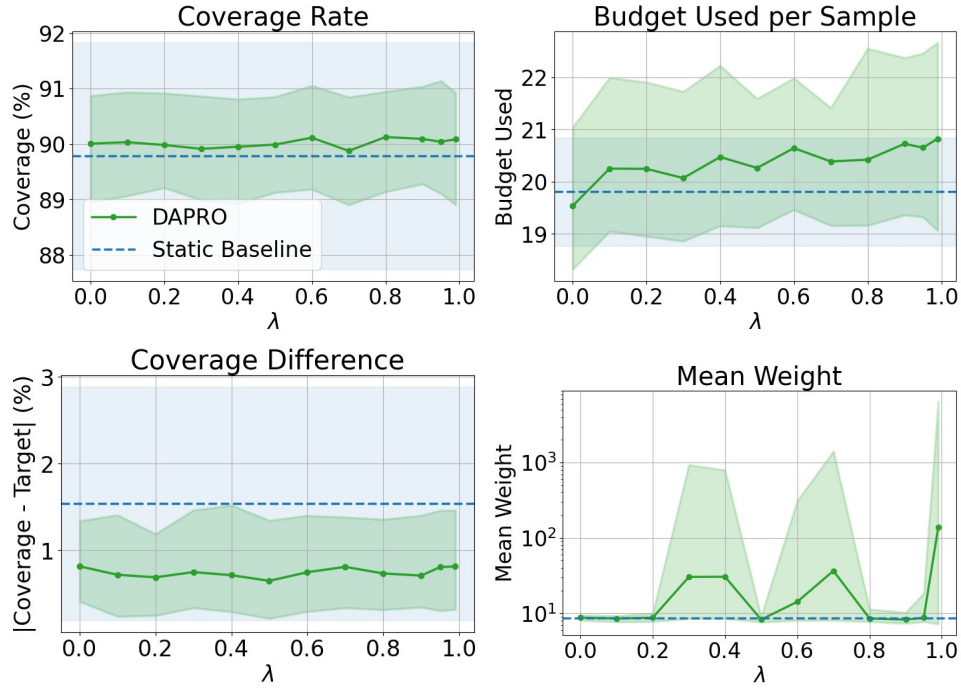


Figure 27: Impact of score informativeness (λ) on empirical coverage, budget consumed per sample, and mean weight on the Toxicity dataset. The Qwen 2.5 14B Instruct model serves as both attacker and target, with a first-split set size of $N_1 = 100$. Scores are corrupted by injecting random noise with level λ . The nominal coverage level is set to $1 - \alpha = 90\%$ and the target budget is 20. Shaded regions denote semi-deviations over 50 random calibration-test splits.

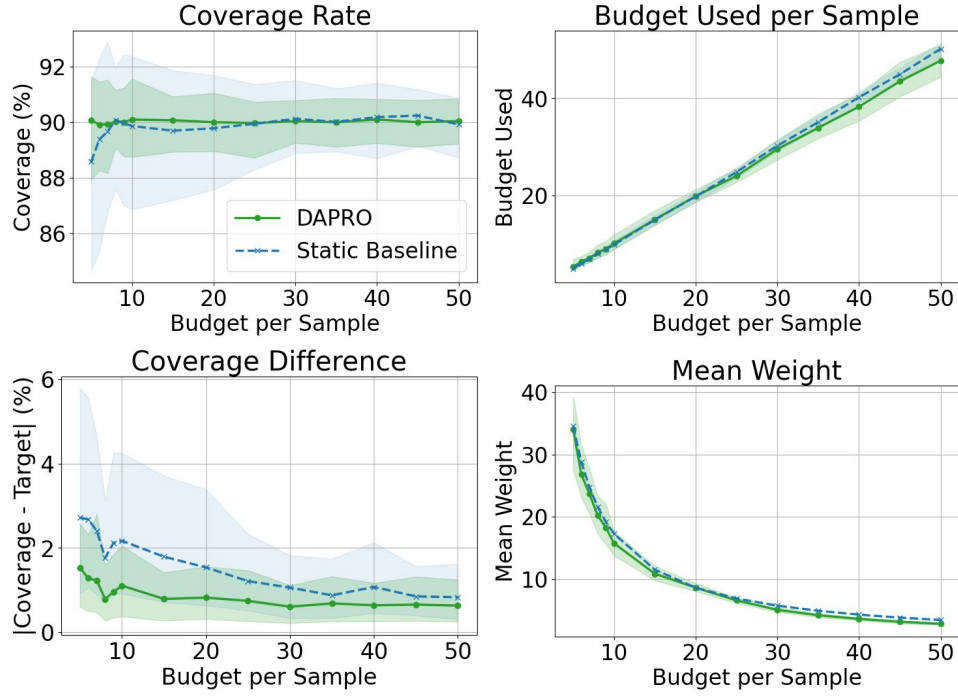


Figure 28: Impact of the nominal budget per sample on empirical coverage, budget consumed per sample, and mean allocation weight on the Toxicity dataset. The Qwen 2.5 14B Instruct model serves as both attacker and target. The target coverage level is $1 - \alpha = 0.90$. Shaded regions denote semi-deviations across 50 random calibration-test splits. DAPRO satisfies both coverage and budget constraints across all scenarios, whereas the static baseline exhibits high variance and coverage deviation under low-budget regimes.